

# Trust Me, I'm a Certificate Authority

**PyCon PT 2026 Tutorial**

**Tiago Montes**



What happens when you browse to **https://example.com/**?



What happens when you browse to **https://example.com/**?

Or when your code interacts with **https://api.example.net/**?

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# Trust Me, I'm a Certificate Authority

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Or when your code interacts with **https://api.example.net/**?

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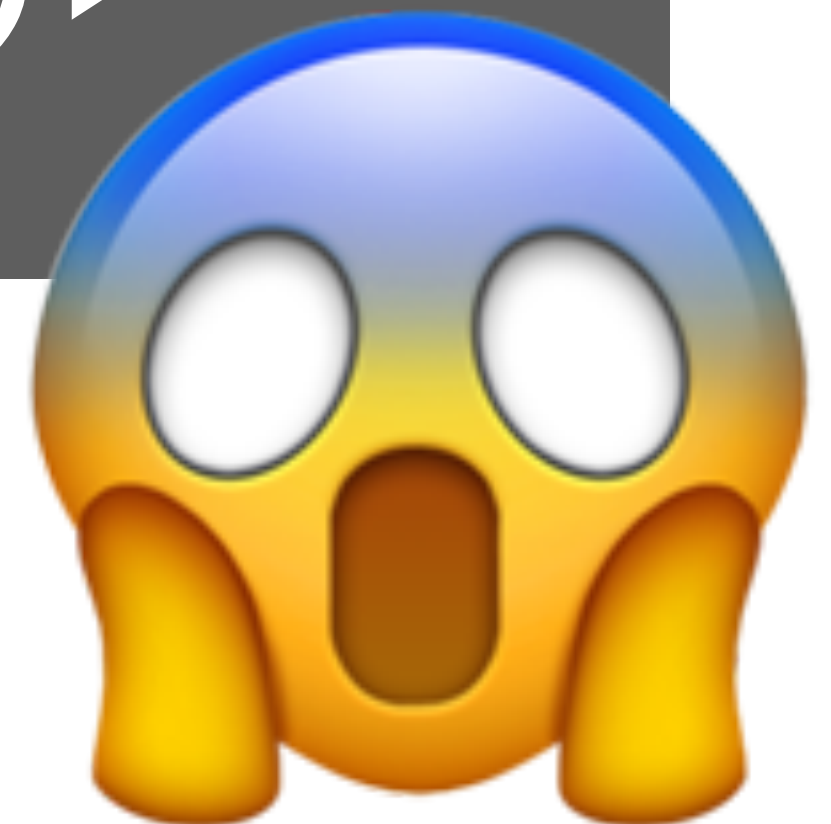




# Trust Me, I'm a Certificate Authority

understand why this shows up in your logs

```
urllib.error.URLError: <urlopen error  
[SSL: CERTIFICATE_VERIFY_FAILED]  
certificate verify failed: unable to get  
local issuer certificate (_ssl.c:1144)>
```



**welcome!**



# Cast of Characters

# Cast of Characters



**Alice**



# Cast of Characters



**Alice**



**Bob**

# Cast of Characters



**Alice wants**

**...to talk to  
Bob securely**



**Bob**



# Cast of Characters



**Alice wants**

...to talk to  
Bob securely



**Bob wants**

...to respond  
to Alice.

# Cast of Characters



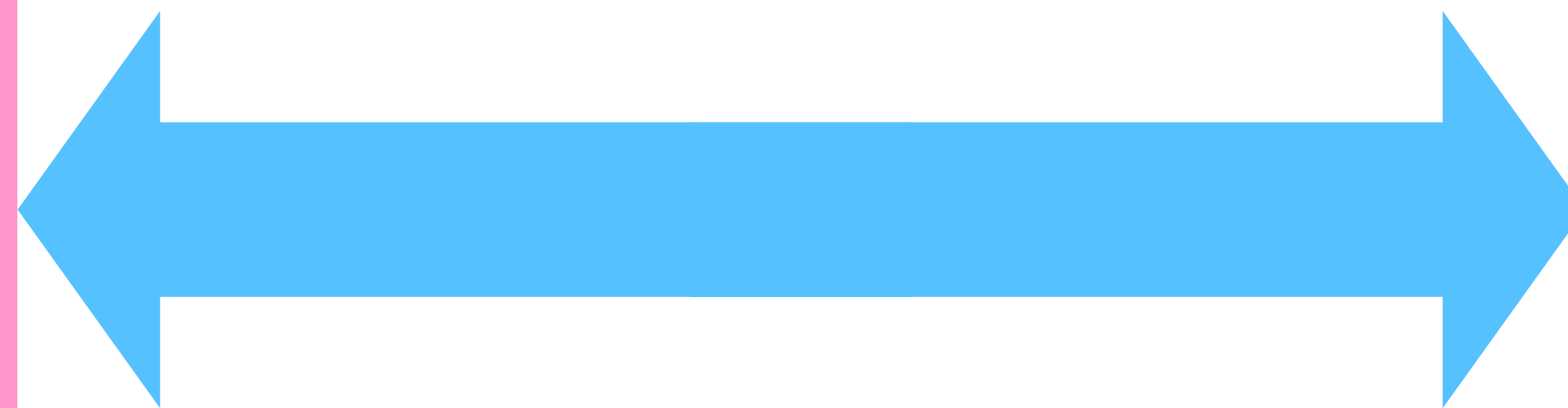
**Alice wants**

**...to talk to  
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**Bob wants**

**...to respond  
to Alice.**





# Cast of Characters



**Alice wants**

...to talk to  
Bob securely

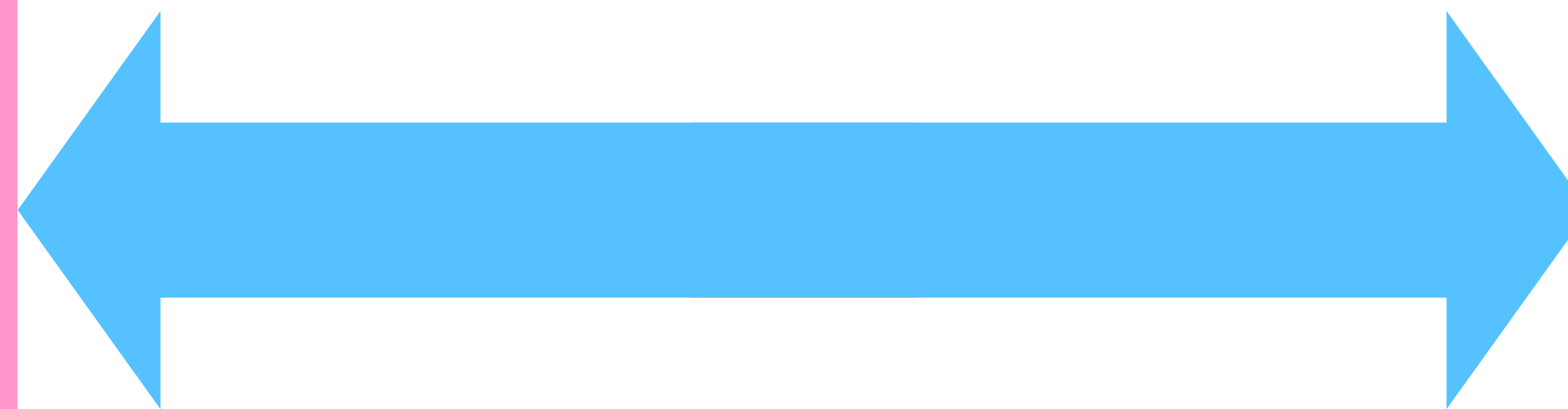


**Eve**



**Bob wants**

...to respond  
to Alice.



# Cast of Characters





# Cast of Characters



**Alice wants**

...to talk to  
Bob securely



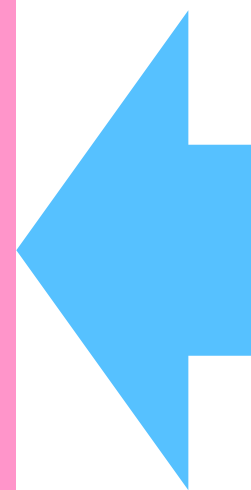
**Eve wants**

...to listen to  
or forge talks.



**Bob wants**

...to respond  
to Alice.



# DISCLAIMER

**THIS IS TEACHING MATERIAL, NOT SECURITY SOFTWARE.**

**IT CUTS CORNERS ON PURPOSE.**

**SEVERAL EXERCISES ARE DELIBERATELY BROKEN.**

**REAL SECURITY IS A GREAT DEAL MORE THAN THIS.**

**USE NONE OF IT FOR ANYTHING THAT MATTERS.**

**NEVER TRUST A CA YOU BUILT HERE.**



# Trust Me, I'm a Certificate Authority

|                     |                                                |
|---------------------|------------------------------------------------|
| <b>RSA</b>          | Keys, encryption, and signatures.              |
| <b>Certificates</b> | Binding keys to identities (yay X.509)         |
| <b>CAs and TLS</b>  | Certificate Authorities, HTTPS clients/servers |
| <b>MITM</b>         | Interception: where trust actually lives       |
| <b>Break Stuff</b>  | Certificates that fail, checks that catch them |

# Setup



# Setup

Python  $\geq$  3.12 and 3.15rc2 works

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Clone a GitHub repository...



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**with uv**

```
$ uv sync
```

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Python  $\geq$  3.12 and 3.15rc2 works

Clone a GitHub repository...

**with uv**

```
$ uv sync
```

**without uv**

```
$ python3 -m venv .venv  
$ source .venv/bin/activate  
$ python3 -m pip install -r requirements.txt
```



# Setup

Python  $\geq$  3.12 and 3.15rc2 works

Clone a GitHub repository...

## with uv

```
$ uv sync
```

## without uv

```
$ python3 -m venv .venv  
$ source .venv/bin/activate  
$ python3 -m pip install -r requirements.txt
```

- Use **python.exe** instead of **python3**.
- Activate with **.venv\Scripts\activate**, instead.

windows

# Setup

Python  $\geq$  3.12 and 3.15rc2 works

Clone a GitHub repository...

## with uv

```
$ uv sync
```

Optionally activate the virtual environment.

## without uv

```
$ python3 -m venv .venv  
$ source .venv/bin/activate  
$ python3 -m pip install -r requirements.txt
```

- Use **python.exe** instead of **python3**.
- Activate with **.venv\Scripts\activate**, instead.

windows



# Setup

Python  $\geq$  3.12 and 3.15rc2 works

Clone a GitHub repository...

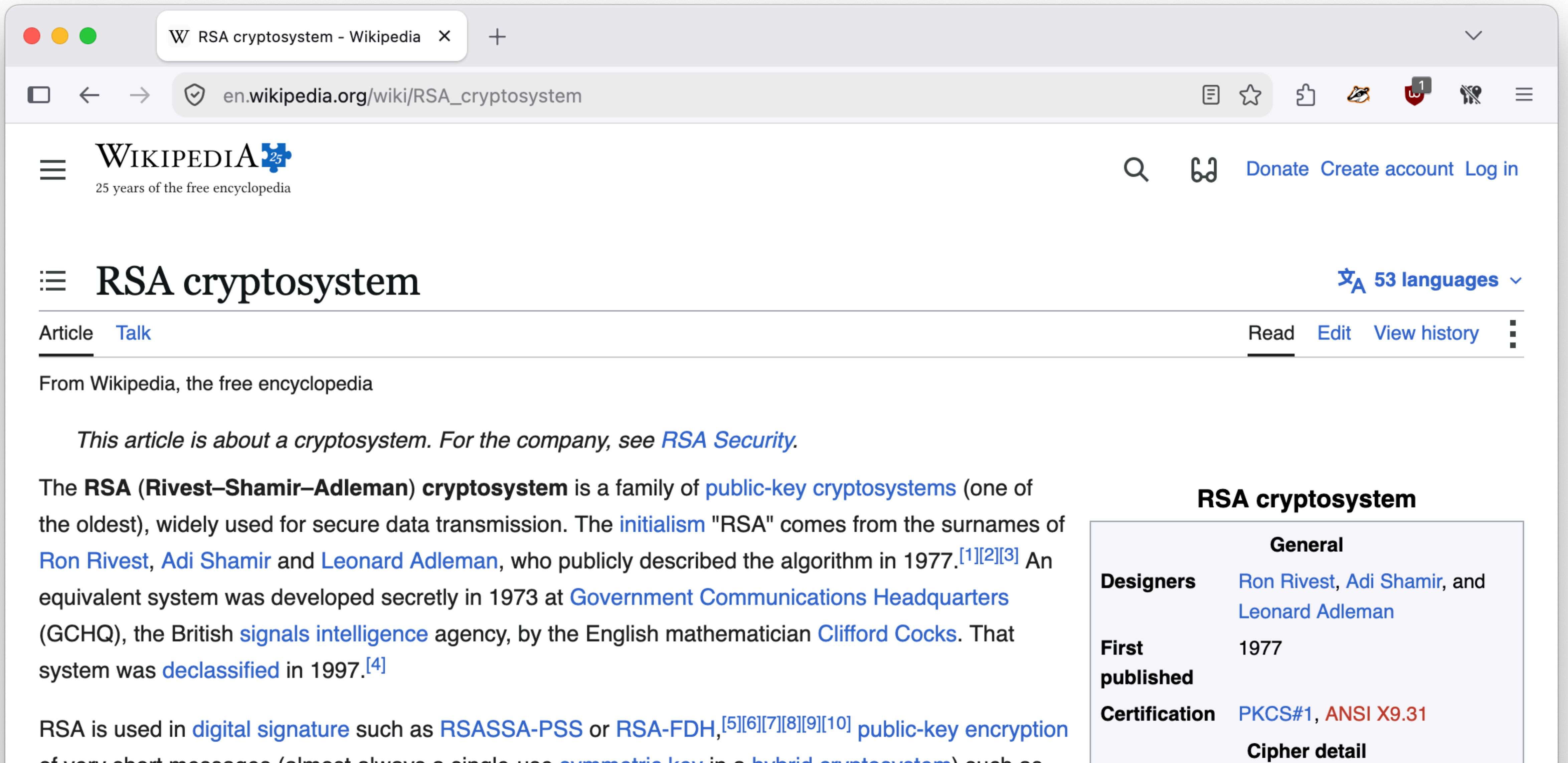
# README.md

...no worries, it's all there!

# **RSA** (Rivest–Shamir–Adleman)



# RSA (Rivest–Shamir–Adleman)



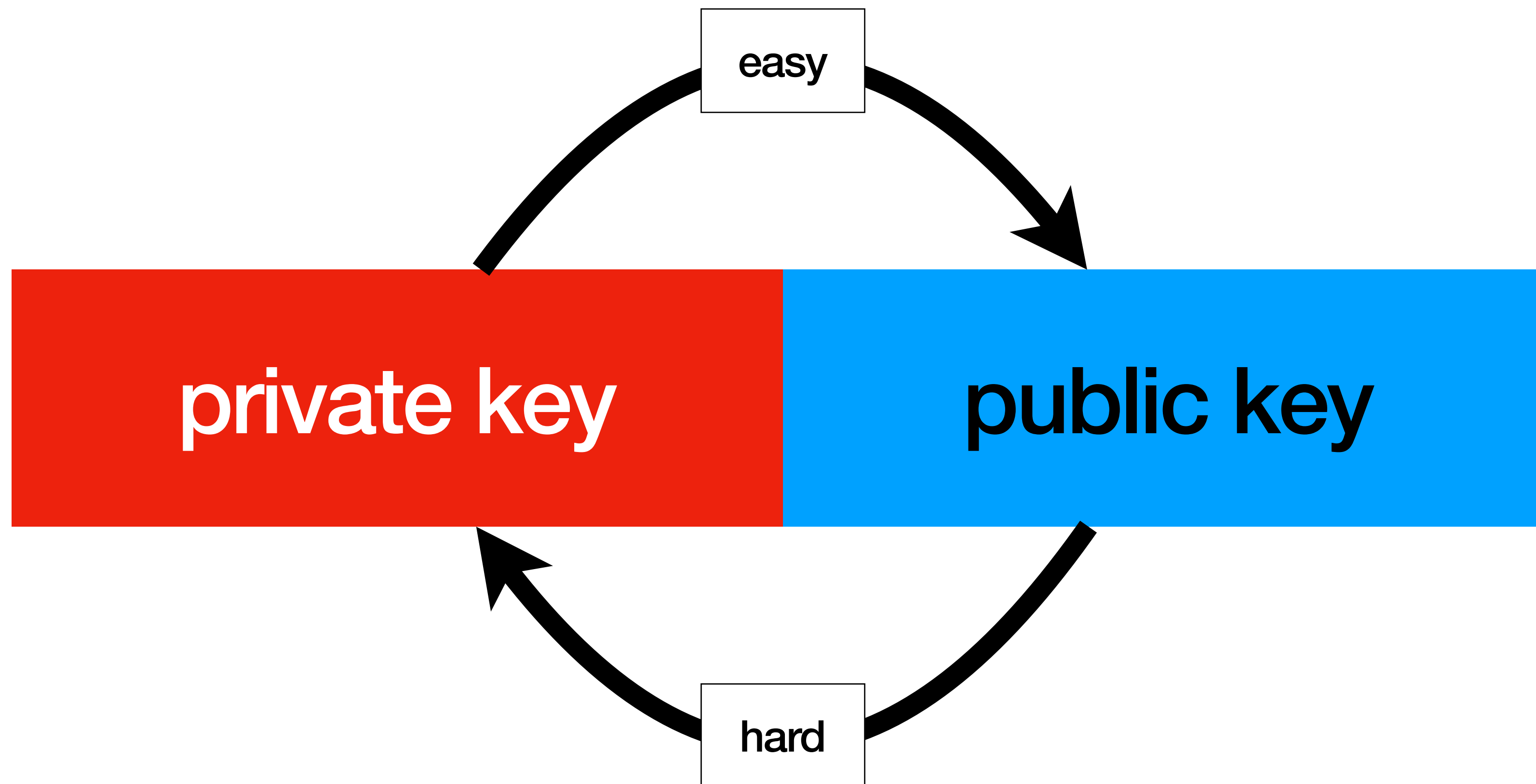
# **RSA key pair**

**private key**

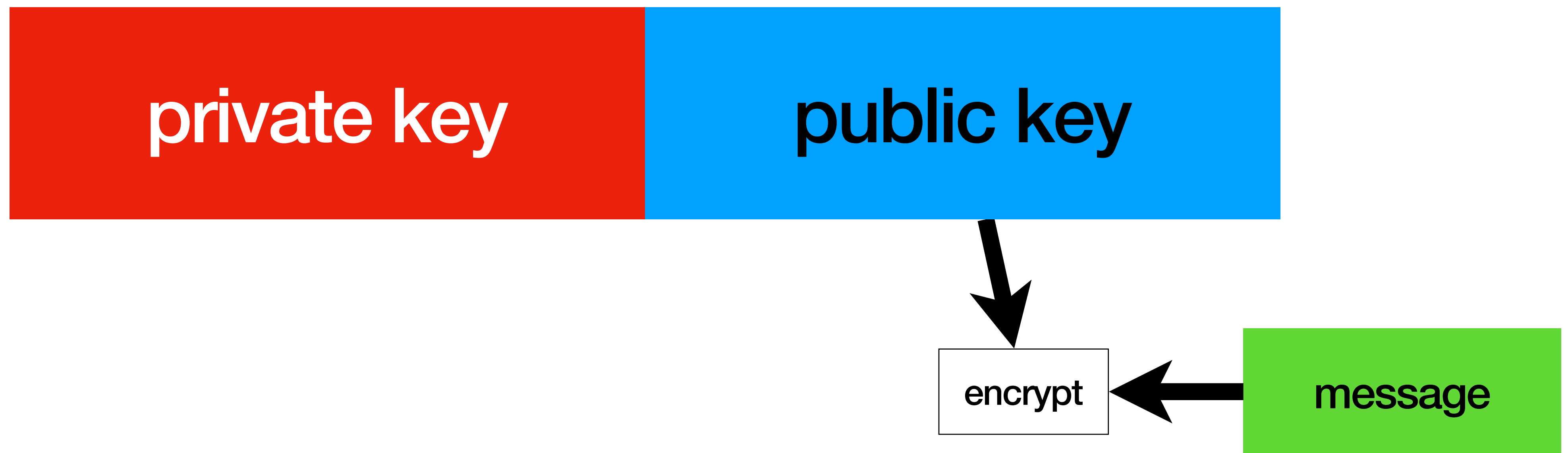
**public key**



# RSA key pair

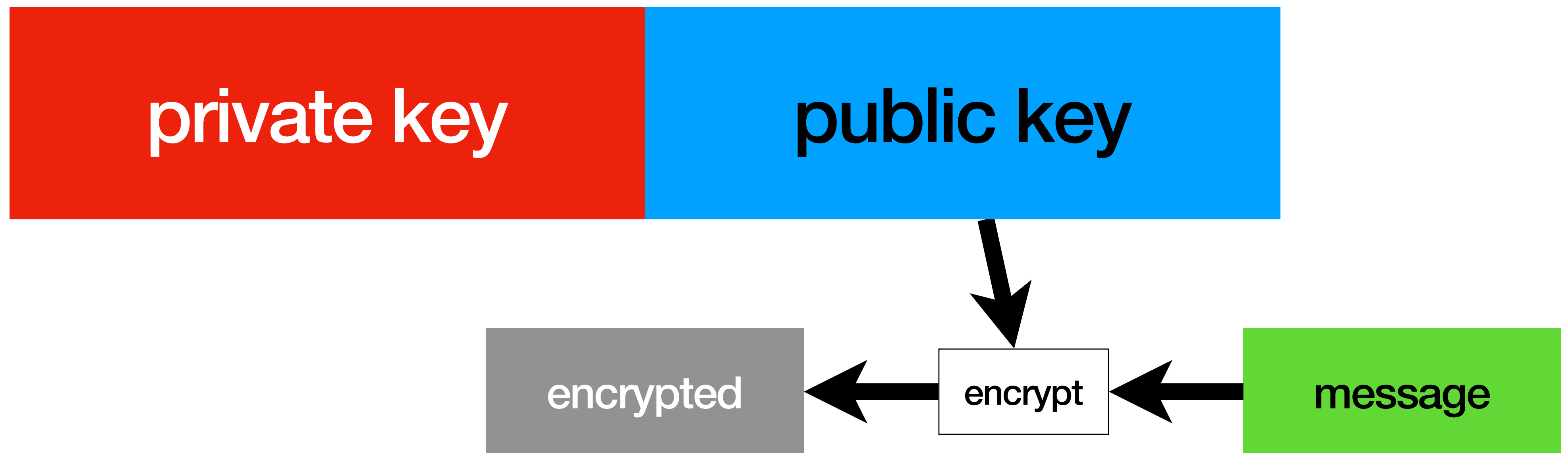


# RSA operations

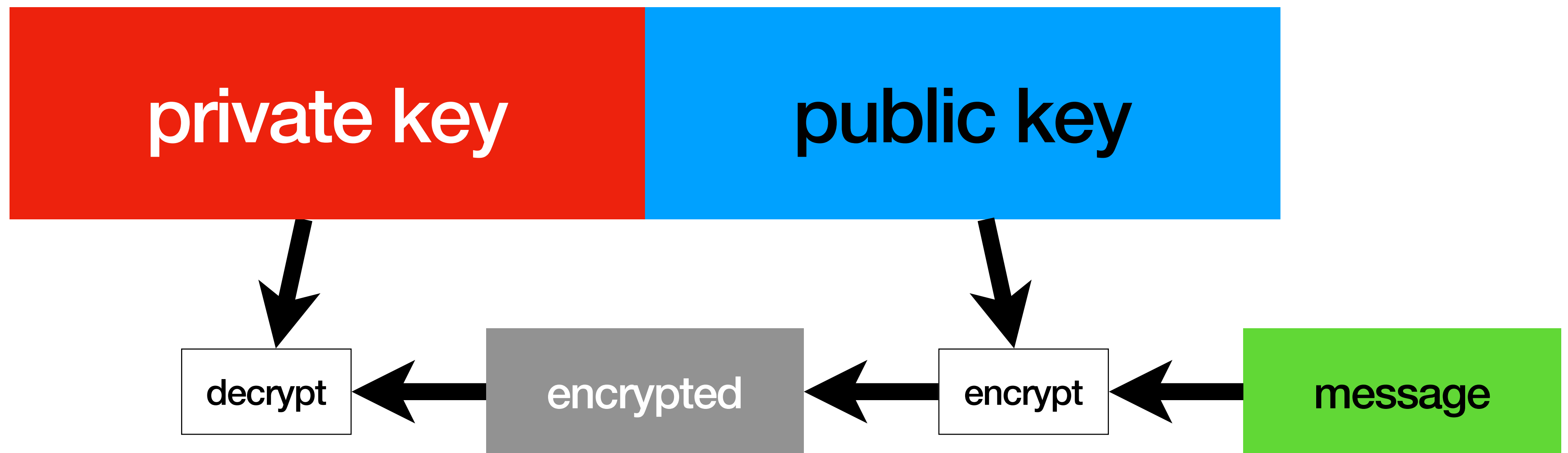




# RSA operations

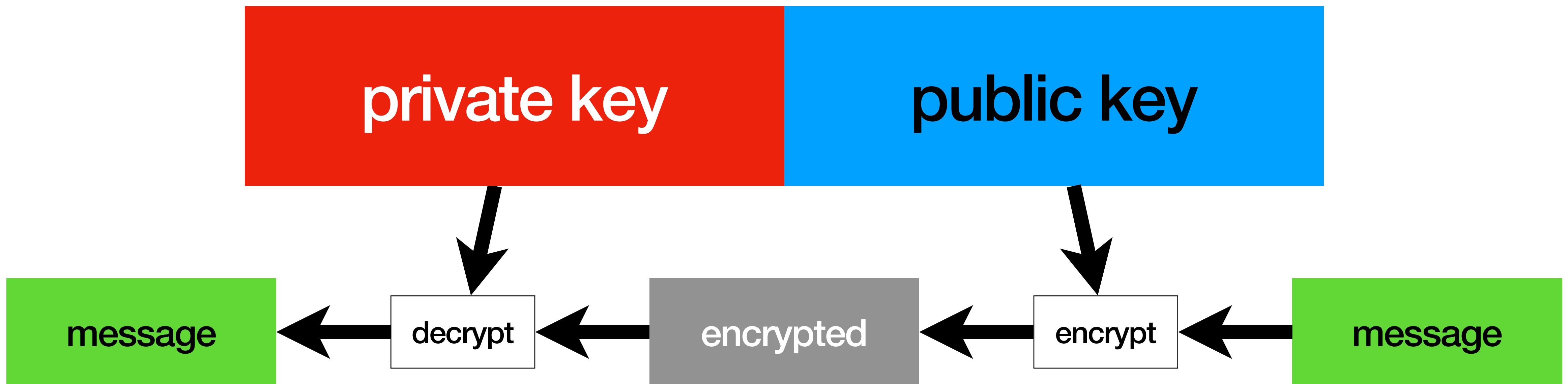


# RSA operations

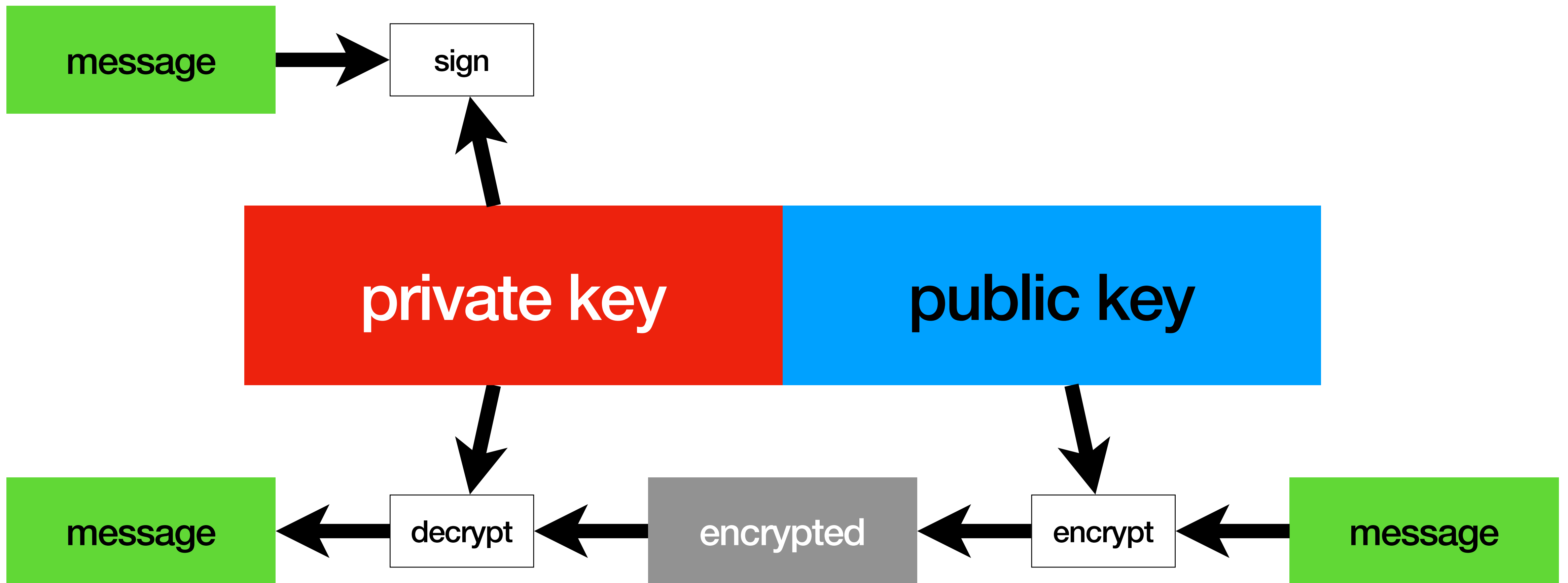




# RSA operations

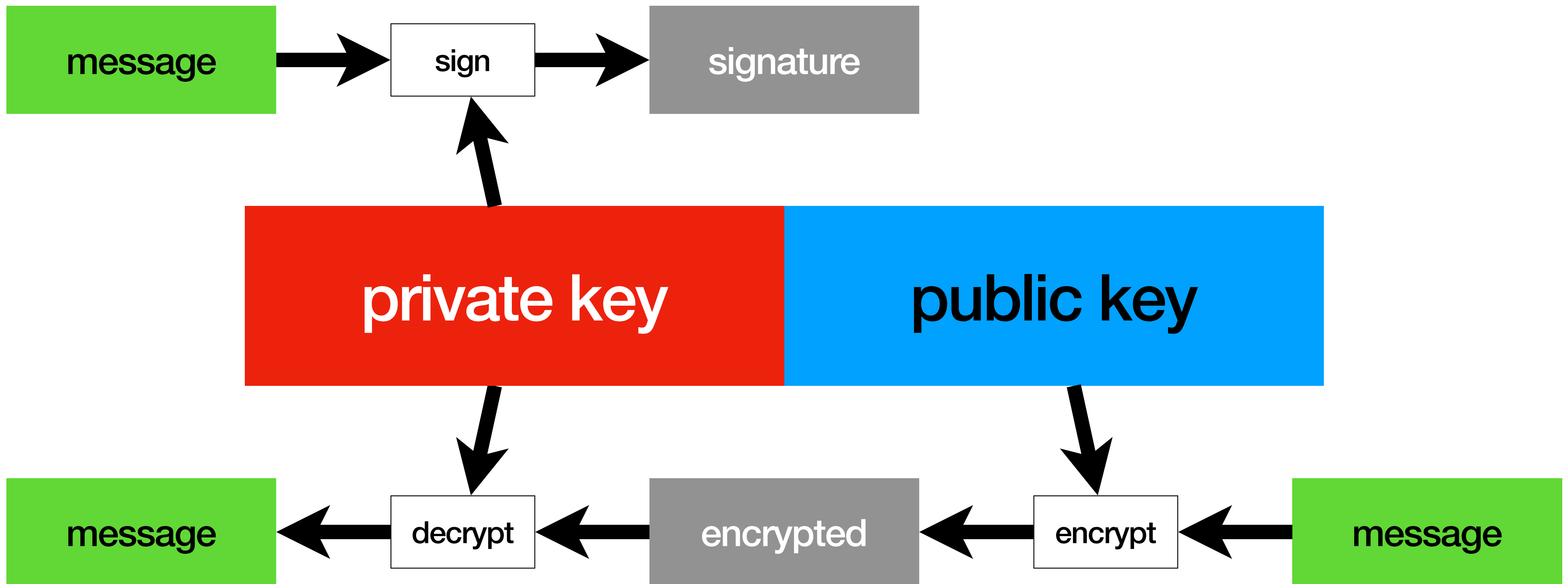


# RSA operations

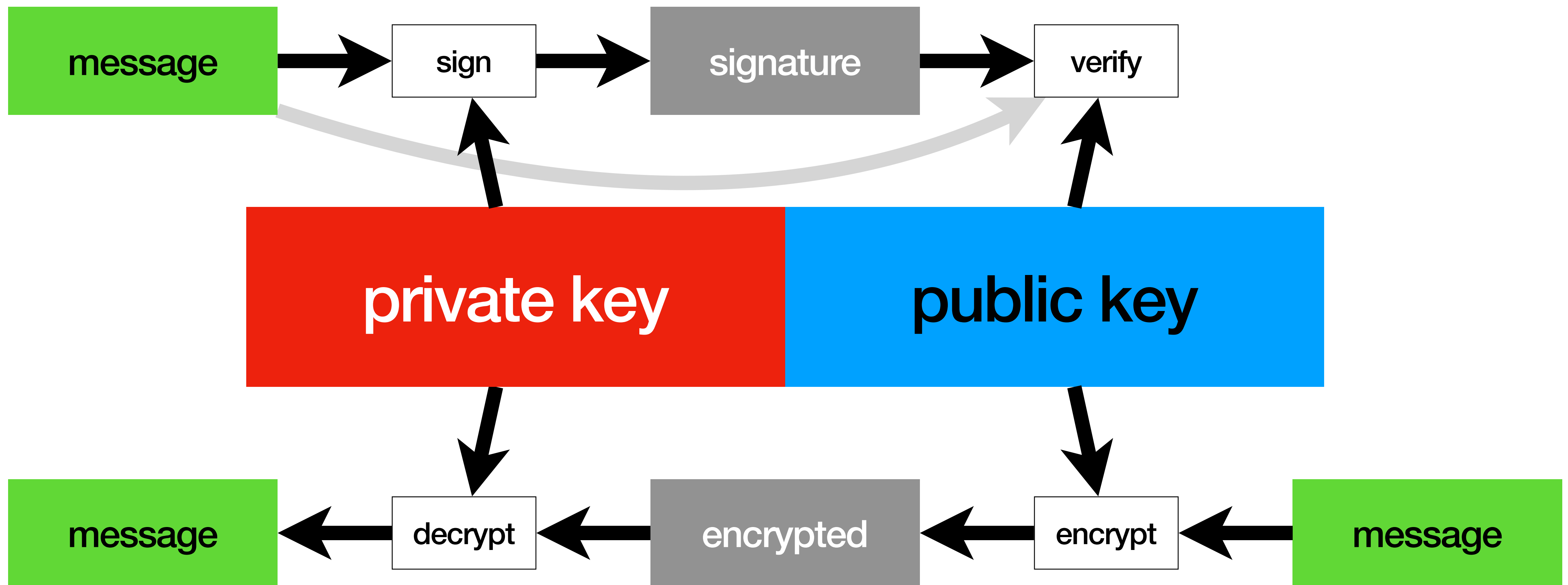




# RSA operations

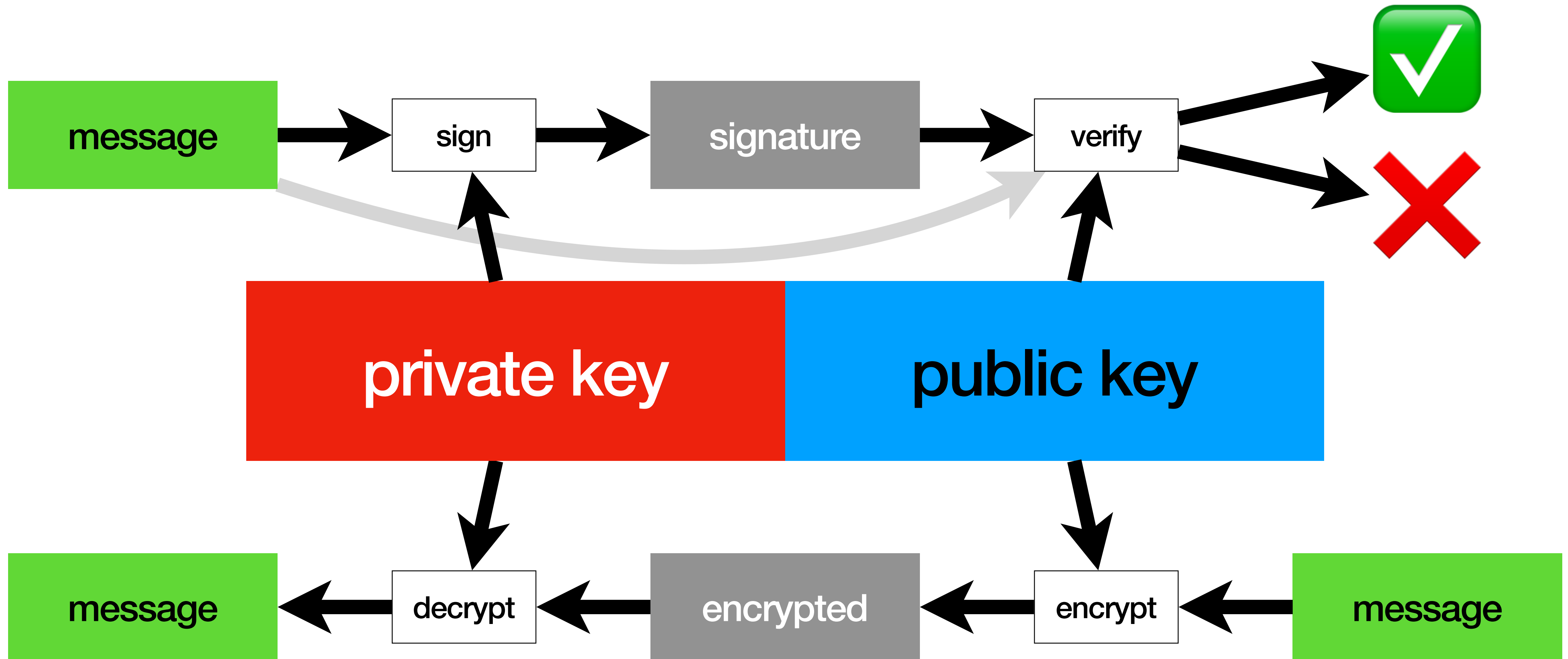


# RSA operations





# RSA operations

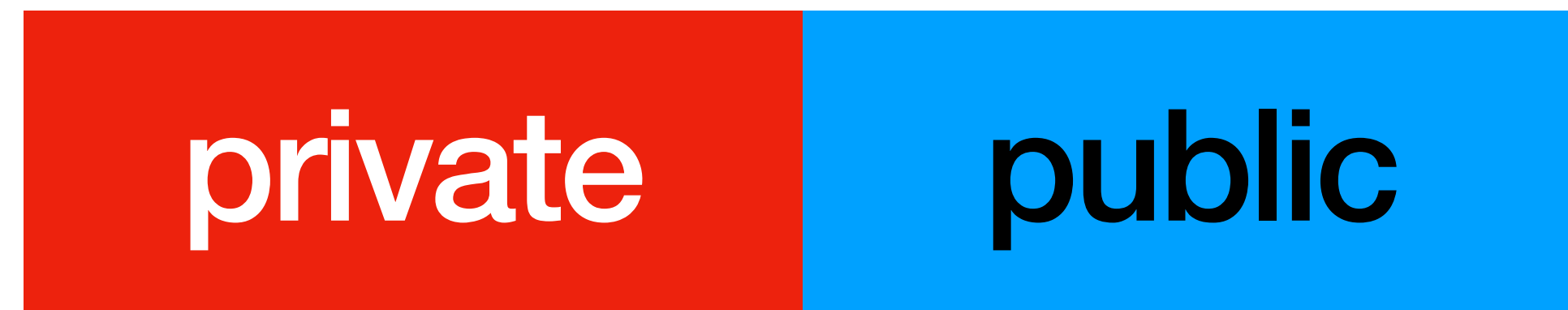


# **RSA private communication**

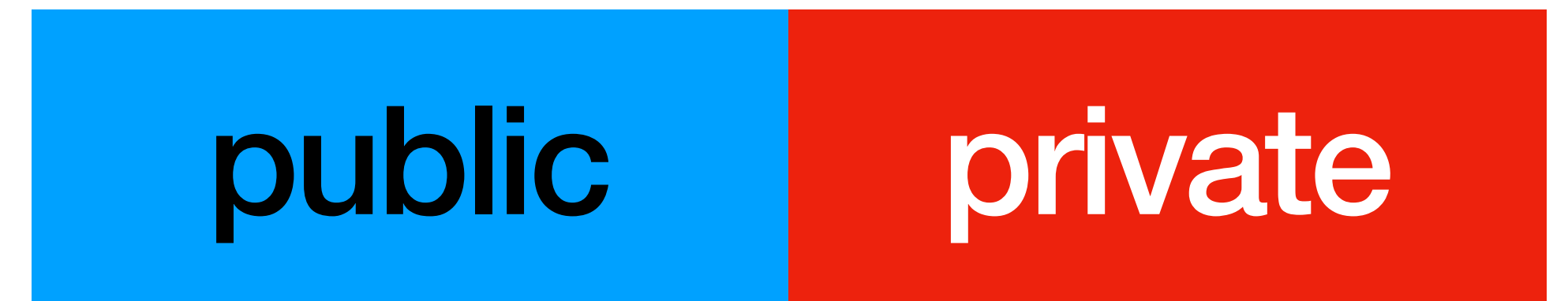


# RSA private communication

Alice

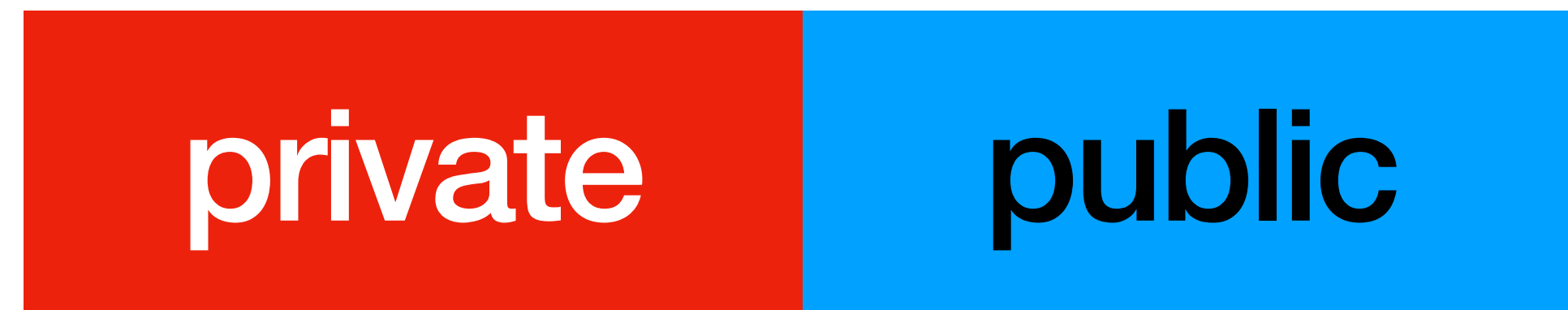


Bob

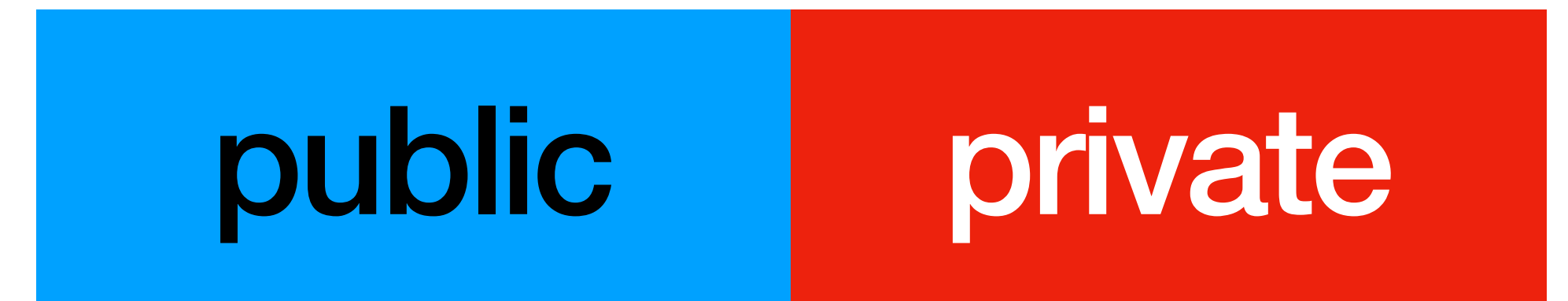


# RSA private communication

Alice



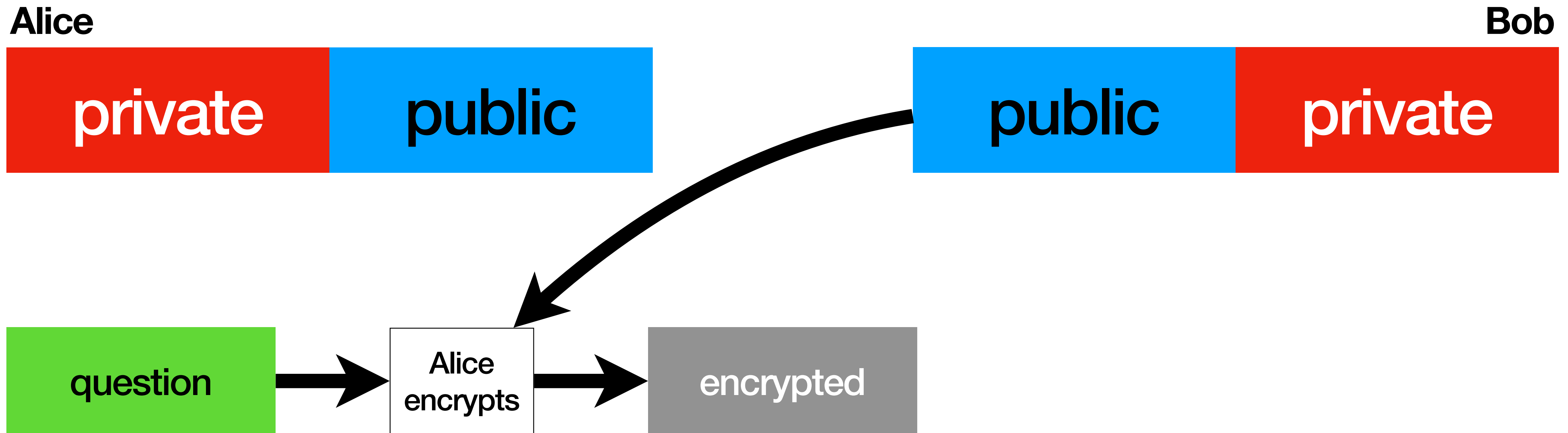
Bob



**Alice** sending a private question to **Bob**

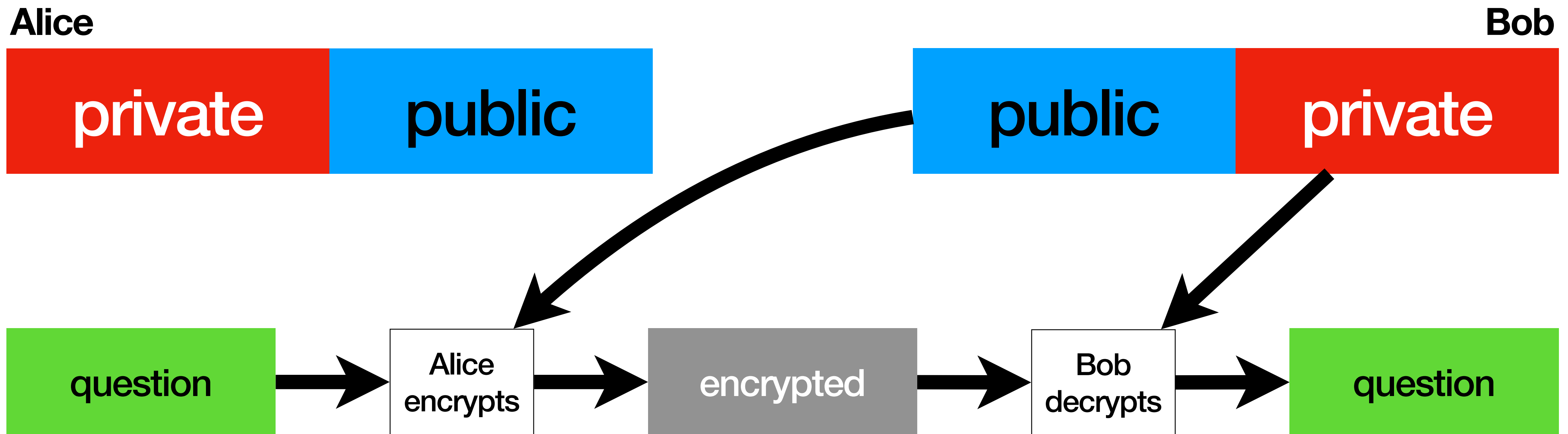


# RSA private communication



**Alice** sending a private question to **Bob**

# RSA private communication

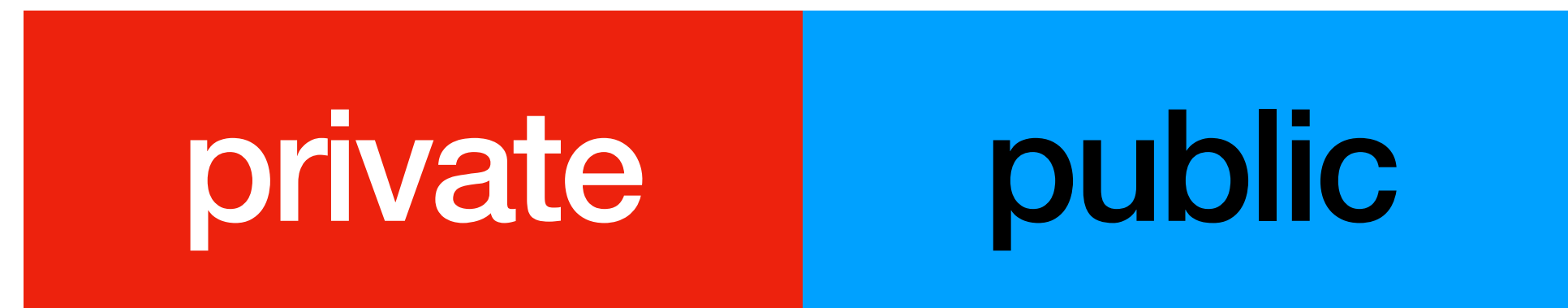


**Alice** sending a private question to **Bob**

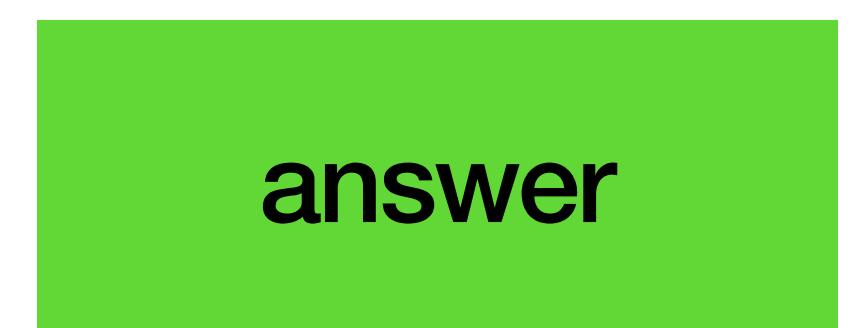
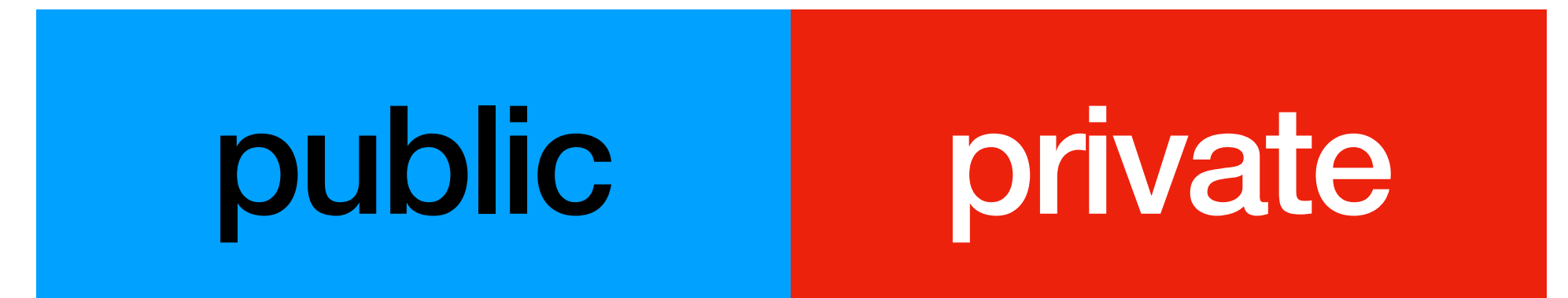


# RSA private communication

Alice

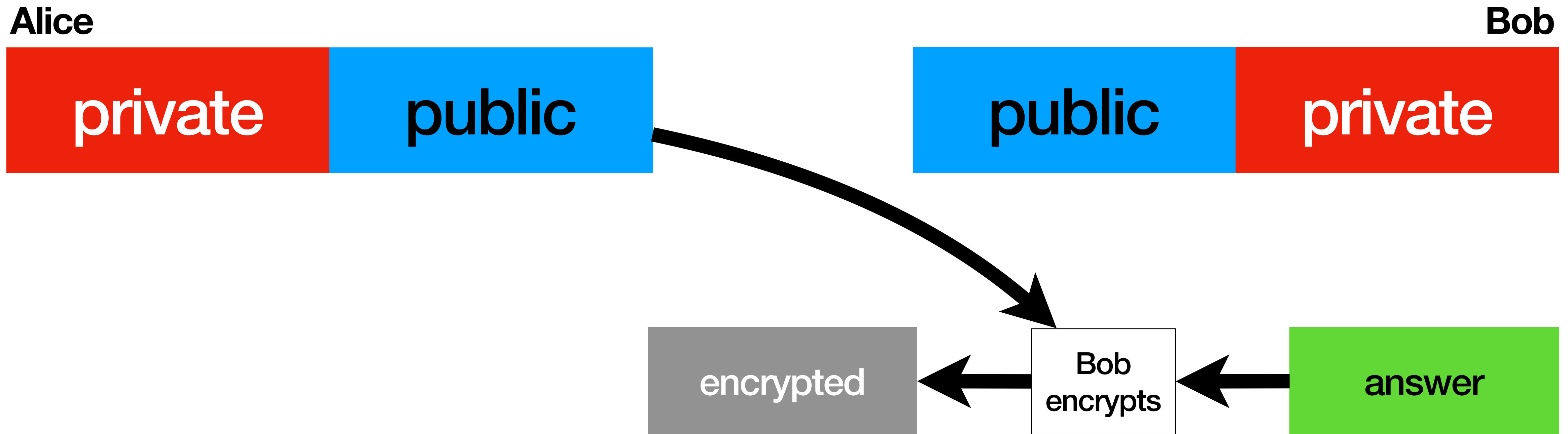


Bob



**Bob** sending a private answer to **Alice**

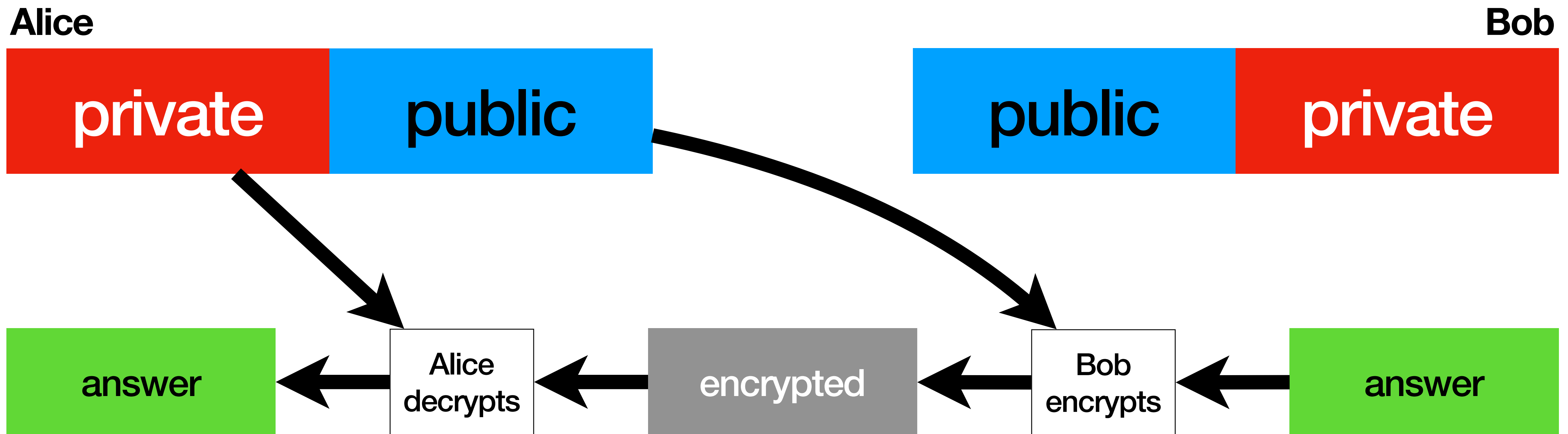
# RSA private communication



**Bob** sending a private answer to **Alice**



# RSA private communication



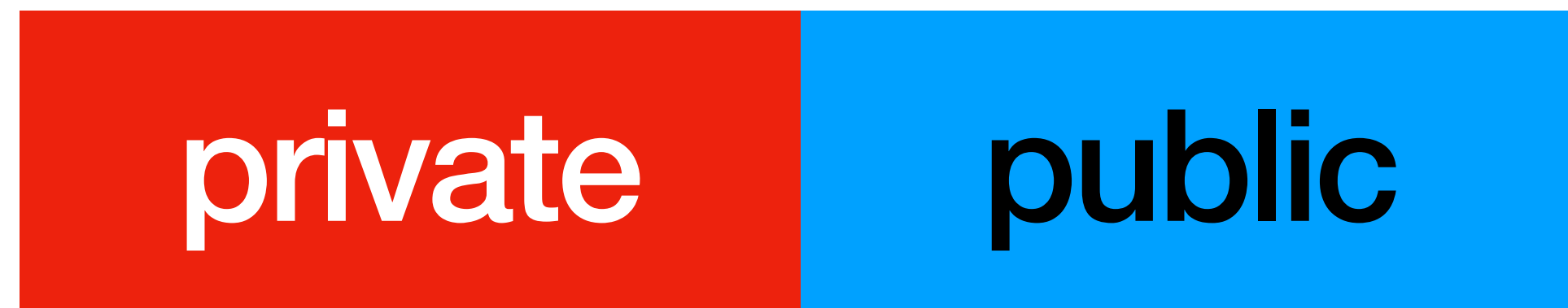
**Bob** sending a private answer to **Alice**

# **RSA signing / authentication**

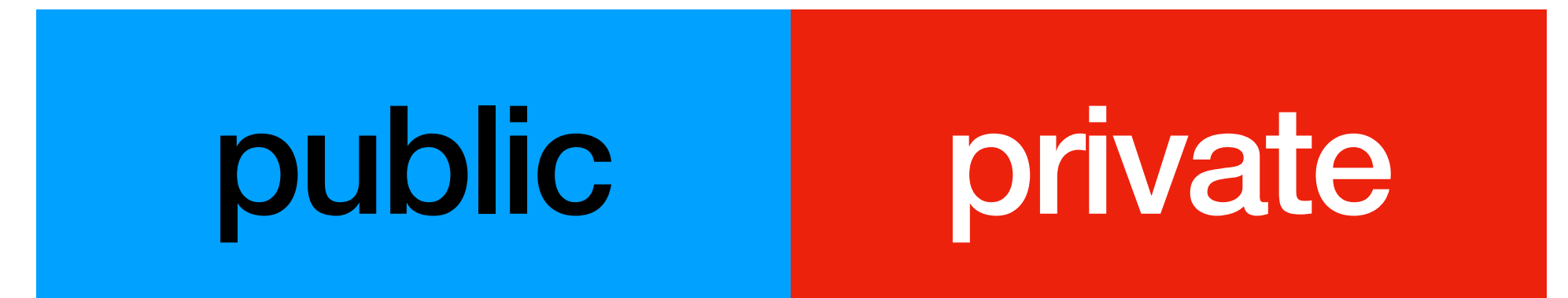


# RSA signing / authentication

Alice

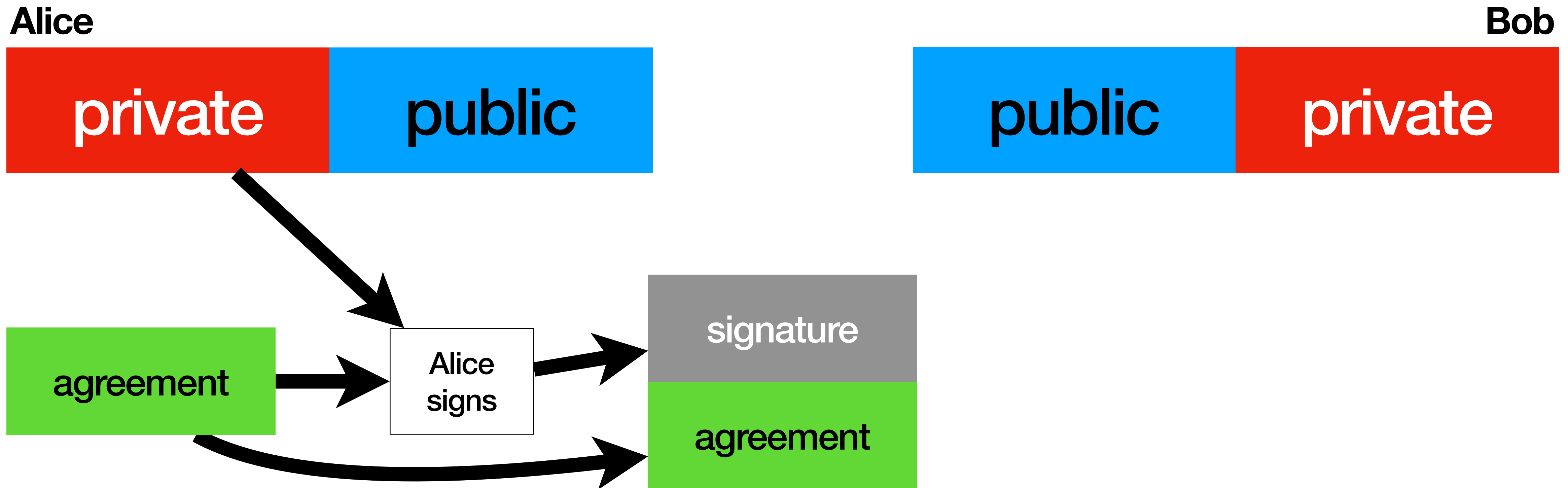


Bob



**Alice** sending an authenticated agreement to **Bob**

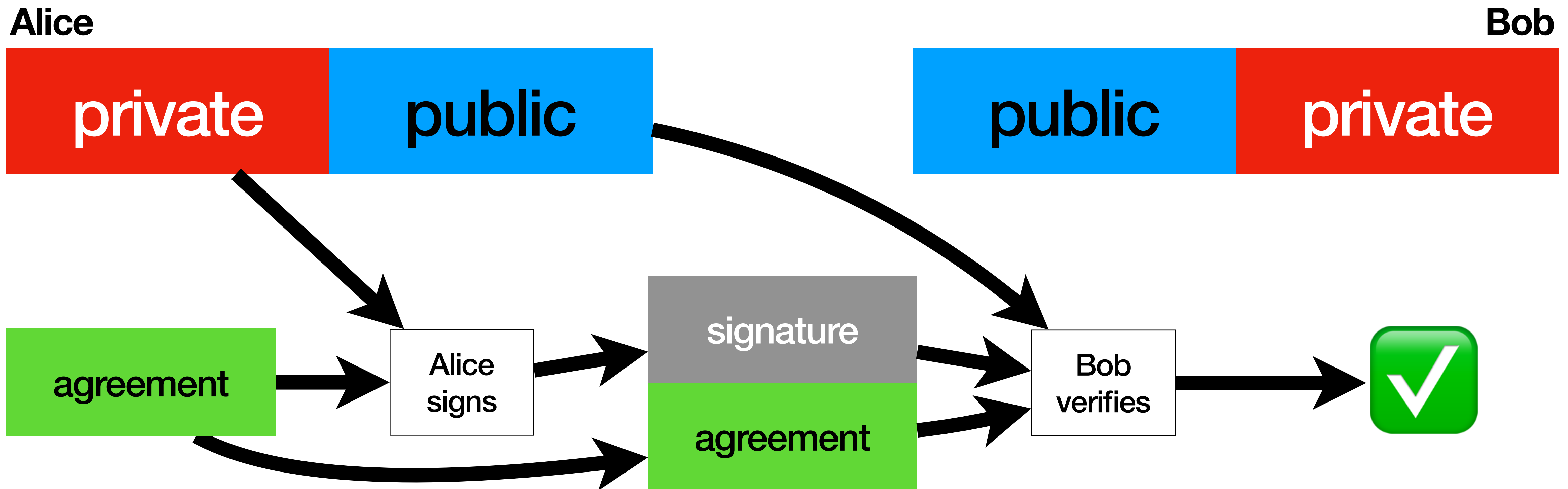
# RSA signing / authentication



**Alice** sending an authenticated agreement to **Bob**



# RSA signing / authentication



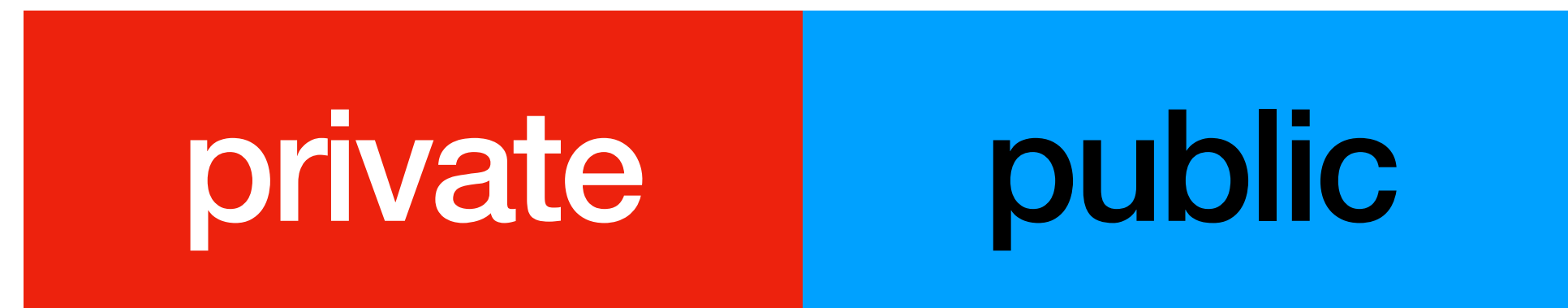
**Alice** sending an authenticated agreement to **Bob**

# **RSA private and authenticated**

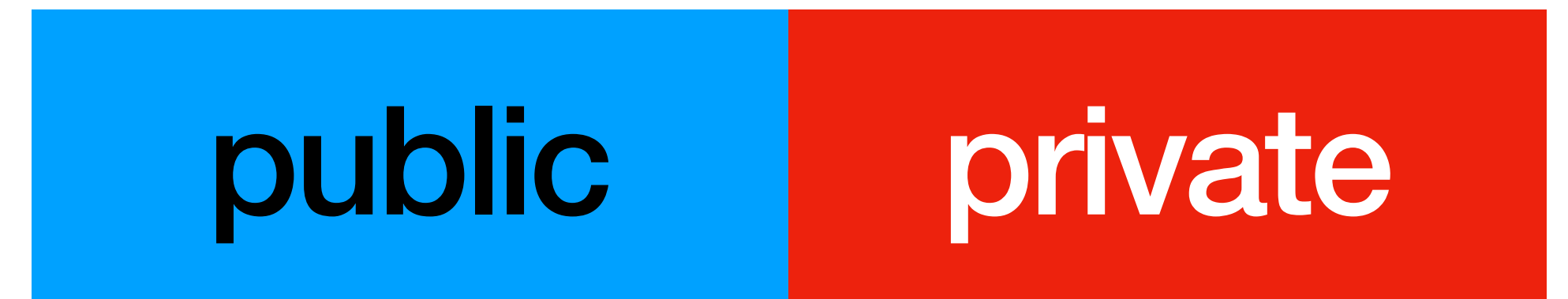


# RSA private and authenticated

Alice

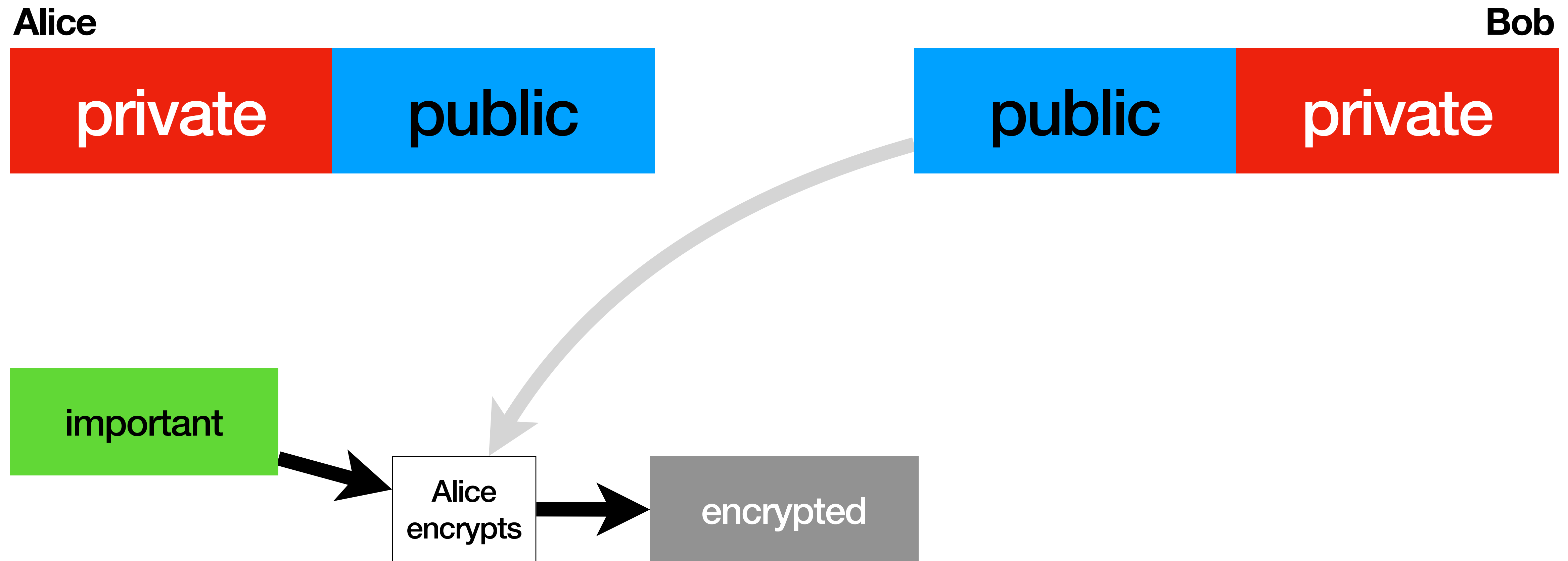


Bob



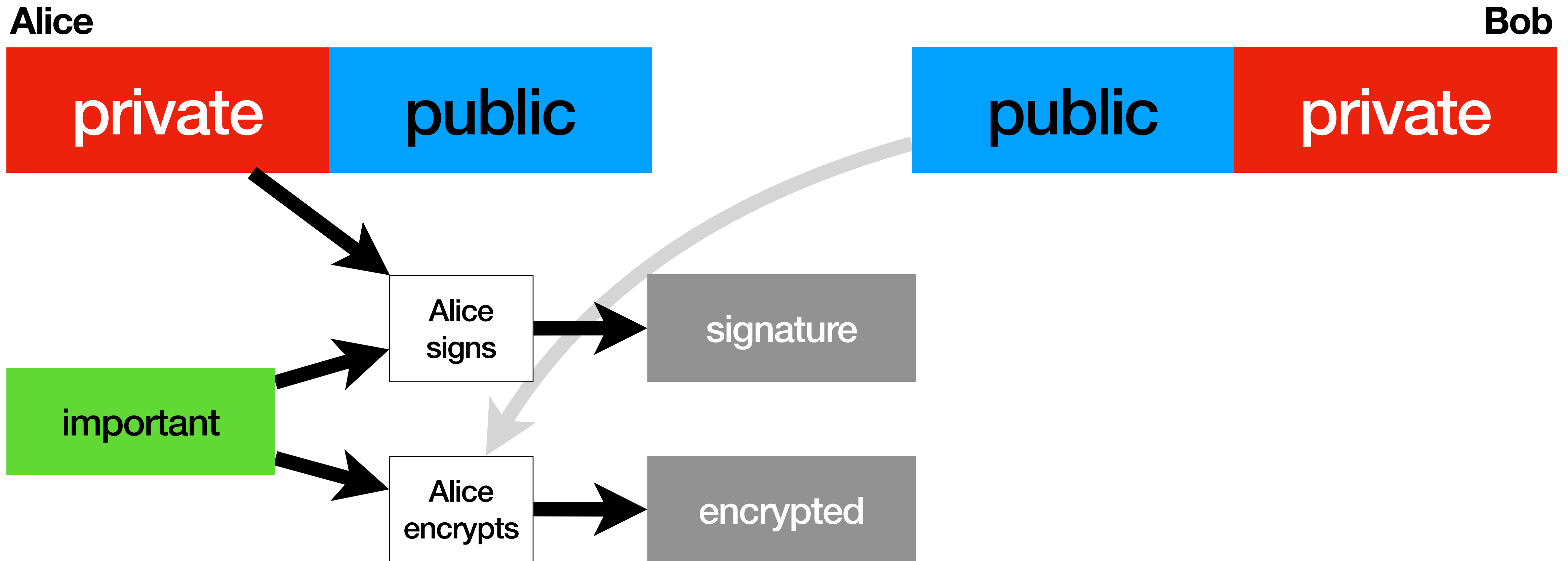
important

# RSA private and authenticated

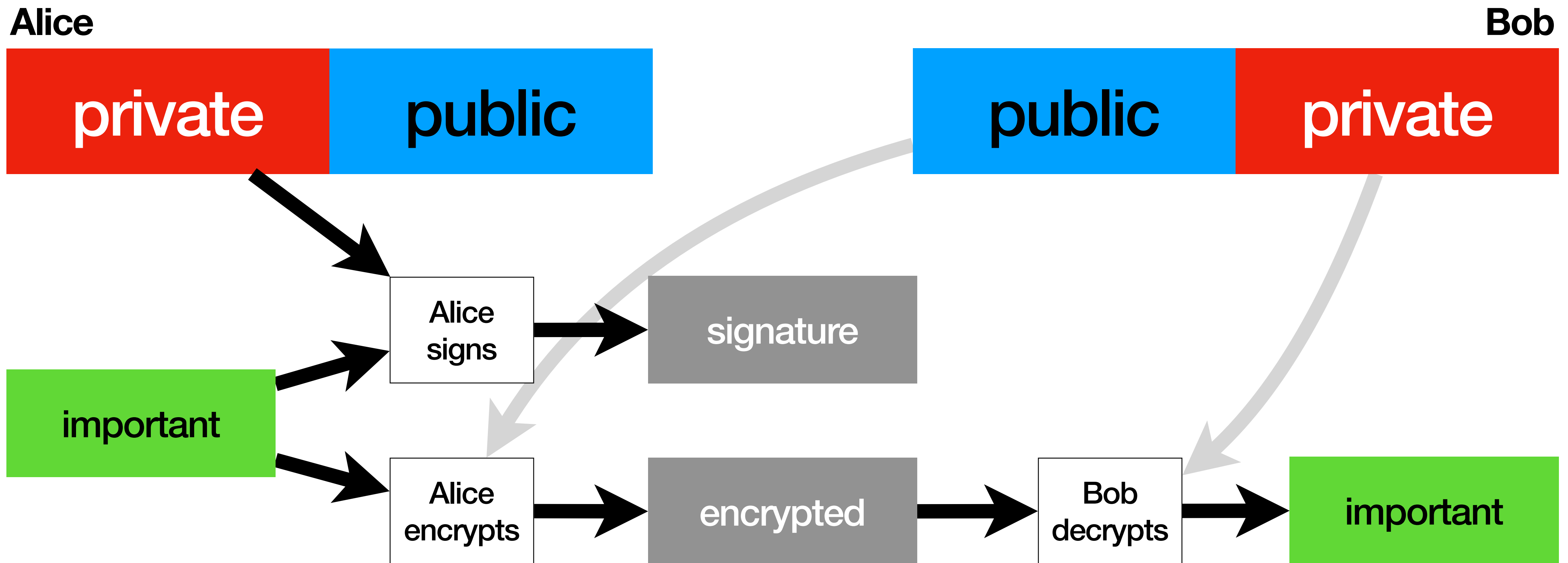




# RSA private and authenticated

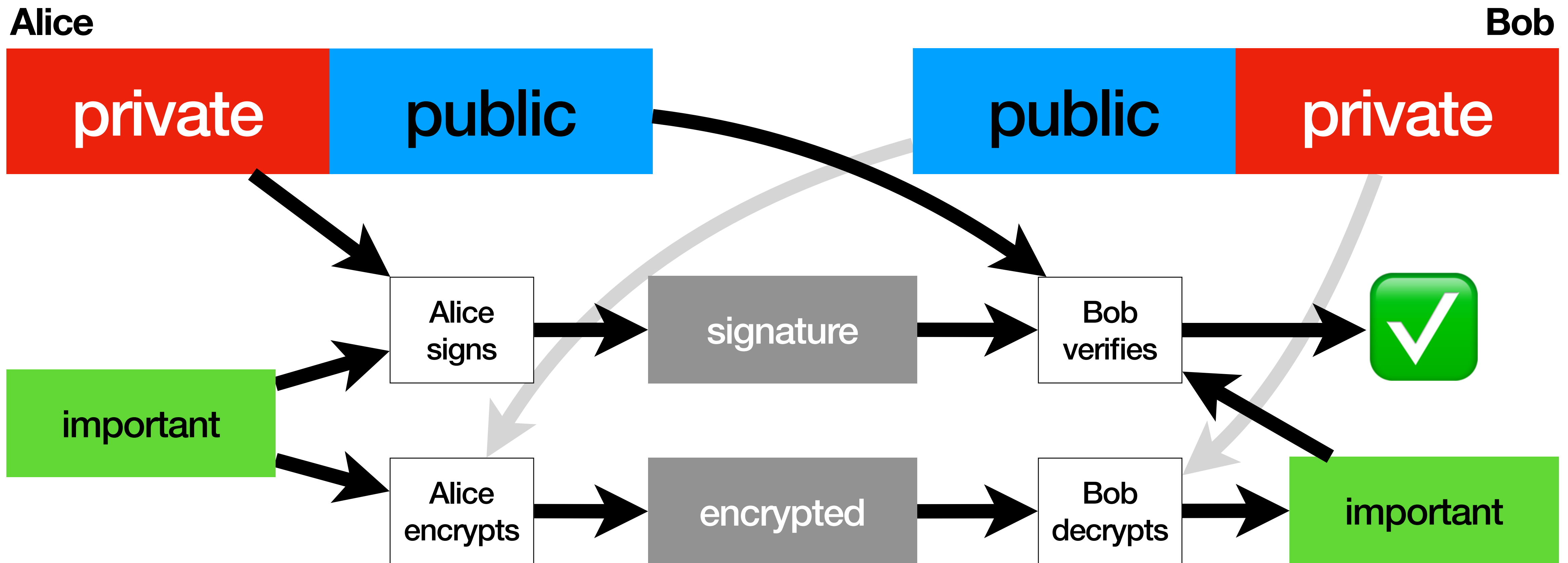


# RSA private and authenticated





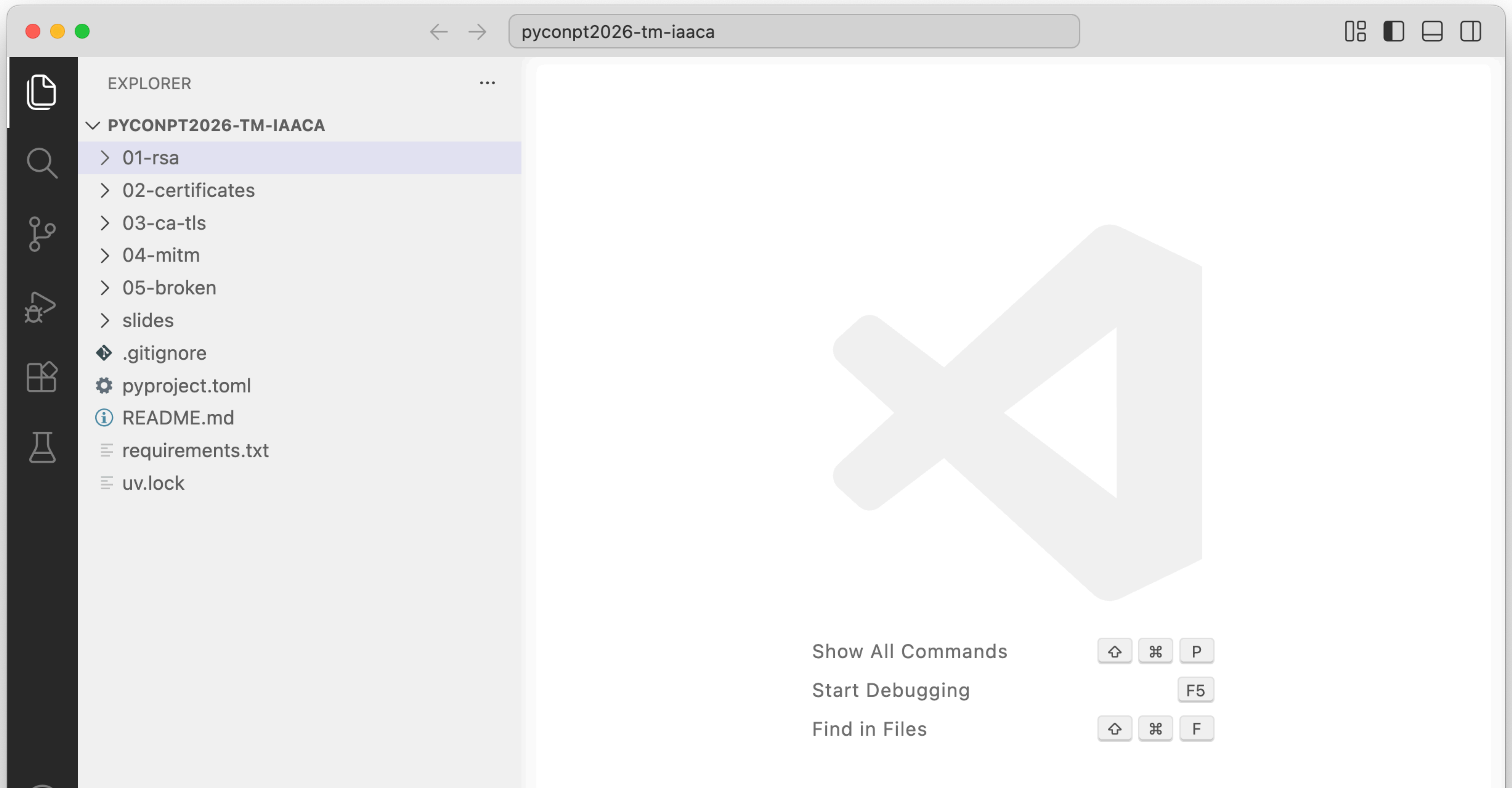
# RSA private and authenticated



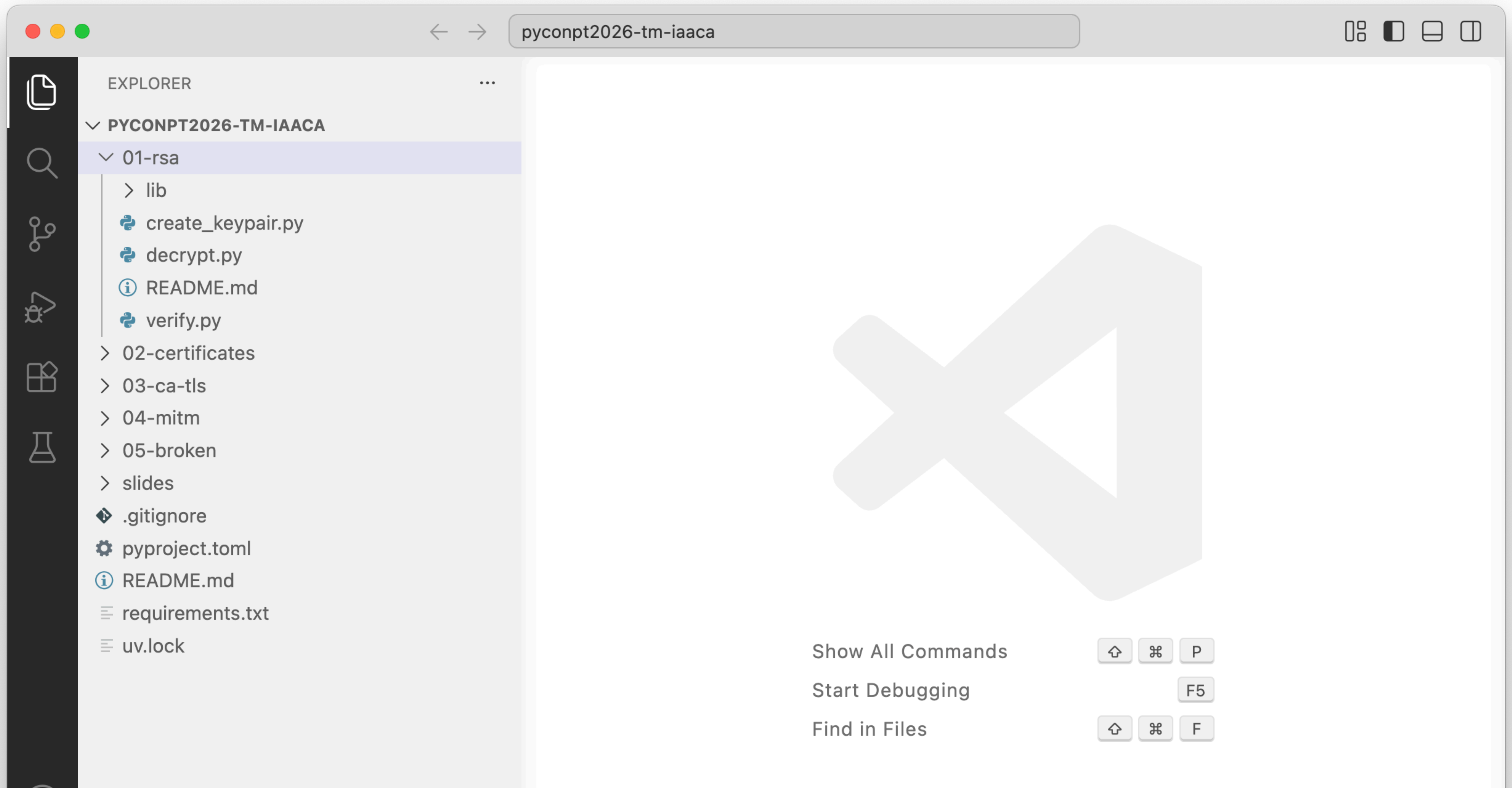
**RSA**



# <https://github.com/tmontes/pyconpt2026-tm-iaaca>

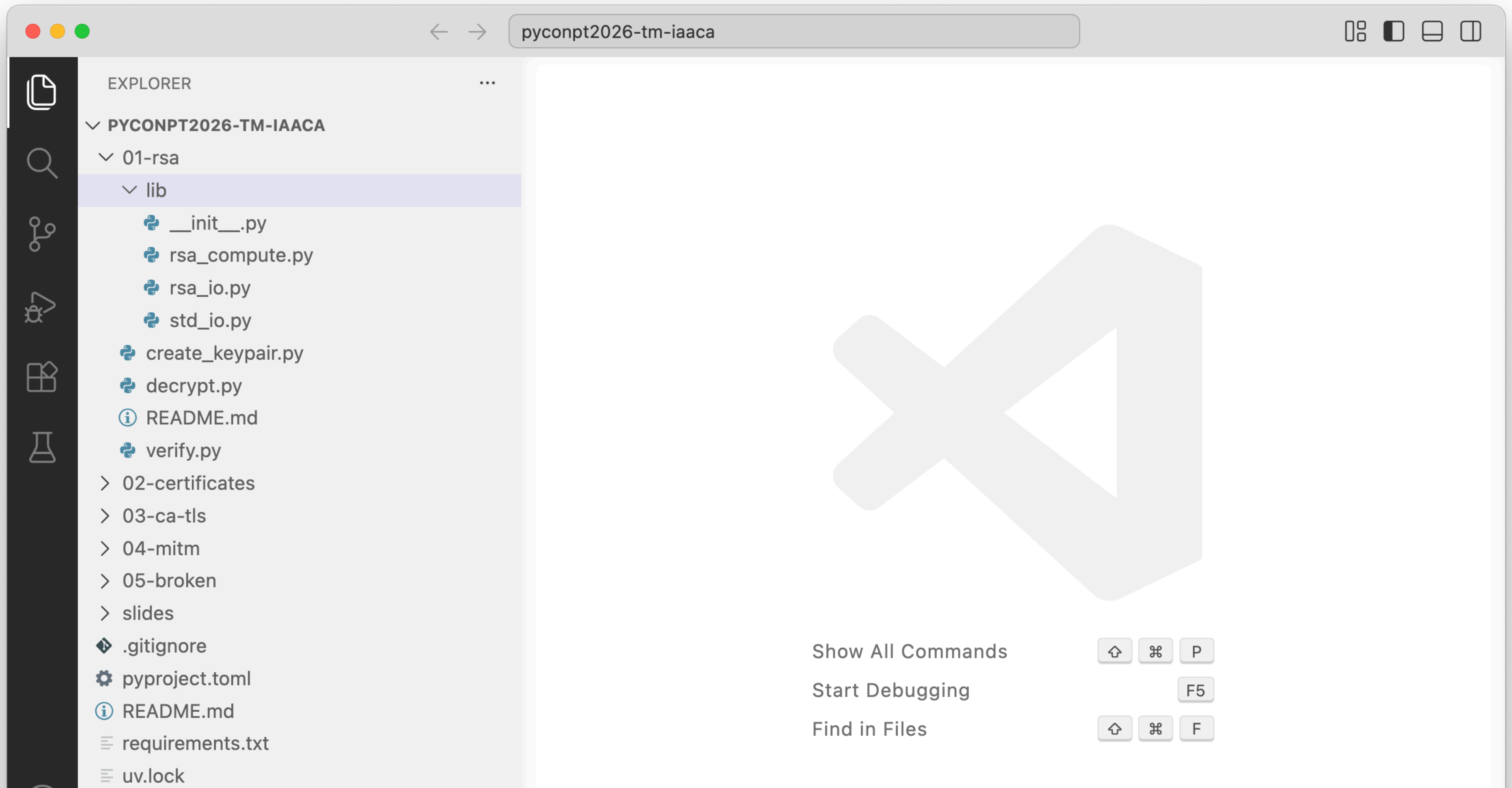


# <https://github.com/tmontes/pyconpt2026-tm-iaaca>





# <https://github.com/tmontes/pyconpt2026-tm-iaaca>



# **RSA**

**<https://github.com/tmontes/pyconpt2026-tm-iaaca>**



# **RSA**

**no way this is enough**

# X.509 Certificates



# X.509 Certificates

**issuer name**

**subject name**

**valid from**

**valid to**

**public key**

**signature**

# X.509 Certificates

**issuer name**  
**subject name**  
**valid from**  
**valid to**  
**public key**  
**signature**

**Issuer** certifies,  
by **signing** the document:



# X.509 Certificates

**issuer name**  
**subject name**  
**valid from**  
**valid to**  
**public key**  
**signature**

**Issuer** certifies,  
by **signing** the document:

- This **public key**

# X.509 Certificates

**issuer name**  
**subject name**  
**valid from**  
**valid to**  
**public key**  
**signature**

**Issuer** certifies,  
by **signing** the document:

- This **public key**
- Belongs to this **Subject**



# X.509 Certificates

**issuer name**  
**subject name**  
**valid from**  
**valid to**  
**public key**  
**signature**

**Issuer** certifies,  
by **signing** the document:

- This **public key**
- Belongs to this **Subject**
- During this **period**

# X.509 Certificates

**issuer name**  
**subject name**  
**valid from**  
**valid to**  
**public key**  
**signature**

*the subject's*

*the issuer's*

**Issuer** certifies,  
by **signing** the document:

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# X.509 Certificates

**issuer name**  
**subject name**  
**valid from**  
**valid to**  
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**signature**

*the subject's*

*the issuer's*

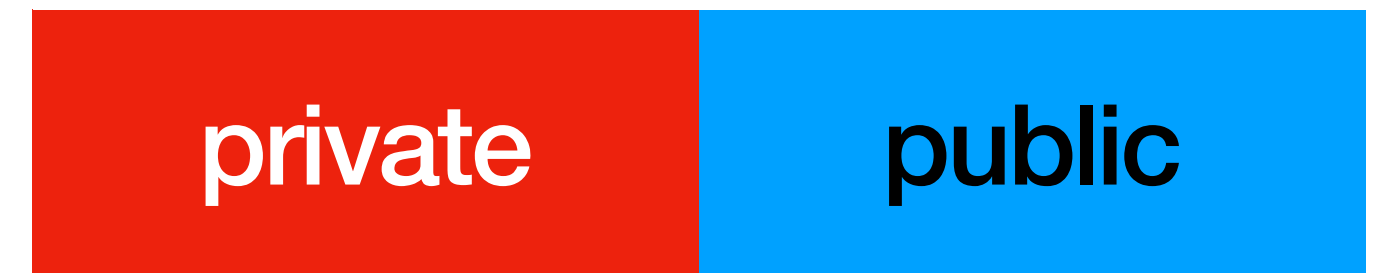
**Issuer** certifies,  
by **signing** the document:

- This **public key**
- Belongs to this **Subject**
- During this **period**

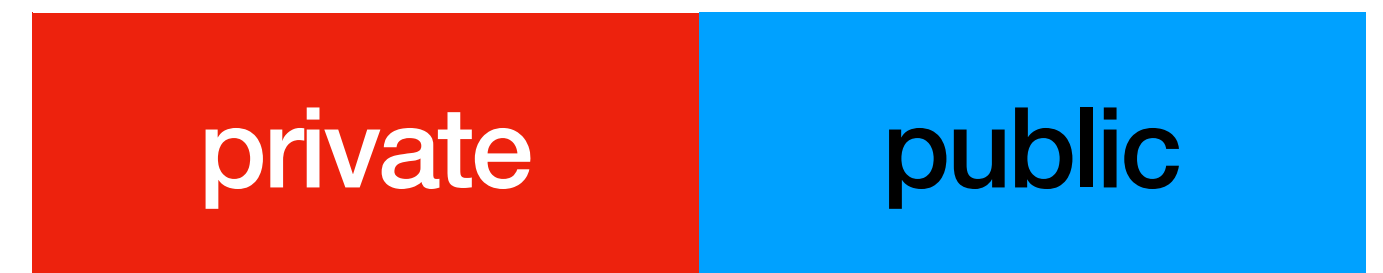
Certificates provide **Identity Assurance**

# X.509 Certificates

**Carol**



**Dave**



**Carol** certifying this public key is **Dave's**



# X.509 Certificates

|              |       |
|--------------|-------|
| issuer name  | Carol |
| subject name | Dave  |
| valid from   | ...   |
| valid to     | ...   |
| public key   |       |
| signature    |       |

Carol

private

public

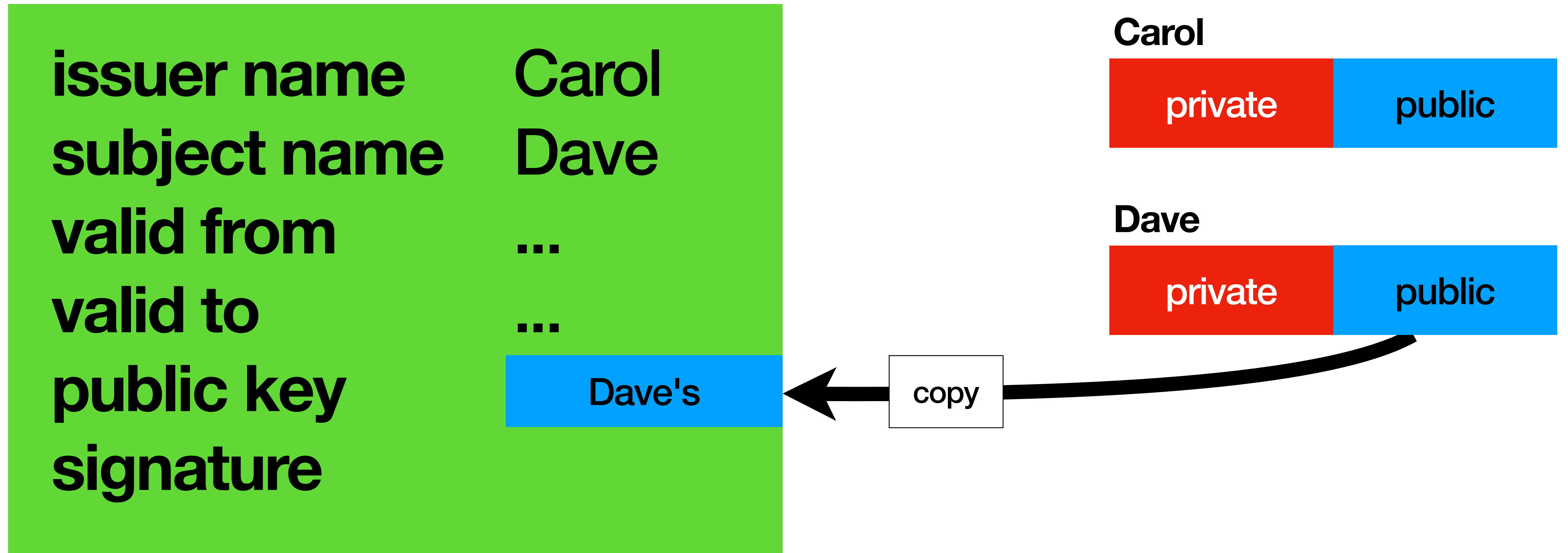
Dave

private

public

**Carol** certifying this public key is **Dave's**

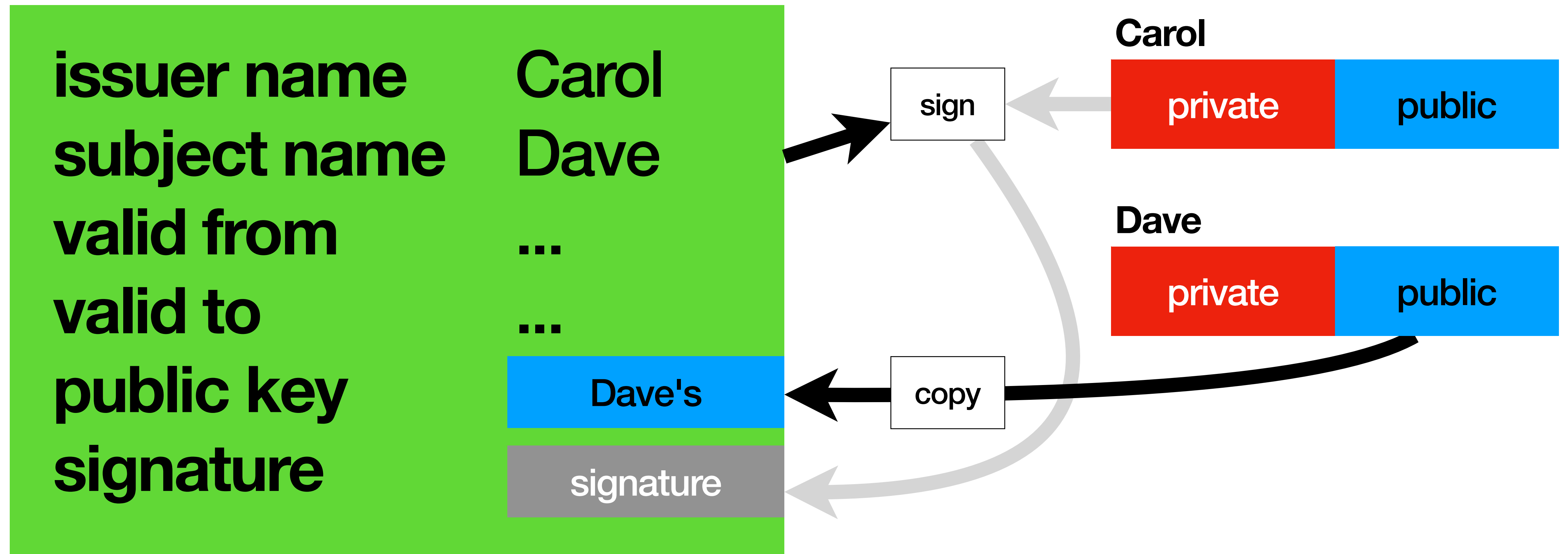
# X.509 Certificates



**Carol** certifying this public key is **Dave's**

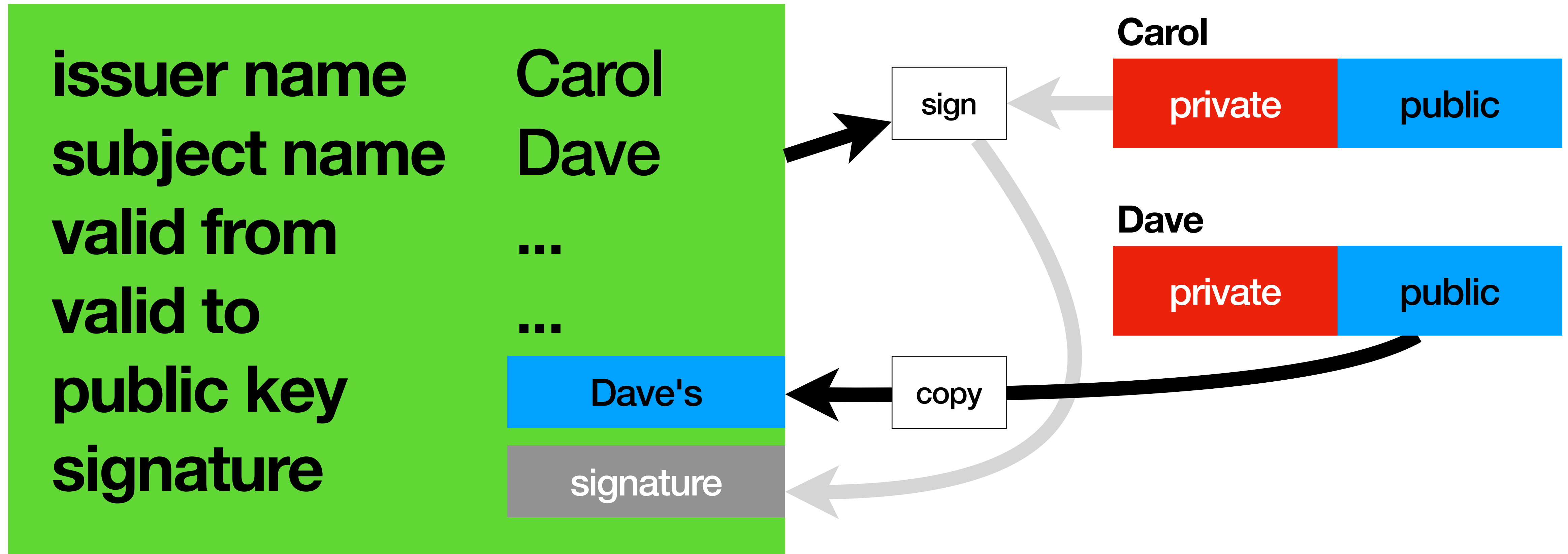


# X.509 Certificates



**Carol** certifying this public key is **Dave's**

# X.509 Certificates



**Certificates** hold no secrets and are **public**.



# **X.509 Self-Signed Certificates**

# X.509 Self-Signed Certificates

|                     |            |
|---------------------|------------|
| <b>issuer name</b>  | <b>Bob</b> |
| <b>subject name</b> | <b>Bob</b> |
| <b>valid from</b>   | <b>...</b> |
| <b>valid to</b>     | <b>...</b> |
| <b>public key</b>   |            |
| <b>signature</b>    |            |

**Bob**

**private**

**public**



# X.509 Self-Signed Certificates

|                     |              |
|---------------------|--------------|
| <b>issuer name</b>  | <b>Bob</b>   |
| <b>subject name</b> | <b>Bob</b>   |
| <b>valid from</b>   | <b>...</b>   |
| <b>valid to</b>     | <b>...</b>   |
| <b>public key</b>   | <b>Bob's</b> |
| <b>signature</b>    |              |

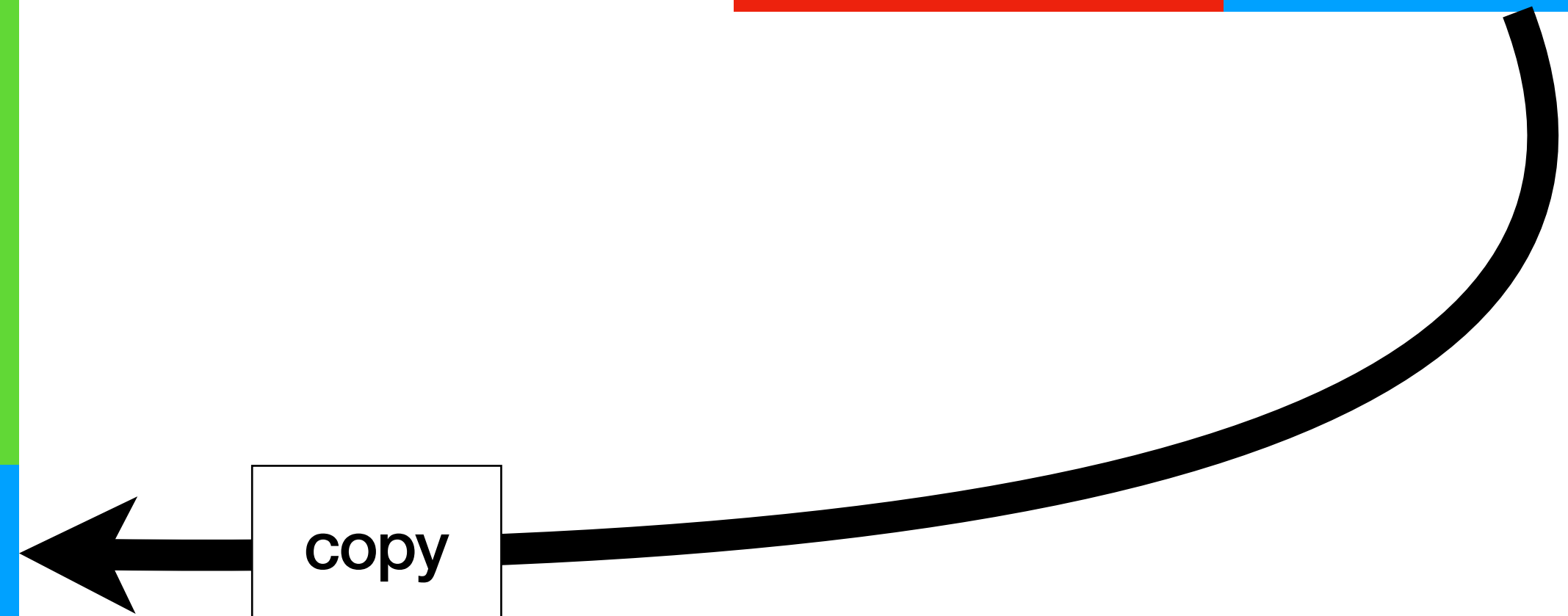
**Bob**

**private**

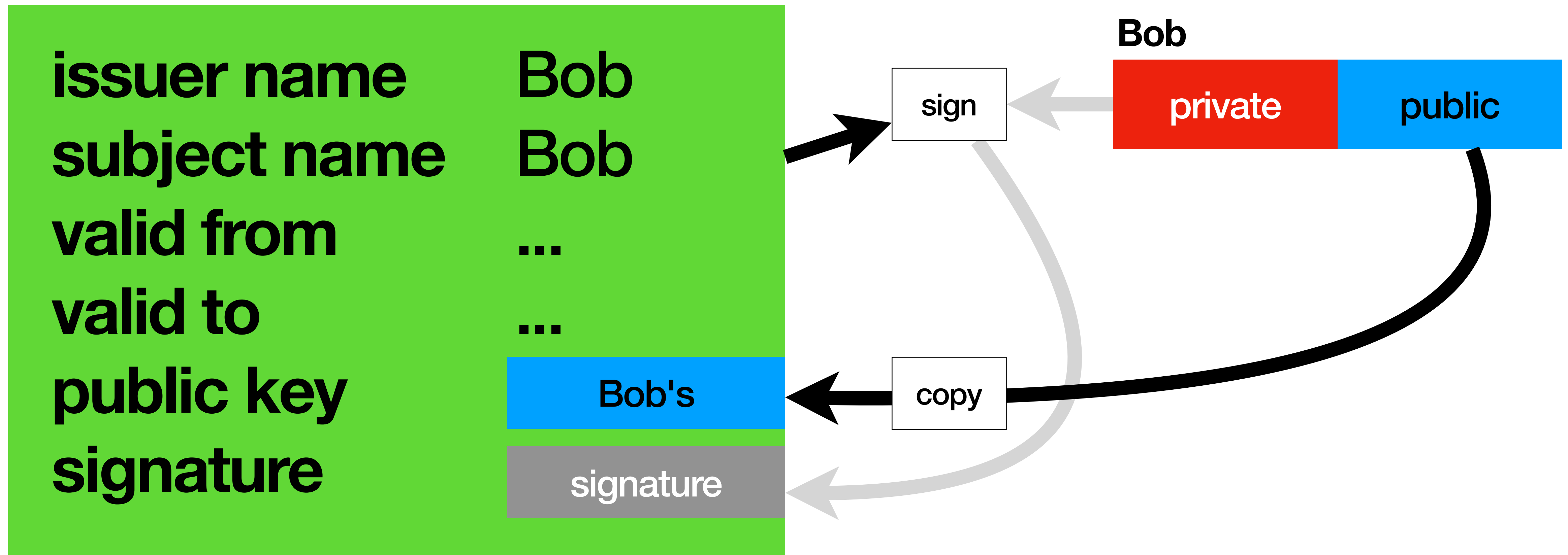
**public**

copy

Bob's

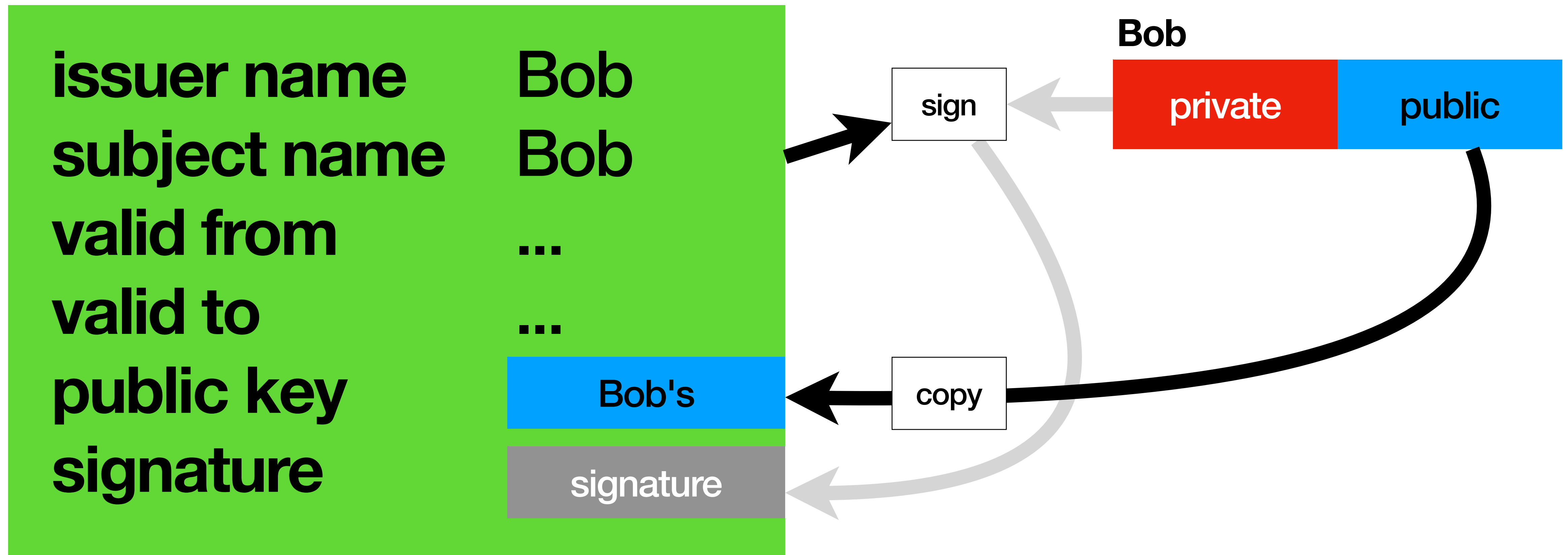


# X.509 Self-Signed Certificates





# X.509 Self-Signed Certificates



Self-Signed Certificates are **auto proclamations of Identity**

# Certificates



# Certificates

**no way self-signed certificates are enough**

# Certificate Authorities



# Certificate Authorities

- Issue **Certificates**, by **signing** them

# Certificate Authorities

- Issue **Certificates**, by **signing** them
- Using their own **private** key

CA

private

public



# Certificate Authorities

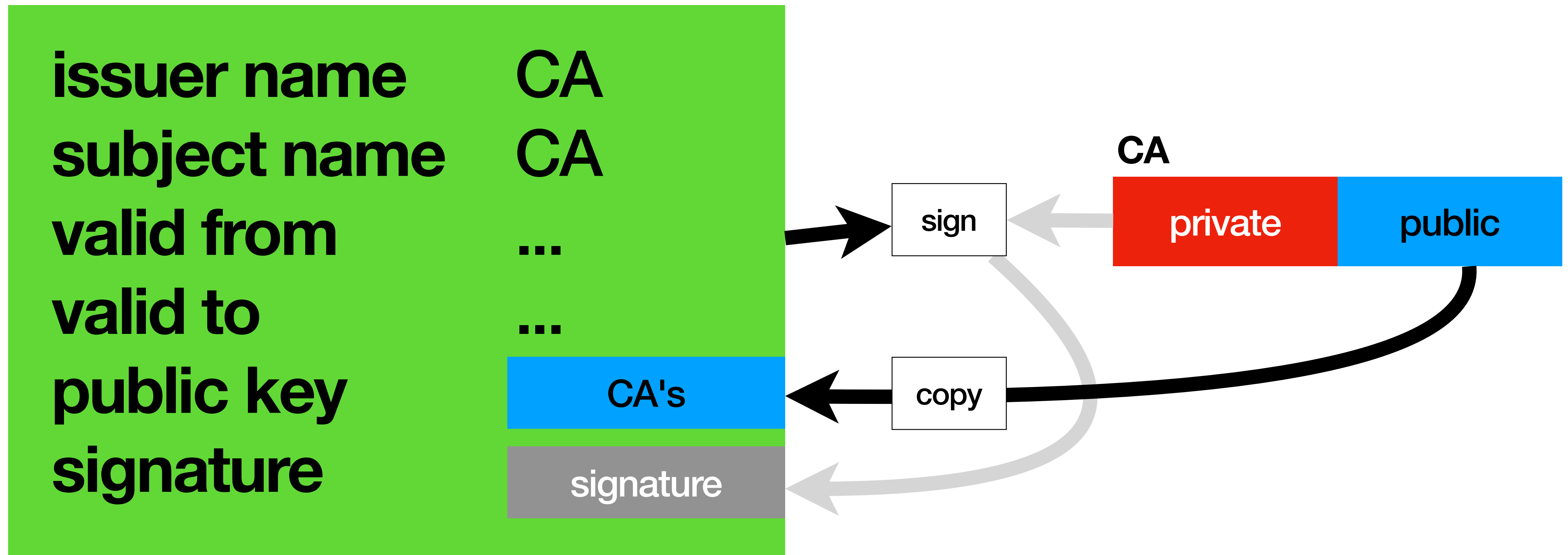
- Issue **Certificates**, by **signing** them
- Using their own **private** key
- And claim their own **Identity** via a **Self-Signed Certificate**

CA

private

public

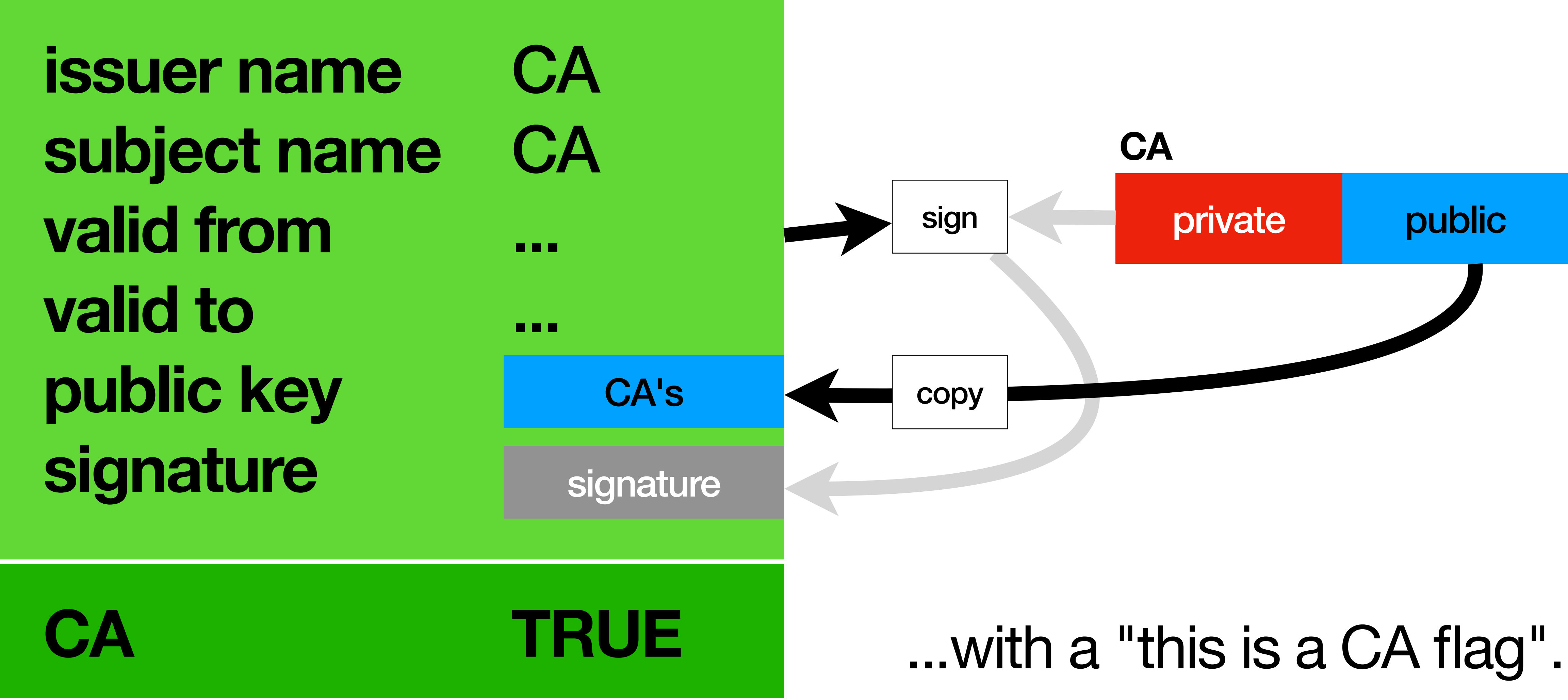
# Certificate Authorities



**Certificate Authorities are built on Self-Signed Certificates**



# Certificate Authorities

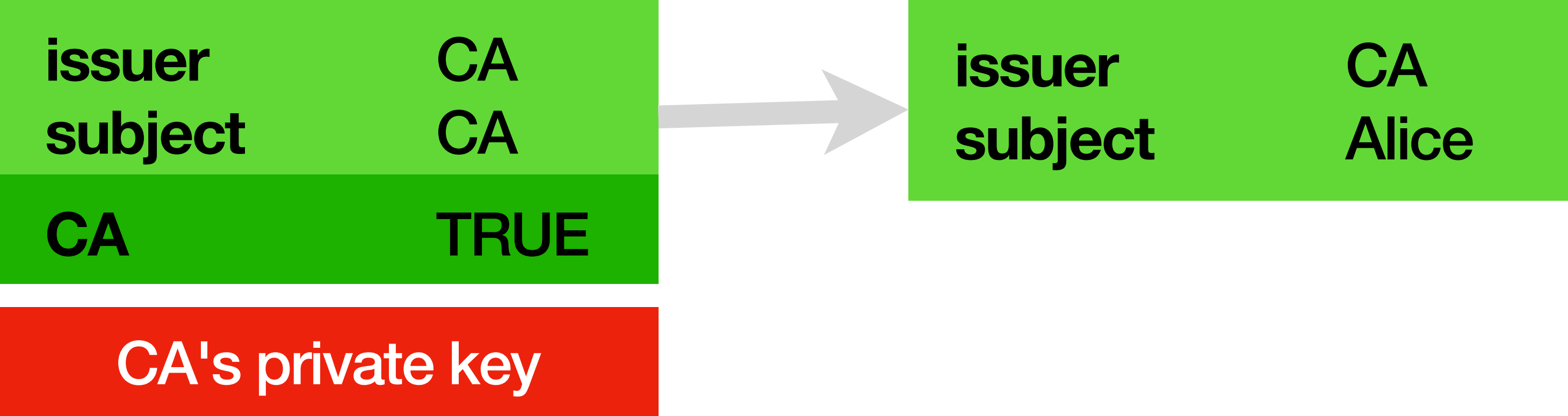


# Certificate Authorities

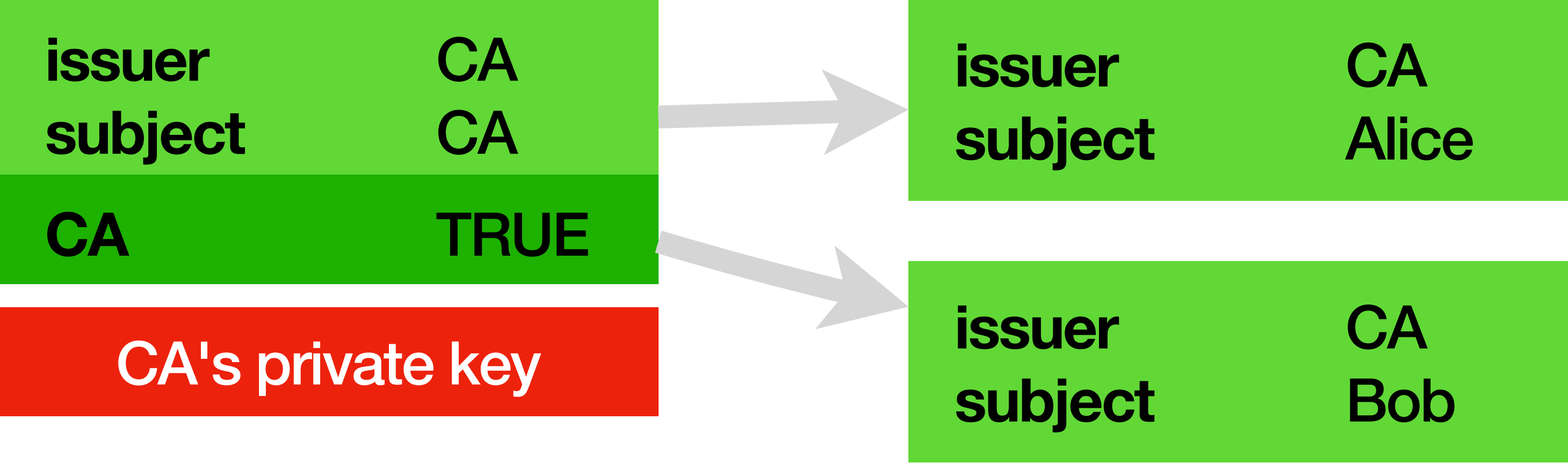
|                  |      |
|------------------|------|
| issuer           | CA   |
| subject          | CA   |
| CA               | TRUE |
| CA's private key |      |



# Certificate Authorities

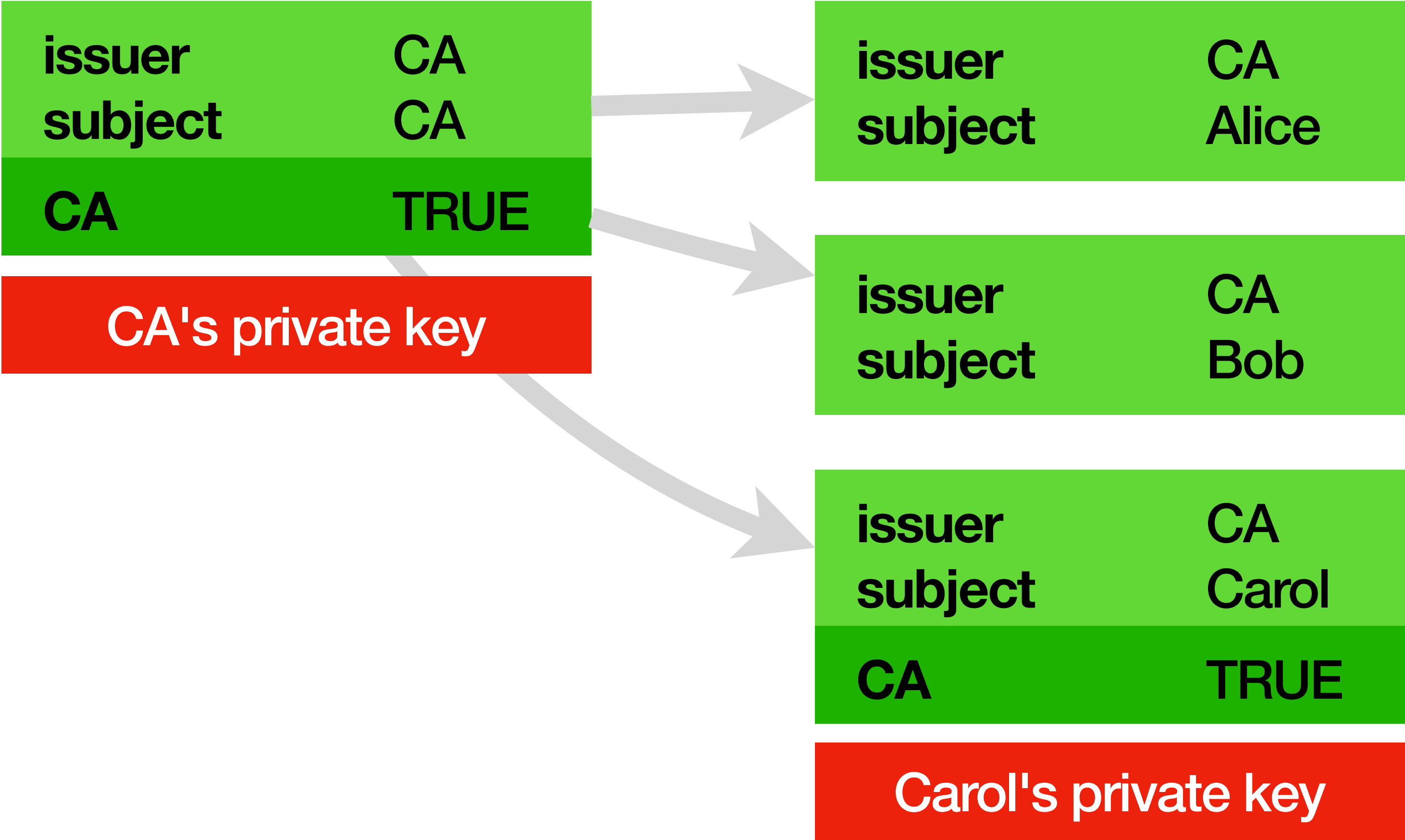


# Certificate Authorities

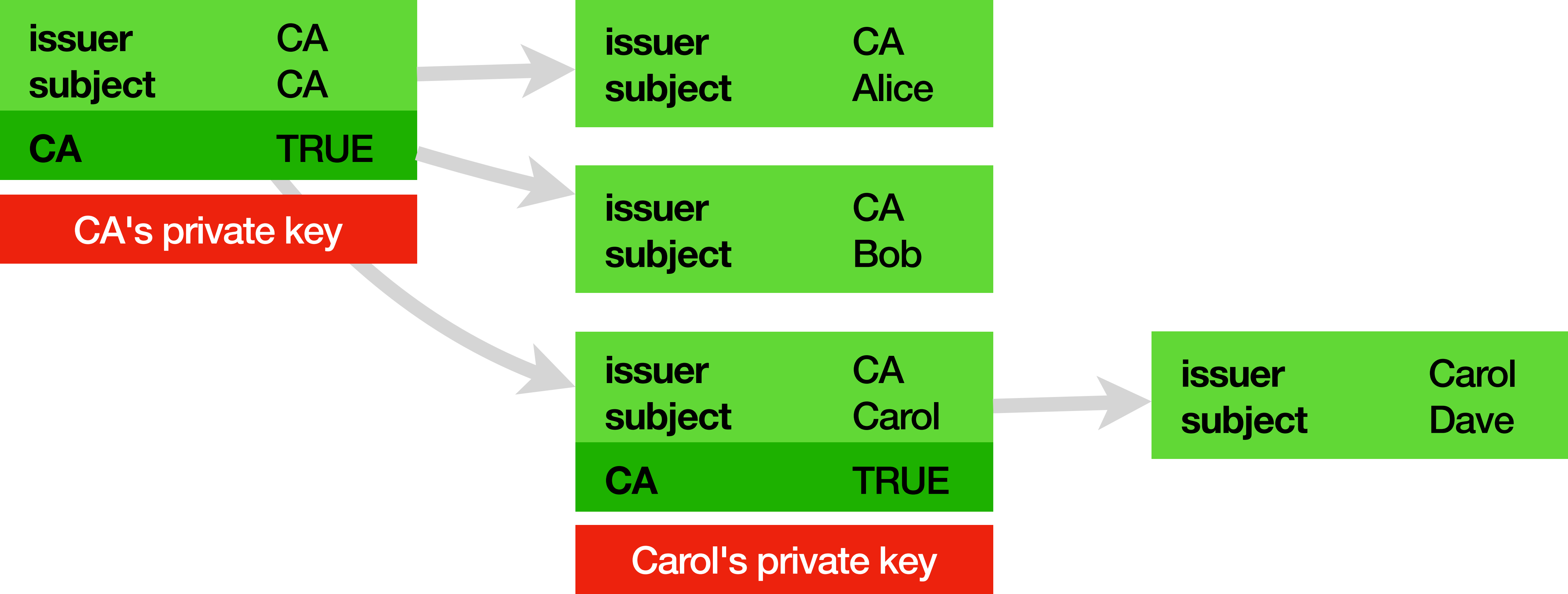




# Certificate Authorities

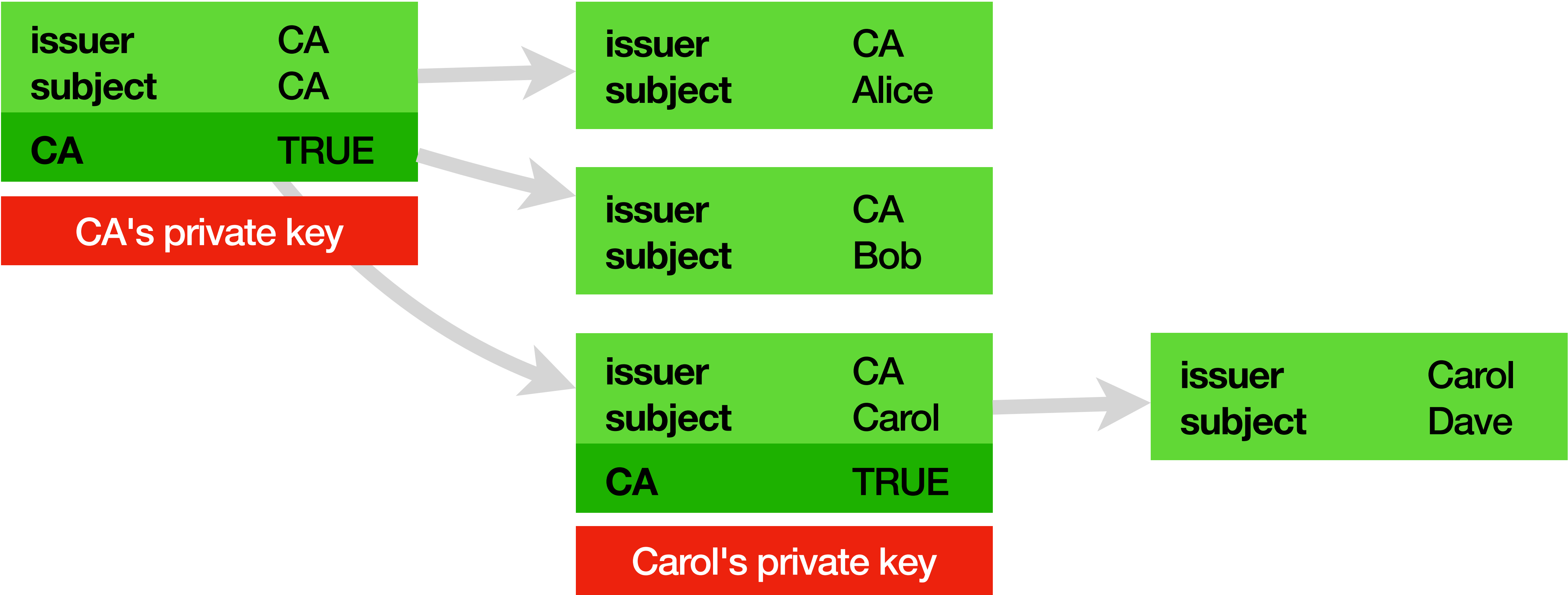


# Certificate Authorities





# Certificate Authorities



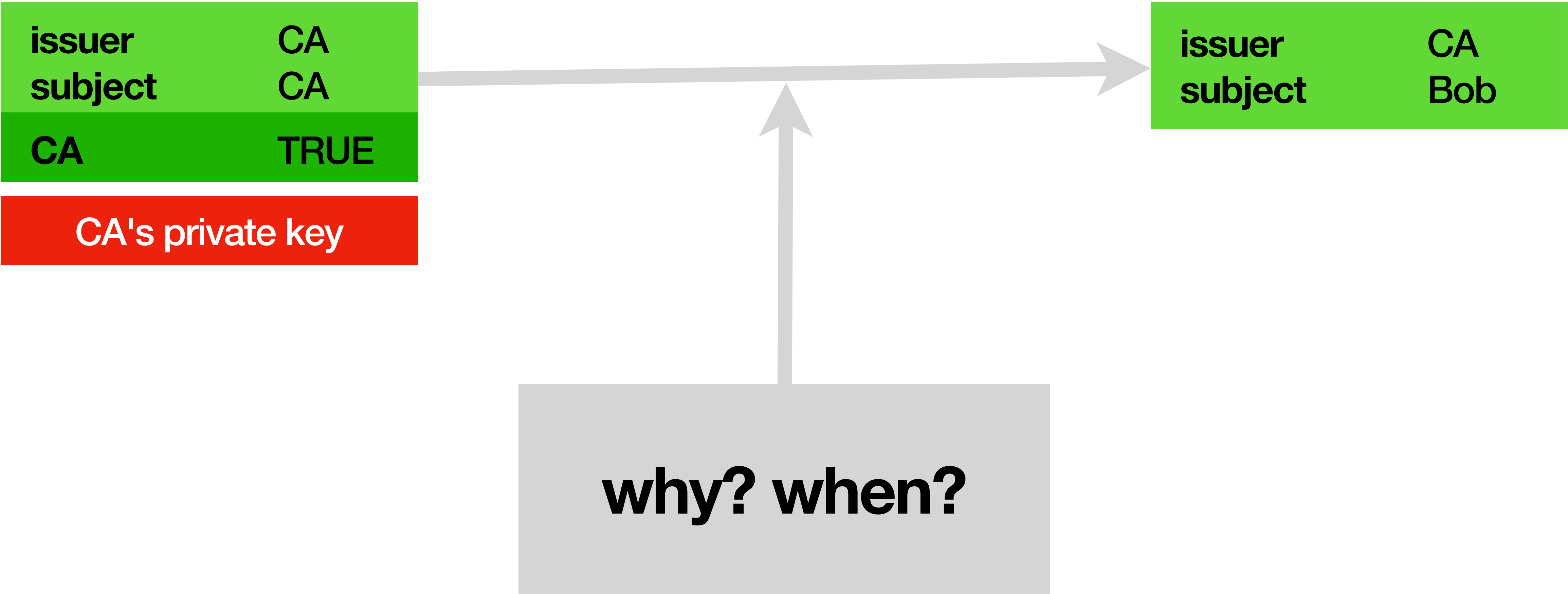
*...turtles all the way down.*

# Certificate Authorities





# Certificate Authorities



# Certificate Authorities

|         |      |
|---------|------|
| issuer  | CA   |
| subject | CA   |
| CA      | TRUE |

CA

|         |        |
|---------|--------|
| private | public |
|---------|--------|

Bob

|         |        |
|---------|--------|
| private | public |
|---------|--------|



# Certificate Authorities

|         |      |
|---------|------|
| issuer  | CA   |
| subject | CA   |
| CA      | TRUE |

CA

|         |        |
|---------|--------|
| private | public |
|---------|--------|

CA's world

public network

Bob

|         |        |
|---------|--------|
| private | public |
|---------|--------|

Bob's world

# Certificate Authorities

|         |      |
|---------|------|
| issuer  | CA   |
| subject | CA   |
| CA      | TRUE |

CA

|         |        |
|---------|--------|
| private | public |
|---------|--------|

CA's world

public network

Bob

|         |        |
|---------|--------|
| private | public |
|---------|--------|

Cert Signing Request

|         |     |
|---------|-----|
| subject | Bob |
|---------|-----|

Bob's world



# Certificate Authorities

|         |      |
|---------|------|
| issuer  | CA   |
| subject | CA   |
| CA      | TRUE |

CA

|         |        |
|---------|--------|
| private | public |
|---------|--------|

CA's world

public network

Bob

|         |        |
|---------|--------|
| private | public |
|---------|--------|

sign

Cert Signing Request

|           |       |
|-----------|-------|
| subject   | Bob   |
| signature | Bob's |

Bob's world

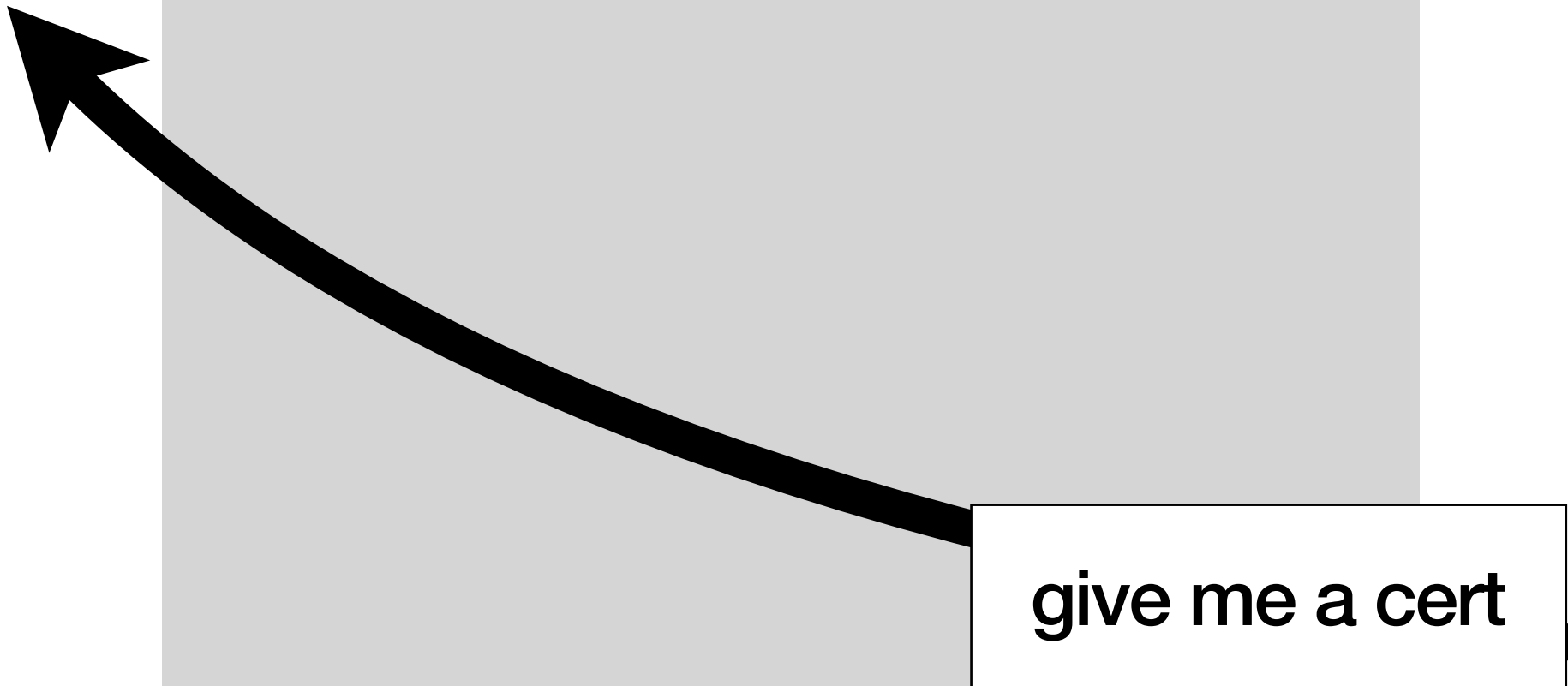
# Certificate Authorities

|         |      |
|---------|------|
| issuer  | CA   |
| subject | CA   |
| CA      | TRUE |

CA

|         |        |
|---------|--------|
| private | public |
|---------|--------|

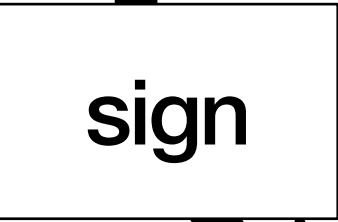
CA's world



public network

Bob

|         |        |
|---------|--------|
| private | public |
|---------|--------|



Cert Signing Request

|           |       |
|-----------|-------|
| subject   | Bob   |
| signature | Bob's |

Bob's world



# Certificate Authorities

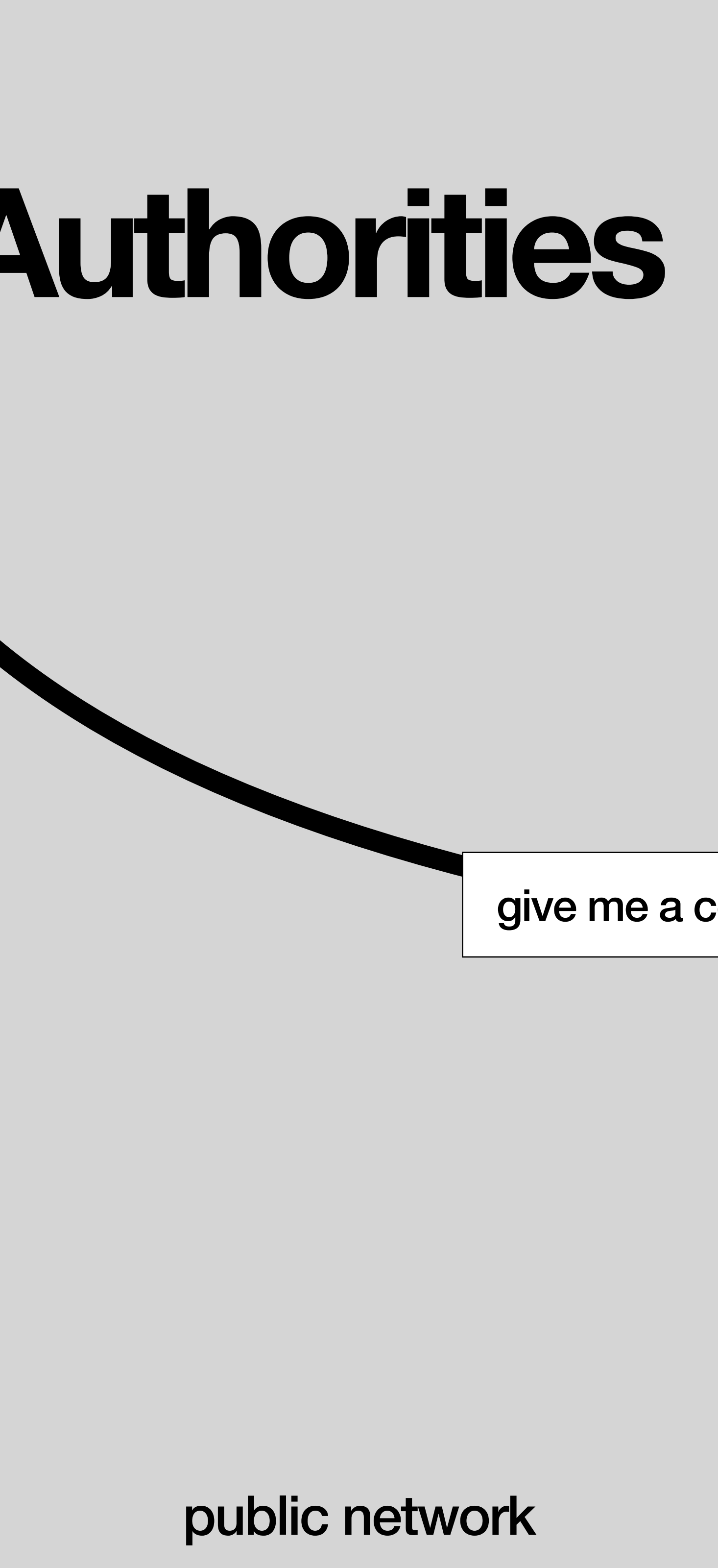
|         |      |
|---------|------|
| issuer  | CA   |
| subject | CA   |
| CA      | TRUE |

CA

|         |        |
|---------|--------|
| private | public |
|---------|--------|

|            |     |
|------------|-----|
| issuer     | CA  |
| subject    | Bob |
| public key |     |
| signature  |     |

CA's world



public network

Bob

|         |        |
|---------|--------|
| private | public |
|---------|--------|

sign

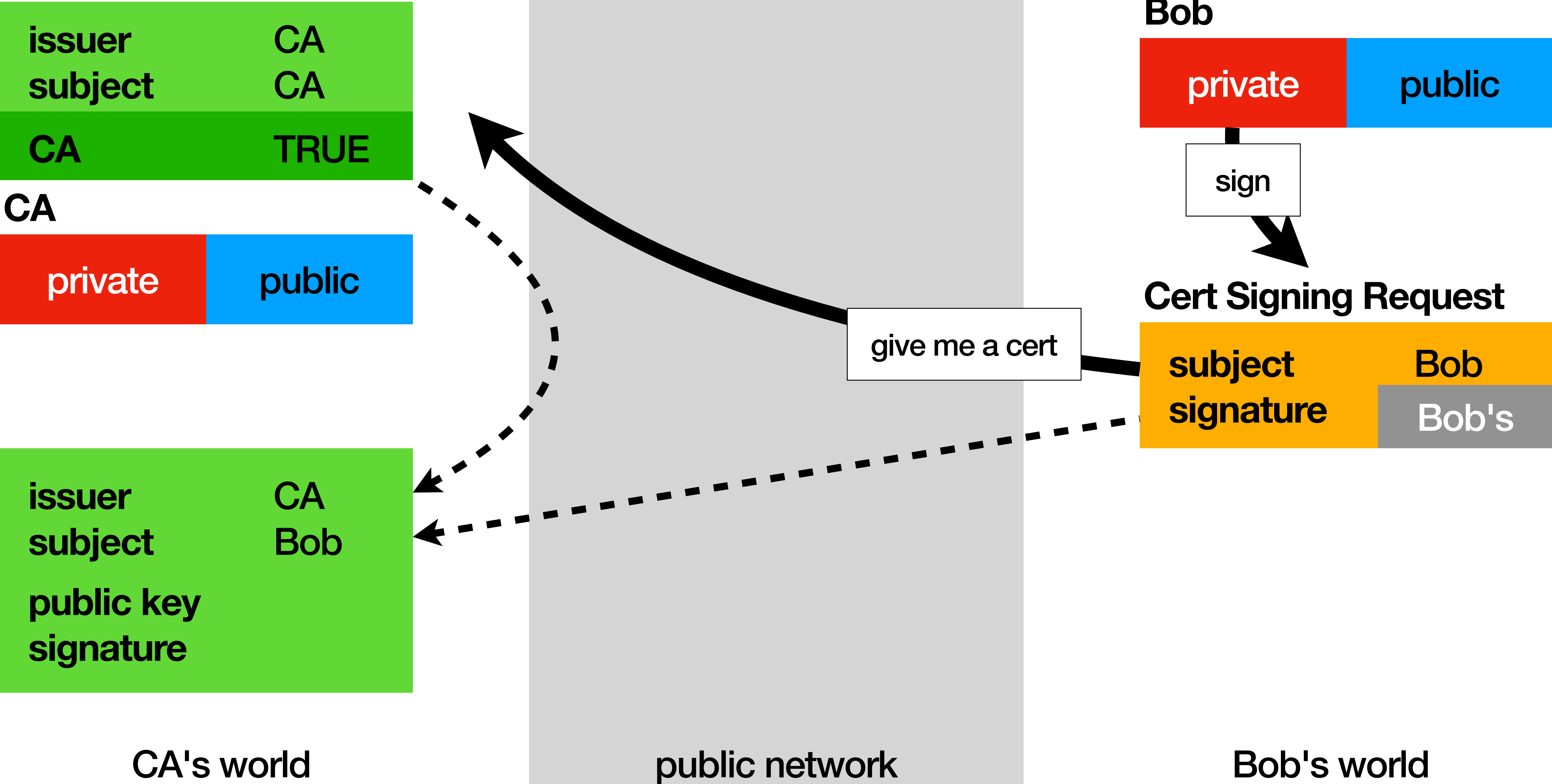
Cert Signing Request

|           |       |
|-----------|-------|
| subject   | Bob   |
| signature | Bob's |

Bob's world

give me a cert

# Certificate Authorities





# Certificate Authorities

|         |      |
|---------|------|
| issuer  | CA   |
| subject | CA   |
| CA      | TRUE |

CA

|         |        |
|---------|--------|
| private | public |
|---------|--------|

|            |       |
|------------|-------|
| issuer     | CA    |
| subject    | Bob   |
| public key | Bob's |
| signature  |       |

CA's world

public network

Bob

|         |        |
|---------|--------|
| private | public |
|---------|--------|

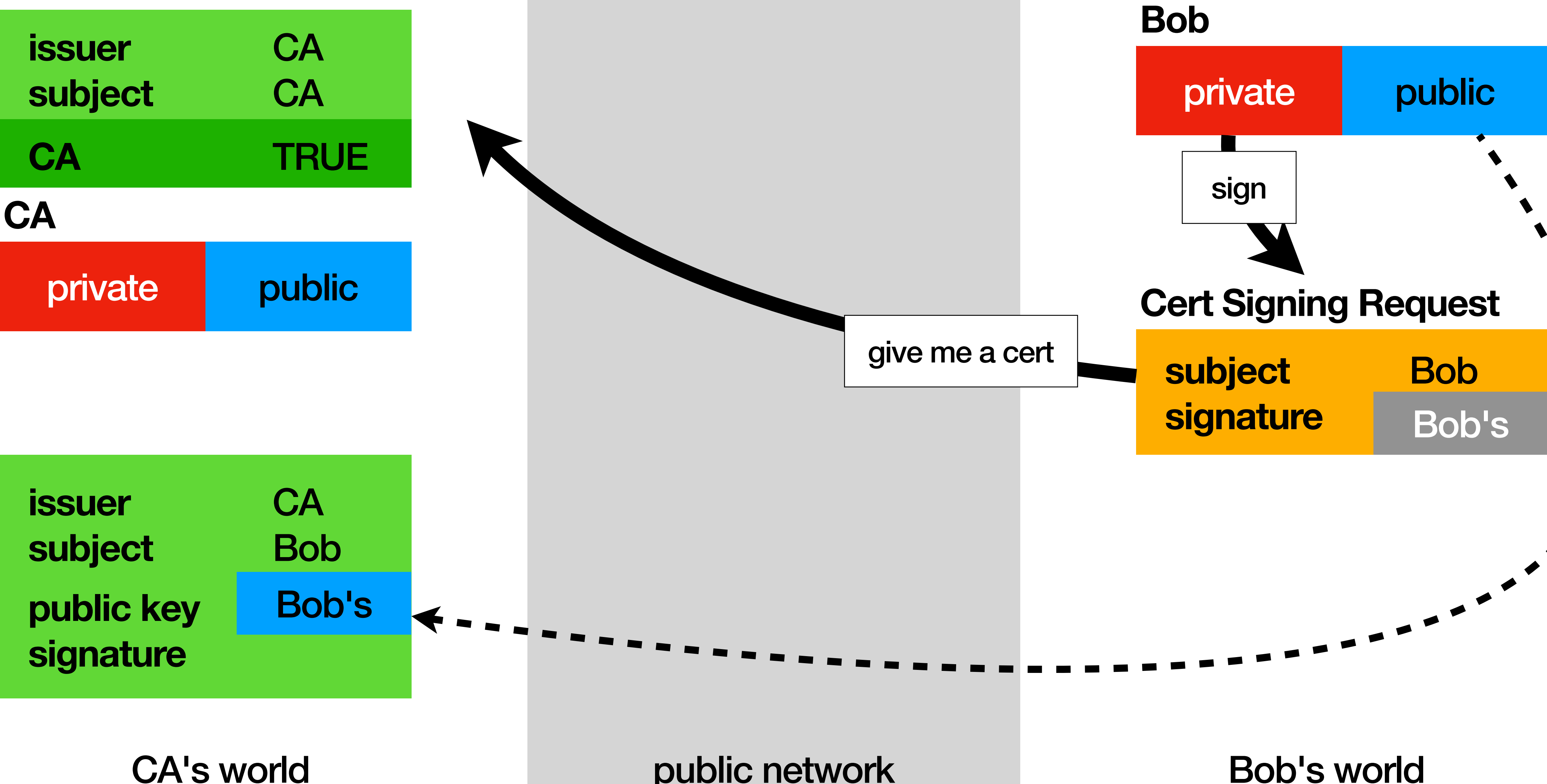
sign

Cert Signing Request

|           |       |
|-----------|-------|
| subject   | Bob   |
| signature | Bob's |

Bob's world

give me a cert



# Certificate Authorities

|         |      |
|---------|------|
| issuer  | CA   |
| subject | CA   |
| CA      | TRUE |

CA

|         |        |
|---------|--------|
| private | public |
|---------|--------|

sign

|            |       |
|------------|-------|
| issuer     | CA    |
| subject    | Bob   |
| public key | Bob's |
| signature  | CA's  |

CA's world

public network

Bob

|         |        |
|---------|--------|
| private | public |
|---------|--------|

sign

Cert Signing Request

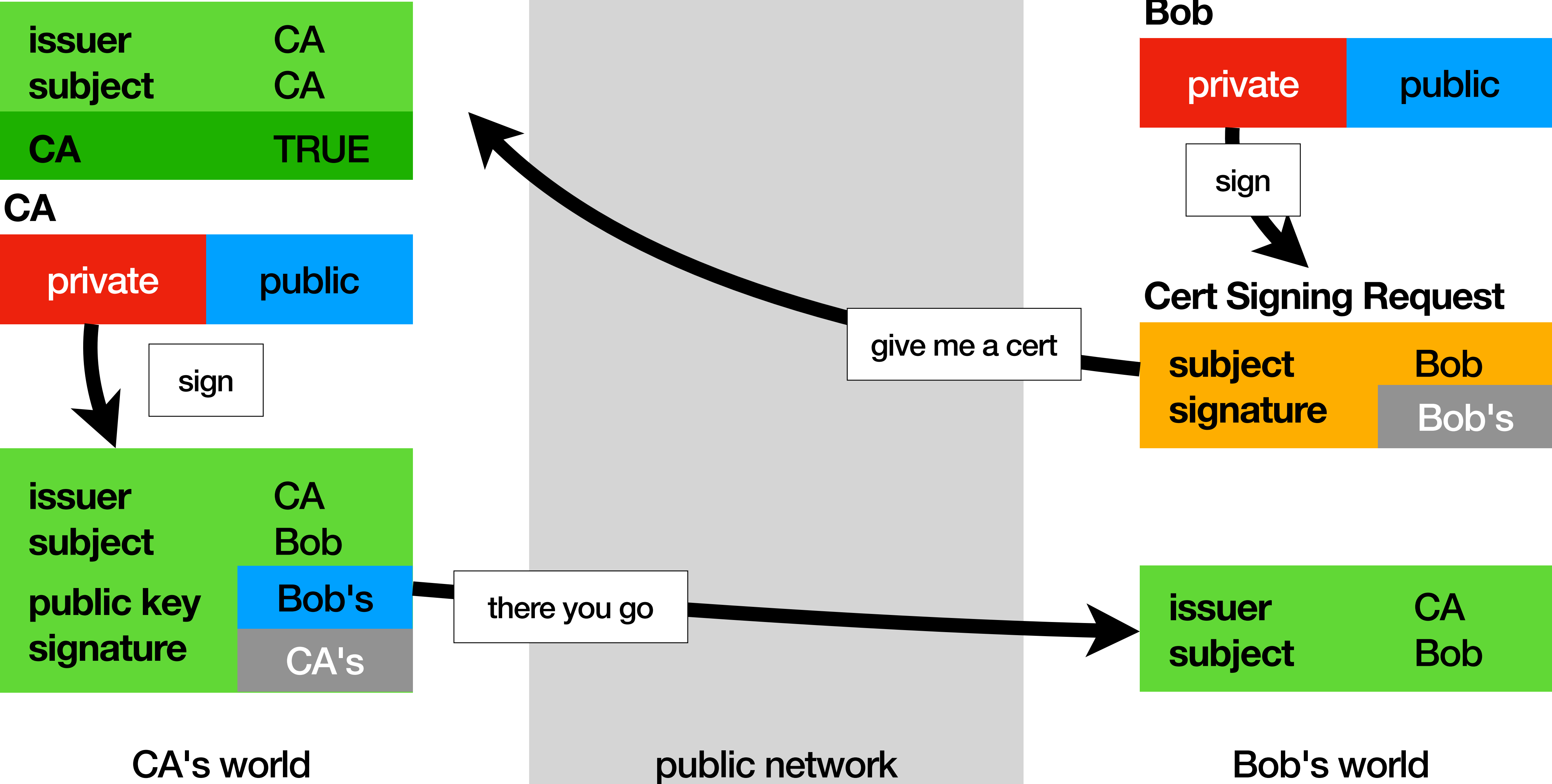
|           |       |
|-----------|-------|
| subject   | Bob   |
| signature | Bob's |

Bob's world

give me a cert



# Certificate Authorities

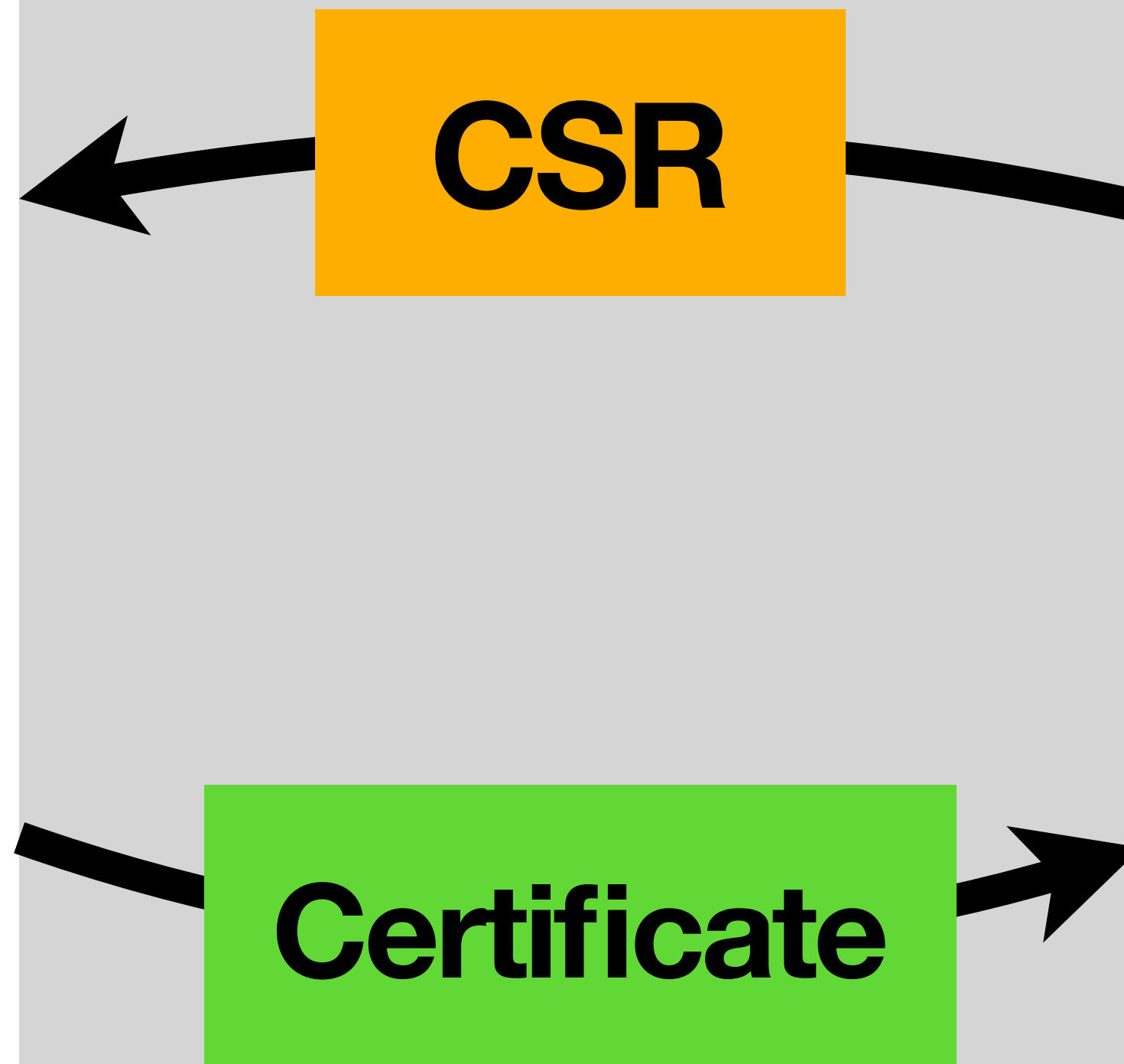


# Certificate Authorities

CA's world

public network

Bob's world





**...and TLS**

# ...and TLS (aka Transport Layer Security)

- Protocol that **wraps other protocols**



# ...and TLS (aka Transport Layer Security)

- Protocol that **wraps other protocols**
- Allows clients to **validate servers using certificates**

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- Protocol that **wraps other protocols**
- Allows clients to **validate servers using certificates**
- Clients connect, server sends certificate, client validates



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- Protocol that **wraps other protocols**
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- Clients connect, server sends certificate, client validates
- **Authenticates server and encrypts traffic**

# ...and TLS (aka Transport Layer Security)

- Protocol that **wraps other protocols**
- Allows clients to **validate servers using certificates**
- Clients connect, server sends certificate, client validates
- **Authenticates** server and **encrypts** traffic  
*traffic encryption is not RSA encryption*



# ...and TLS (aka Transport Layer Security)

- Protocol that **wraps other protocols**
- Allows clients to **validate servers using certificates**
- Clients connect, server sends certificate, client validates
- **Authenticates** server and **encrypts** traffic  
*traffic encryption is not RSA encryption*

*in **m(utual)TLS** clients also use certificates, server validated*

# TLS Server Certificates



# TLS Server Certificates

Bob serves a web app at  
**<https://example.com>**.

# TLS Server Certificates

Bob serves a web app at  
**`https://example.com`**.

Needs a **server certificate**.



# TLS Server Certificates

|              |           |
|--------------|-----------|
| issuer name  | CA        |
| subject name | Bob       |
| valid from   | ...       |
| valid to     | ...       |
| public key   | Bob's     |
| signature    | signature |

Bob serves a web app at **<https://example.com>**.

Needs a **server certificate**.

# TLS Server Certificates

|              |           |
|--------------|-----------|
| issuer name  | CA        |
| subject name | Bob       |
| valid from   | ...       |
| valid to     | ...       |
| public key   | Bob's     |
| signature    | signature |

Bob serves a web app at **`https://example.com`**.

Needs a **server certificate**.

This is **not enough**.



# TLS Server Certificates

|                     |                  |
|---------------------|------------------|
| <b>issuer name</b>  | <b>CA</b>        |
| <b>subject name</b> | <b>Bob</b>       |
| <b>valid from</b>   | <b>...</b>       |
| <b>valid to</b>     | <b>...</b>       |
| <b>public key</b>   | <b>Bob's</b>     |
| <b>signature</b>    | <b>signature</b> |

|            |                    |
|------------|--------------------|
| <b>SAN</b> | <b>example.com</b> |
|------------|--------------------|

Bob serves a web app at **<https://example.com>**.

Needs a **server certificate**.

# TLS Server Certificates

|              |           |
|--------------|-----------|
| issuer name  | CA        |
| subject name | Bob       |
| valid from   | ...       |
| valid to     | ...       |
| public key   | Bob's     |
| signature    | signature |

|     |             |
|-----|-------------|
| SAN | example.com |
|-----|-------------|

Bob serves a web app at **<https://example.com>**.

Needs a **server certificate**.

Server certificates include **Subject Alternative Name**.



# TLS Server Certificates

|              |           |
|--------------|-----------|
| issuer name  | CA        |
| subject name | Bob       |
| valid from   | ...       |
| valid to     | ...       |
| public key   | Bob's     |
| signature    | signature |

|     |             |
|-----|-------------|
| SAN | example.com |
|-----|-------------|

Bob serves a web app at **`https://example.com`**.

Needs a **server certificate**.

Server certificates include **Subject Alternative Name**.

Also used in client validation.

# TLS Server Certificates

|              |           |
|--------------|-----------|
| issuer name  | CA        |
| subject name | Bob       |
| valid from   | ...       |
| valid to     | ...       |
| public key   | Bob's     |
| signature    | signature |

|     |             |
|-----|-------------|
| SAN | example.com |
|-----|-------------|

## Client Validation



# TLS Server Certificates

|                     |                  |
|---------------------|------------------|
| <b>issuer name</b>  | <b>CA</b>        |
| <b>subject name</b> | <b>Bob</b>       |
| <b>valid from</b>   | <b>...</b>       |
| <b>valid to</b>     | <b>...</b>       |
| <b>public key</b>   | <b>Bob's</b>     |
| <b>signature</b>    | <b>signature</b> |

|            |                    |
|------------|--------------------|
| <b>SAN</b> | <b>example.com</b> |
|------------|--------------------|

## Client Validation

- Signature is valid.

# TLS Server Certificates

|              |           |
|--------------|-----------|
| issuer name  | CA        |
| subject name | Bob       |
| valid from   | ...       |
| valid to     | ...       |
| public key   | Bob's     |
| signature    | signature |

|     |             |
|-----|-------------|
| SAN | example.com |
|-----|-------------|

## Client Validation

- Signature is valid.
- Valid from/to is ok.



# TLS Server Certificates

|                     |                  |
|---------------------|------------------|
| <b>issuer name</b>  | <b>CA</b>        |
| <b>subject name</b> | <b>Bob</b>       |
| <b>valid from</b>   | <b>...</b>       |
| <b>valid to</b>     | <b>...</b>       |
| <b>public key</b>   | <b>Bob's</b>     |
| <b>signature</b>    | <b>signature</b> |

|            |                    |
|------------|--------------------|
| <b>SAN</b> | <b>example.com</b> |
|------------|--------------------|

## Client Validation

- Signature is valid.
- Valid from/to is ok.
- SAN matches.

# TLS Server Certificates

|              |           |
|--------------|-----------|
| issuer name  | CA        |
| subject name | Bob       |
| valid from   | ...       |
| valid to     | ...       |
| public key   | Bob's     |
| signature    | signature |

|     |             |
|-----|-------------|
| SAN | example.com |
|-----|-------------|

## Client Validation

- Signature is valid.
- Valid from/to is ok.
- SAN matches.
- Issued by a CA I trust.  
*or by a sub-CA of a CA I trust*



# TLS Server Certificates

|              |           |
|--------------|-----------|
| issuer name  | CA        |
| subject name | Bob       |
| valid from   | ...       |
| valid to     | ...       |
| public key   | Bob's     |
| signature    | signature |

|     |             |
|-----|-------------|
| SAN | example.com |
|-----|-------------|

## Client Validation

- Signature is valid.
- Valid from/to is ok.
- SAN matches.
- Issued by a CA I trust.  
*or by a sub-CA of a CA I trust*

*...there is much more!*

**CAs & TLS**



# **CAs & TLS**

**who you trust dictates everything**

**MITM** (aka Man-in-the-middle)



# MITM (aka Man-in-the-middle)

web application at  
<http://example.com>

Alice's client

public network

Bob's server

# MITM (aka Man-in-the-middle)

HTTP client (no TLS)

web application at  
<http://example.com>

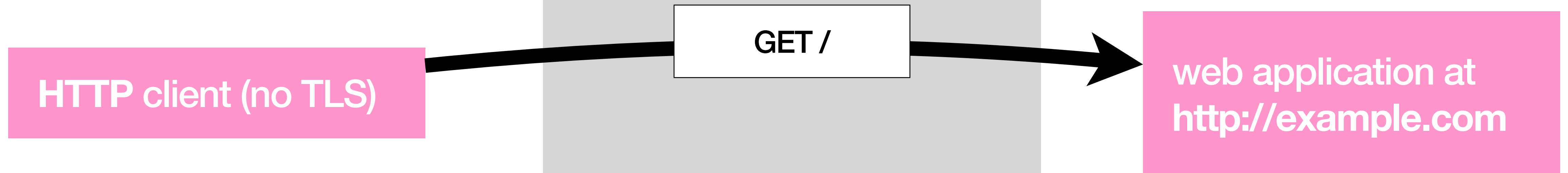
Alice's client

public network

Bob's server



# MITM (aka Man-in-the-middle)

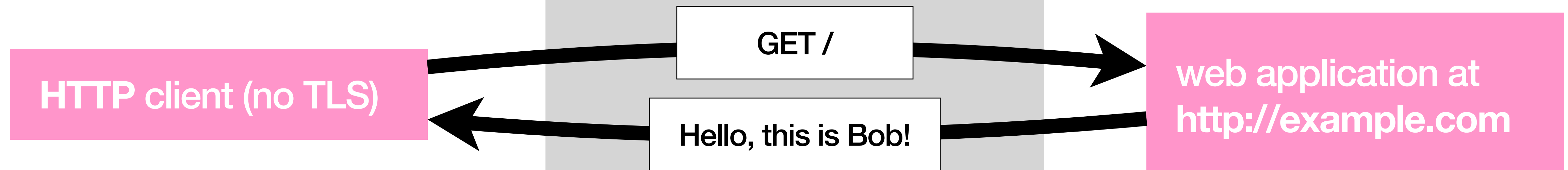


Alice's client

public network

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web application at  
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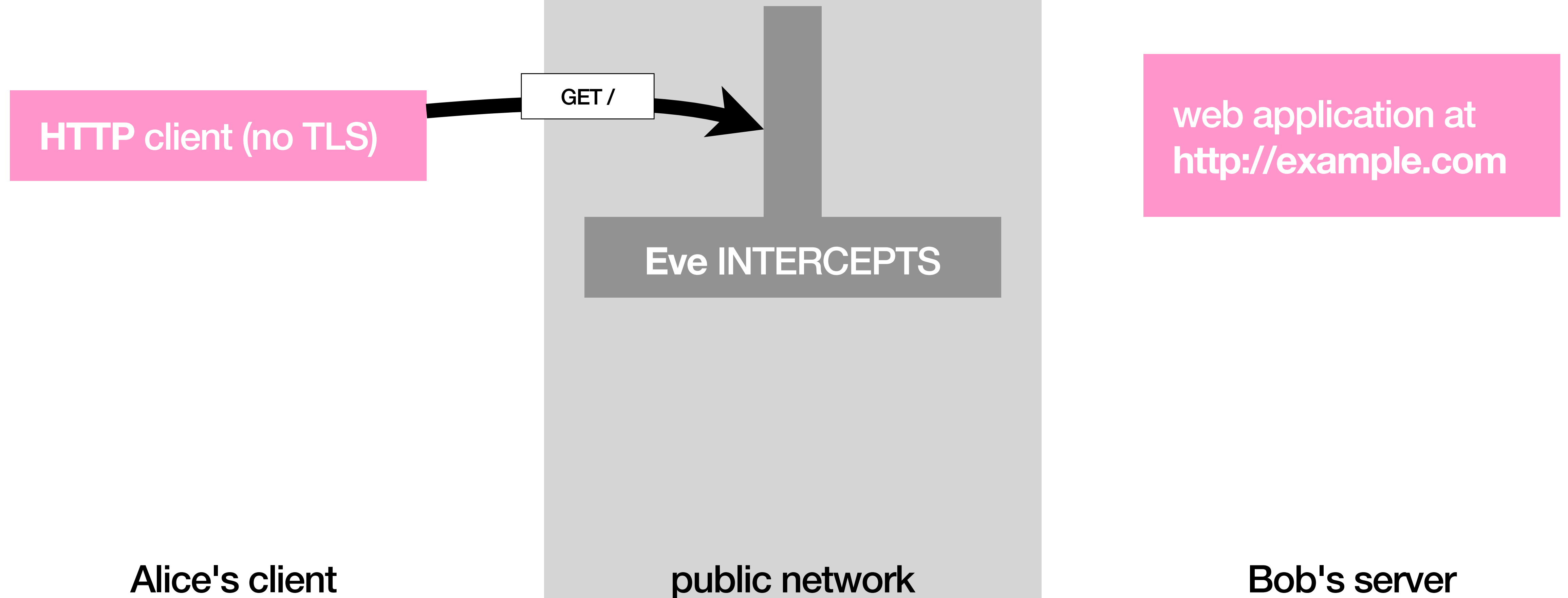
Eve INTERCEPTS

Alice's client

public network

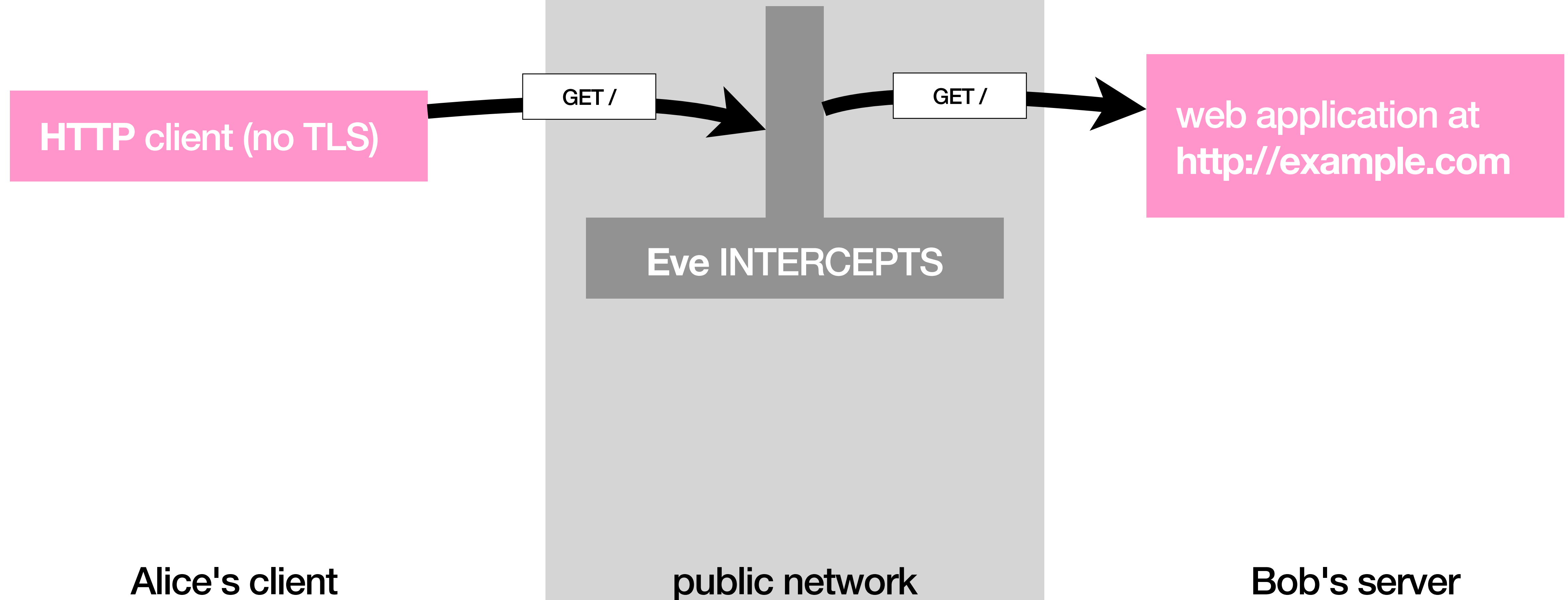
Bob's server

# MITM (aka Man-in-the-middle)

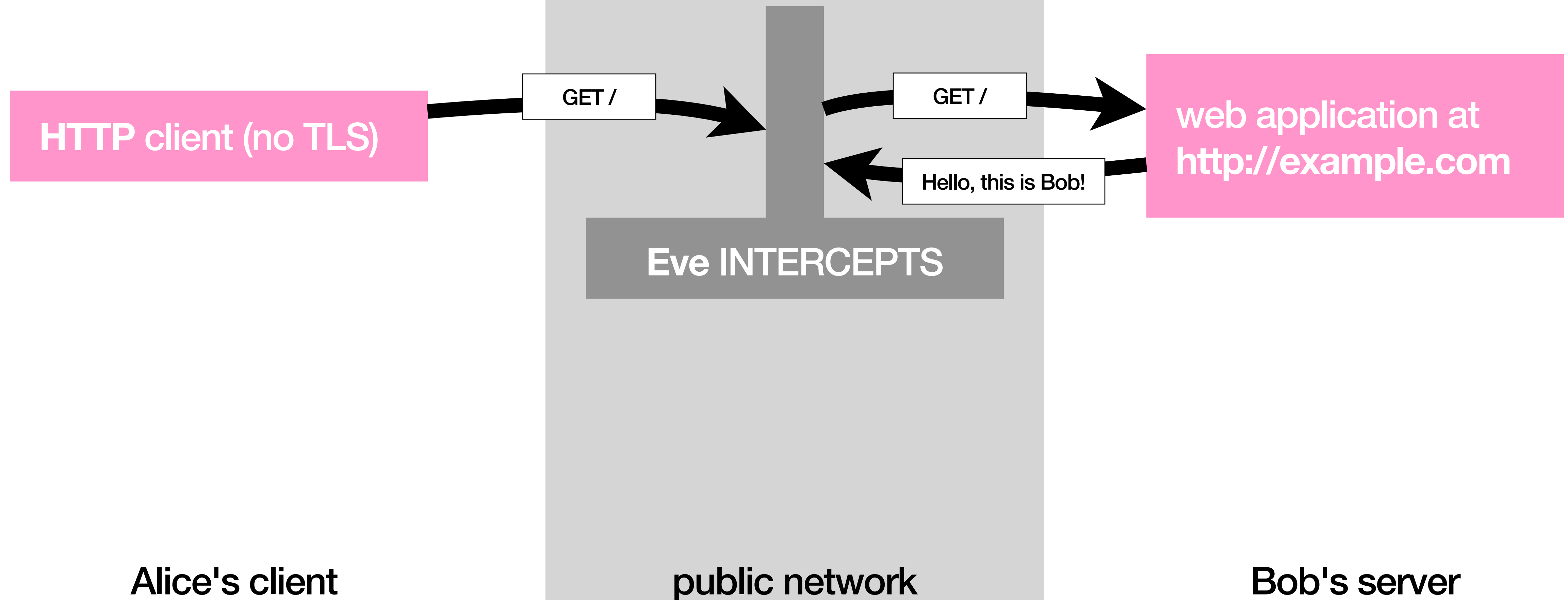




# MITM (aka Man-in-the-middle)

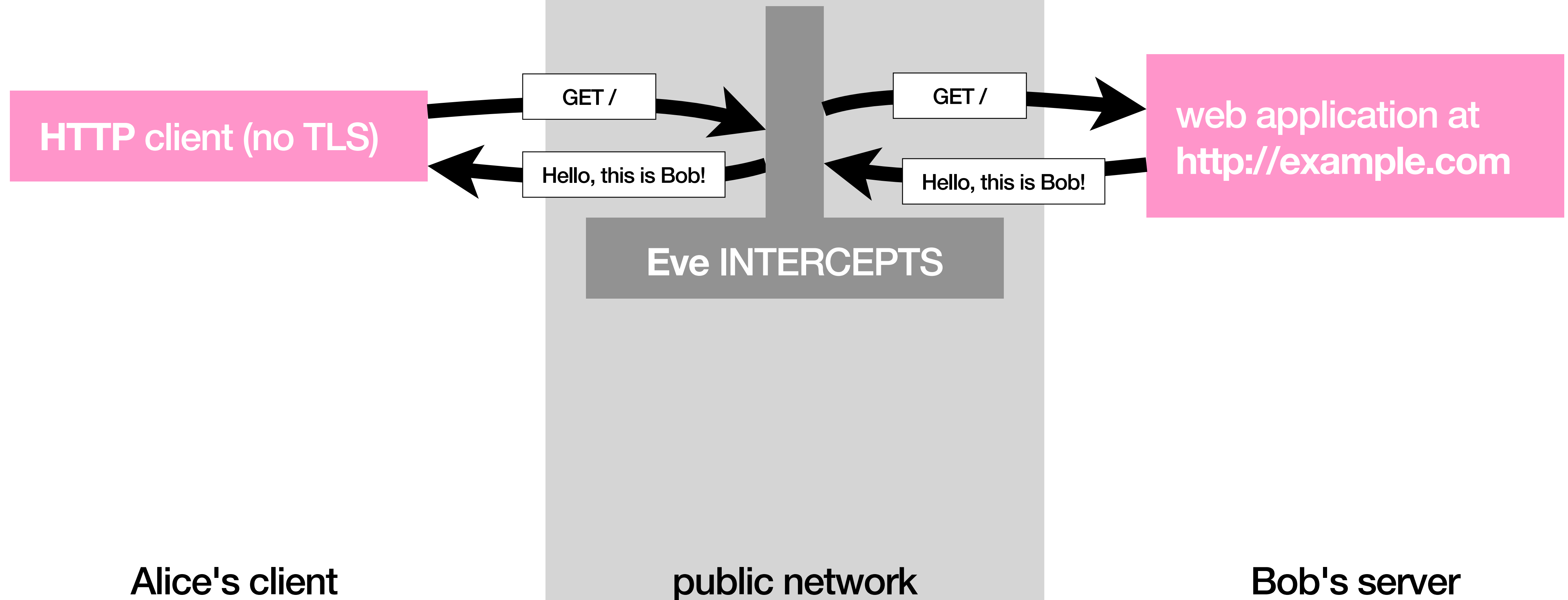


# MITM (aka Man-in-the-middle)





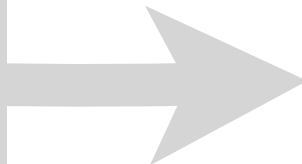
# MITM (aka Man-in-the-middle)



# MITM (aka Man-in-the-middle)

HTTPS client

|         |      |
|---------|------|
| issuer  | CA   |
| subject | CA   |
| CA      | TRUE |



|         |             |
|---------|-------------|
| issuer  | CA          |
| subject | Bob         |
| SAN     | example.com |

web application at  
<https://example.com>

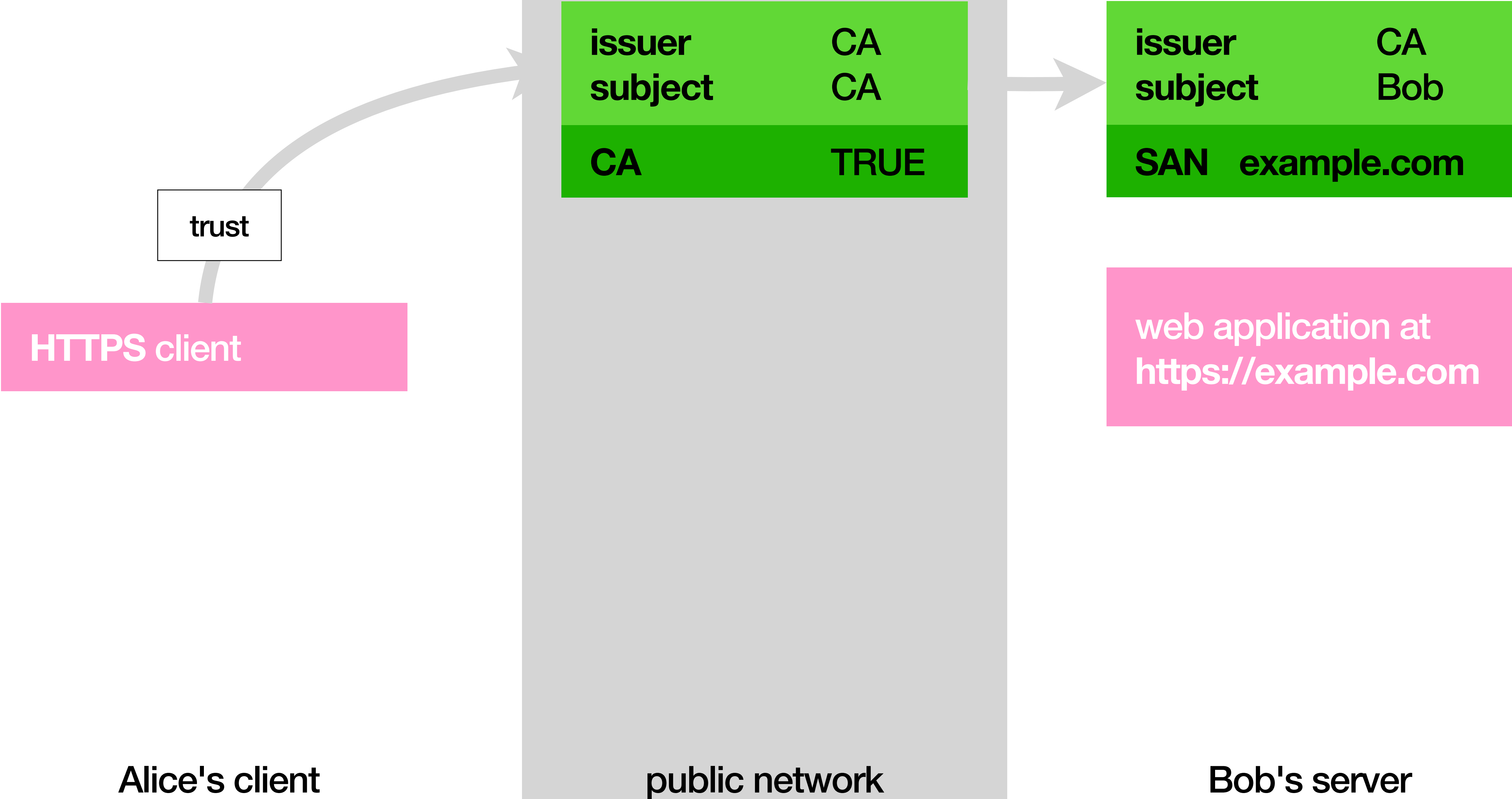
Alice's client

public network

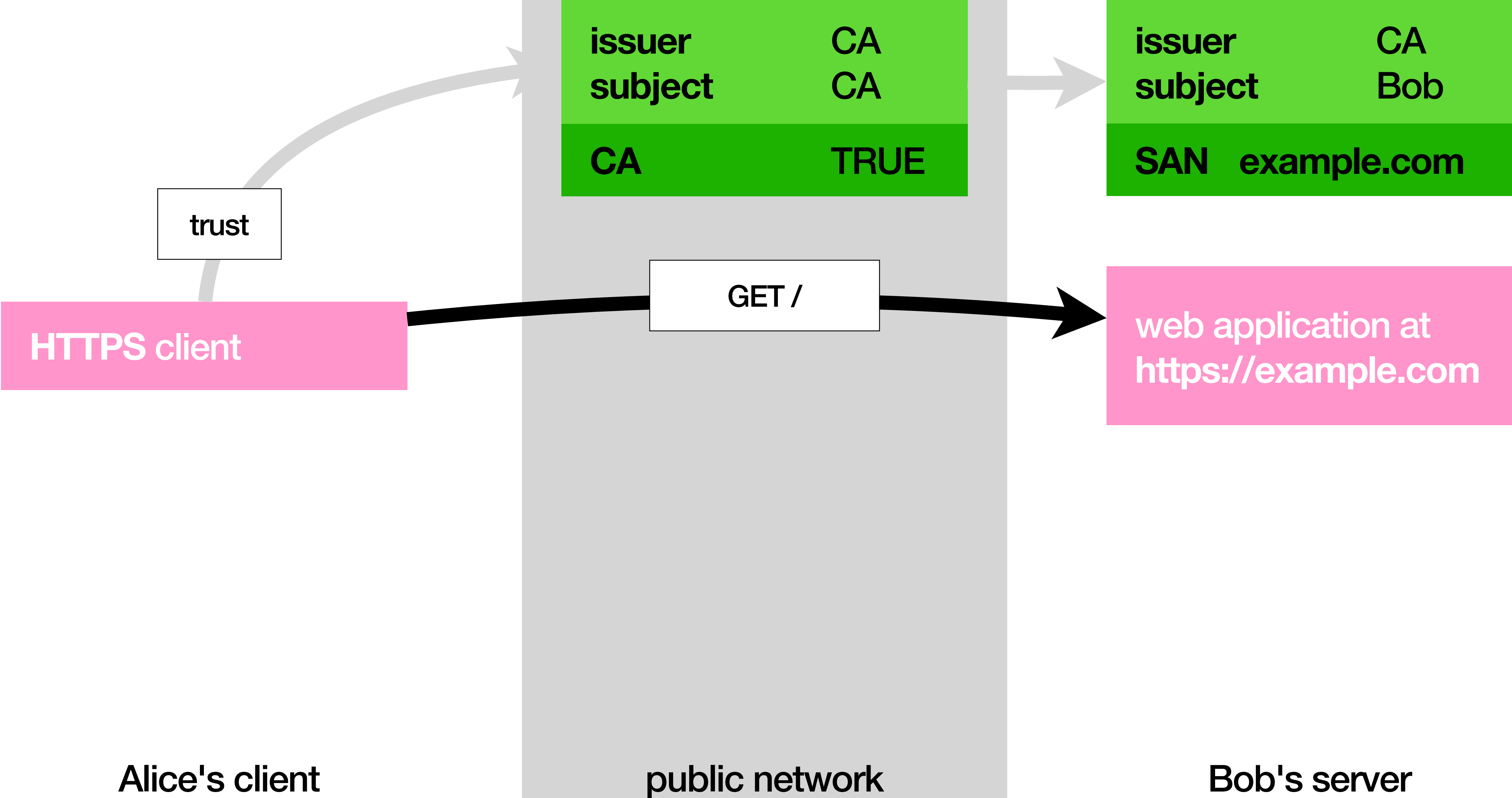
Bob's server



# MITM (aka Man-in-the-middle)

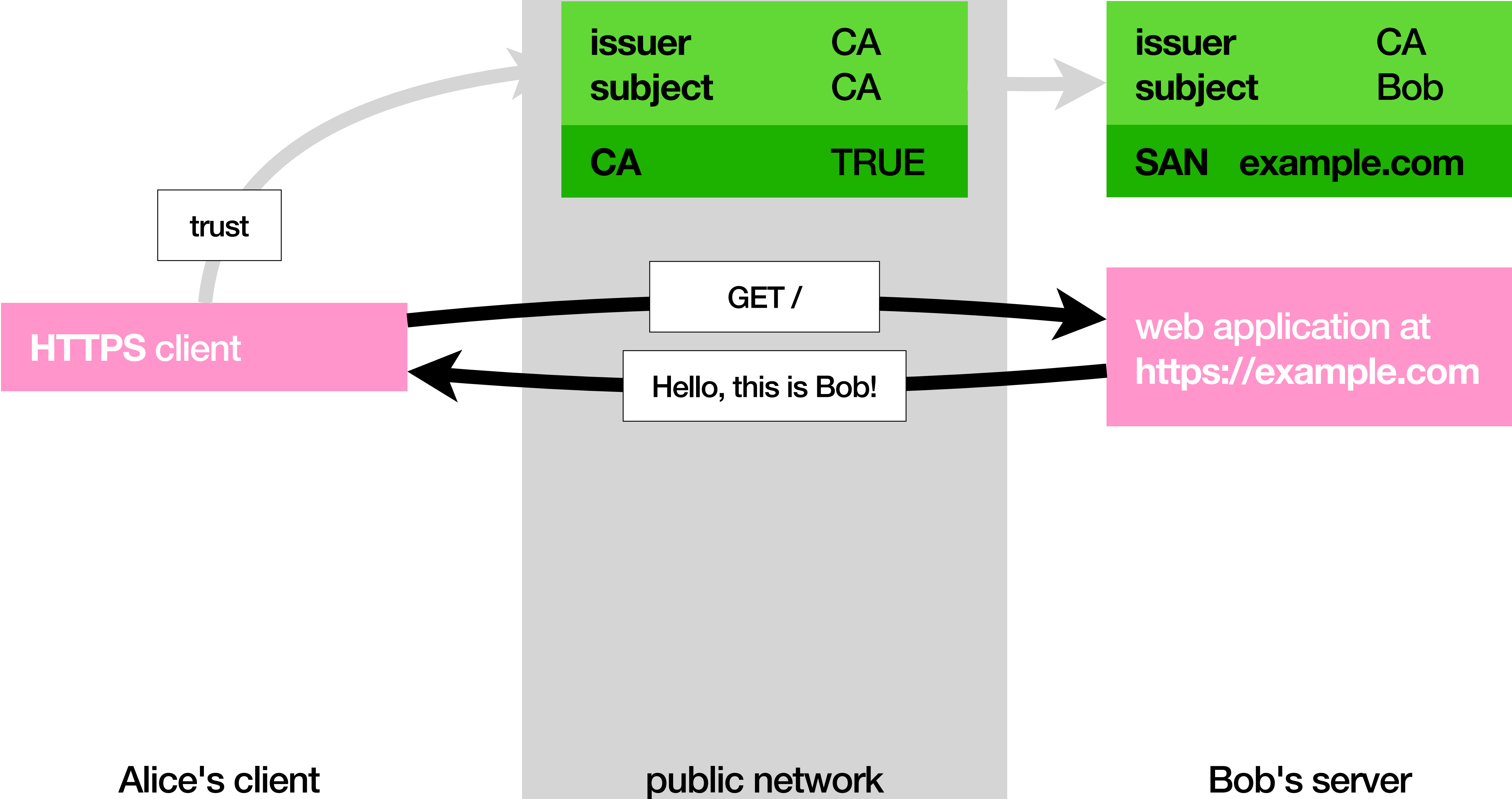


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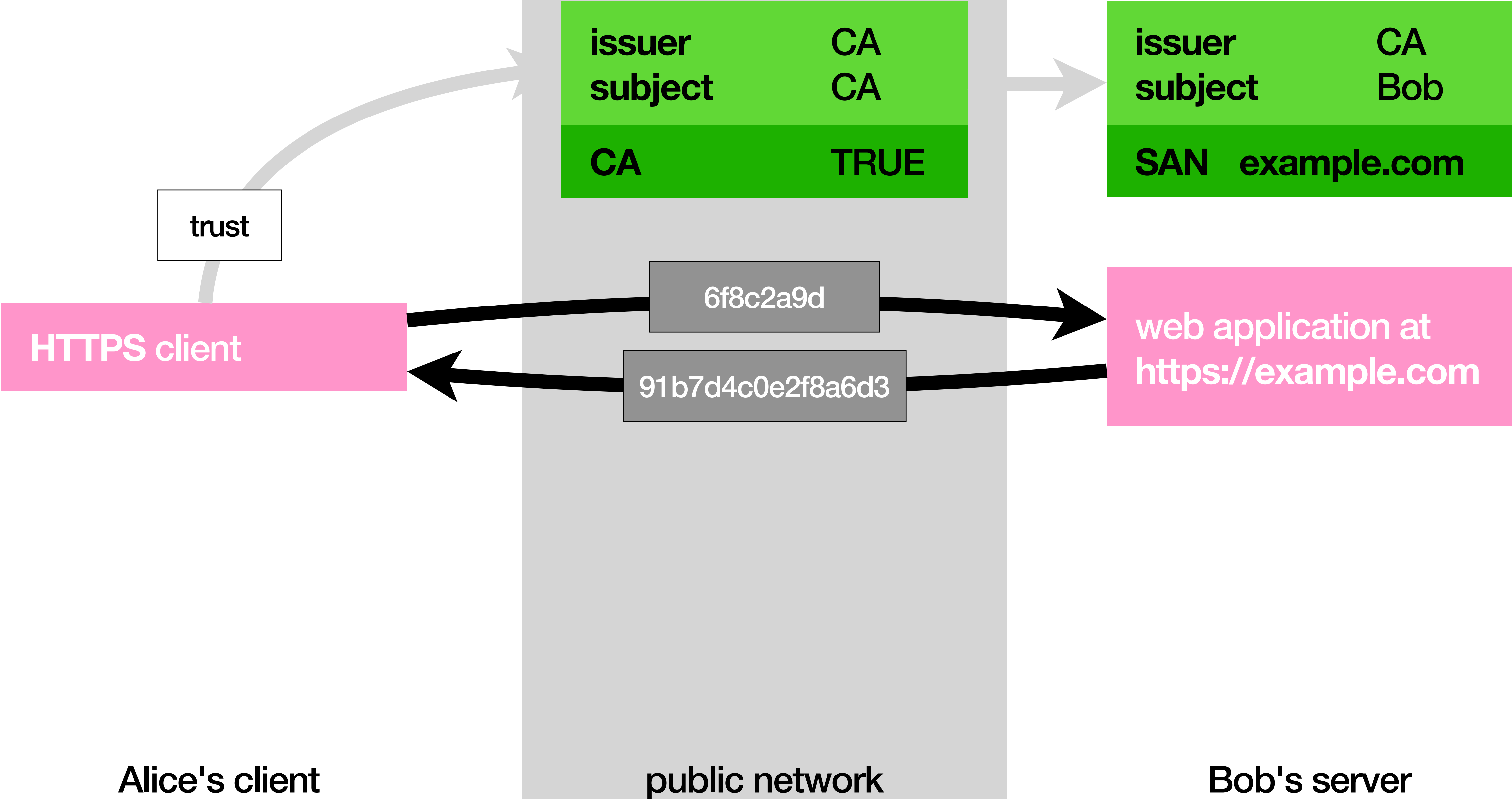




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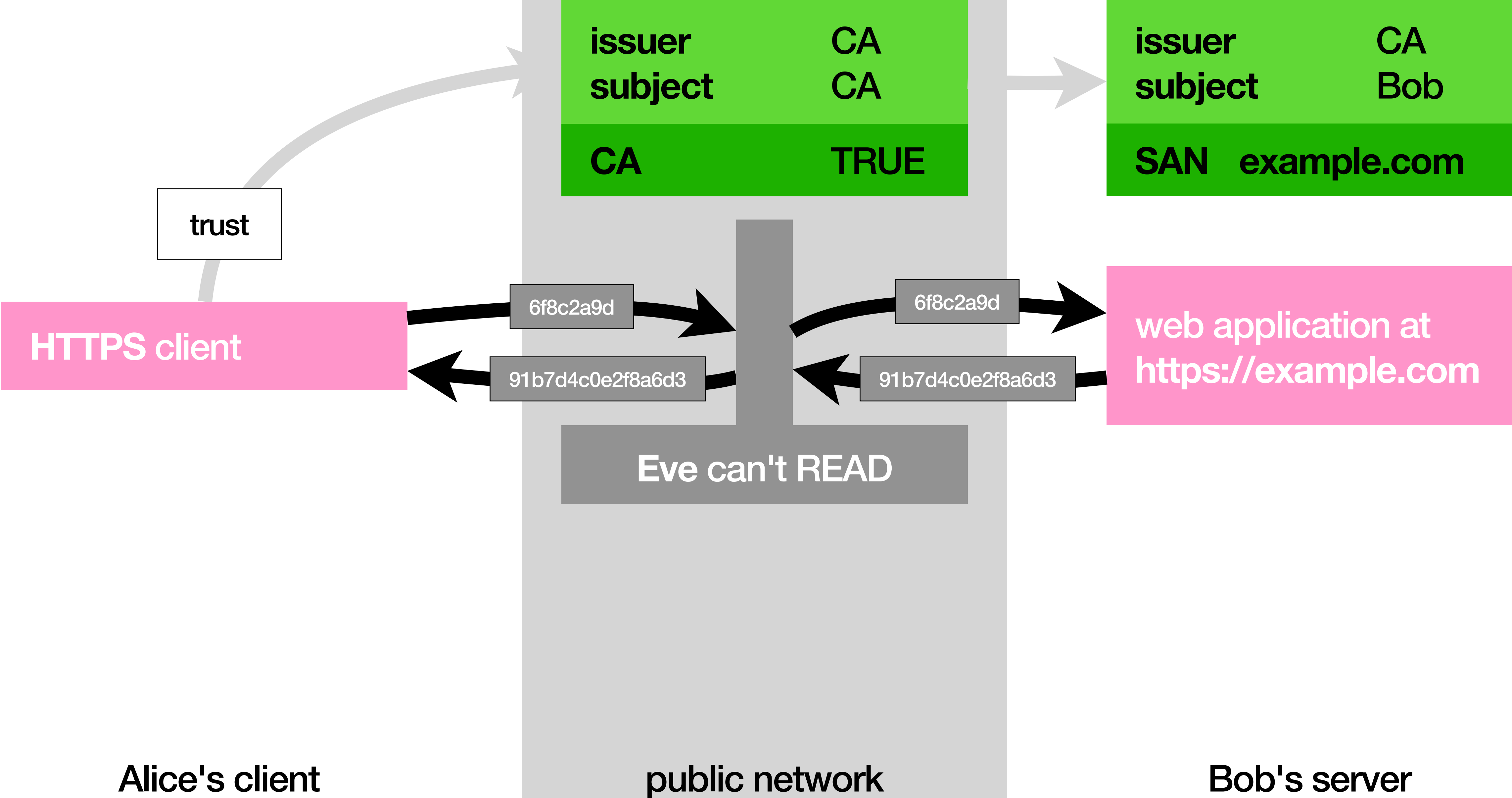


# MITM (aka Man-in-the-middle)

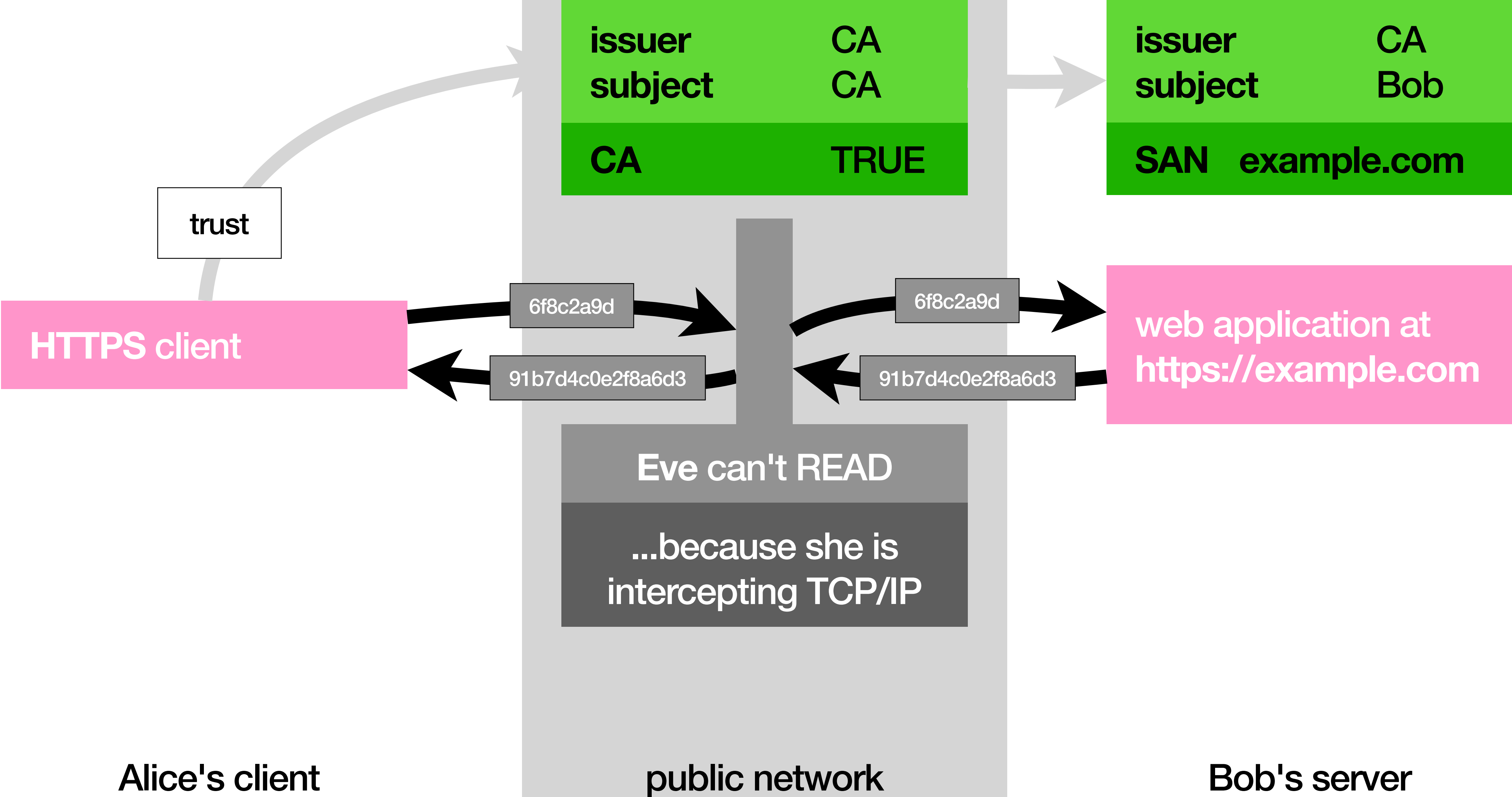




# MITM (aka Man-in-the-middle)

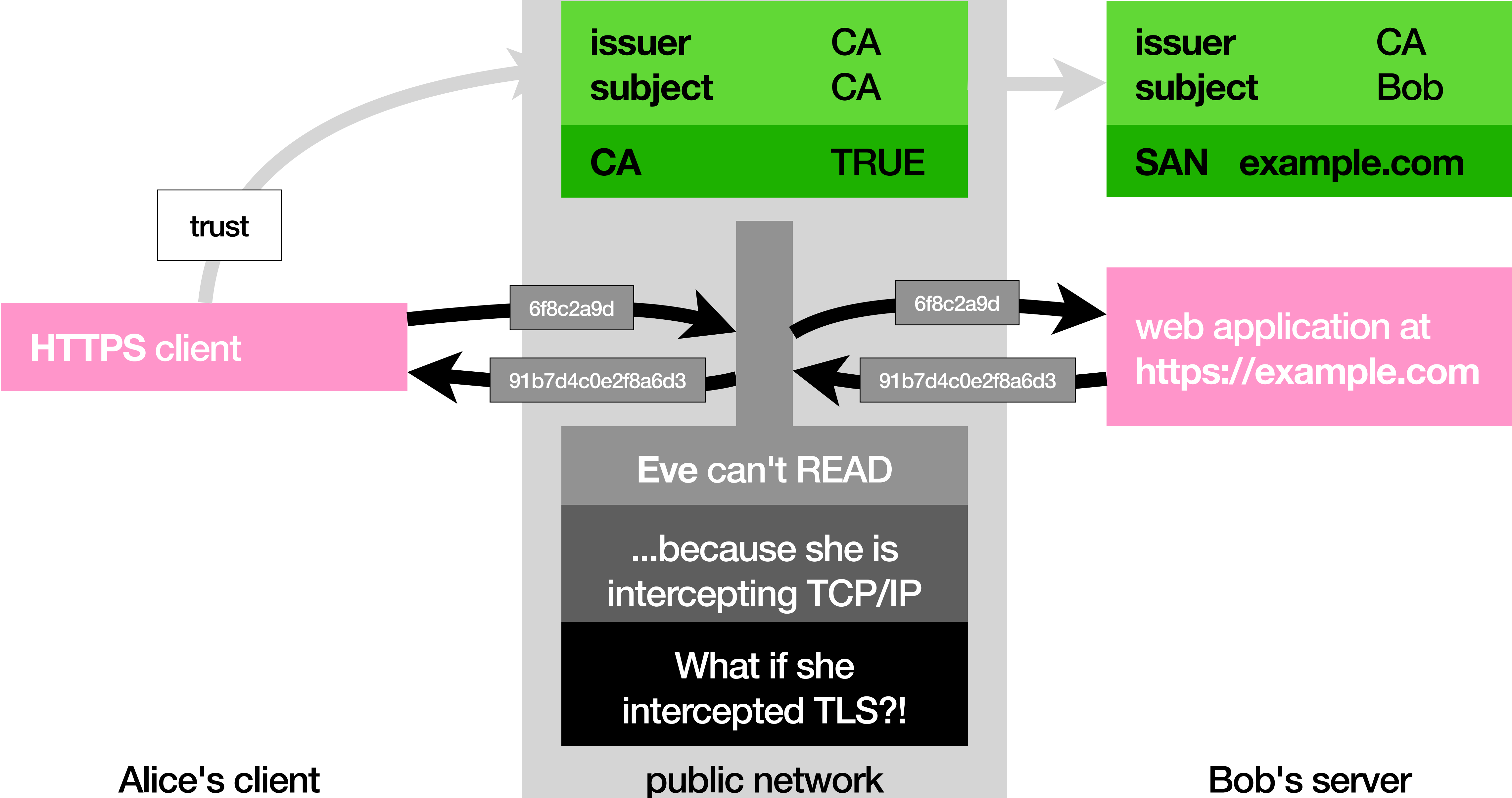


# MITM (aka Man-in-the-middle)





# MITM (aka Man-in-the-middle)



# MITM (aka Man-in-the-middle)

|         |      |
|---------|------|
| issuer  | CA   |
| subject | CA   |
| CA      | TRUE |

|             |     |
|-------------|-----|
| issuer      | CA  |
| subject     | Bob |
| example.com |     |

trust

HTTPS client

web application at  
<https://example.com>

Alice's client

public network

Bob's server



# MITM (aka Man-in-the-middle)

|         |      |
|---------|------|
| issuer  | CA   |
| subject | CA   |
| CA      | TRUE |

|             |     |
|-------------|-----|
| issuer      | CA  |
| subject     | Bob |
| example.com |     |



trust

HTTPS client

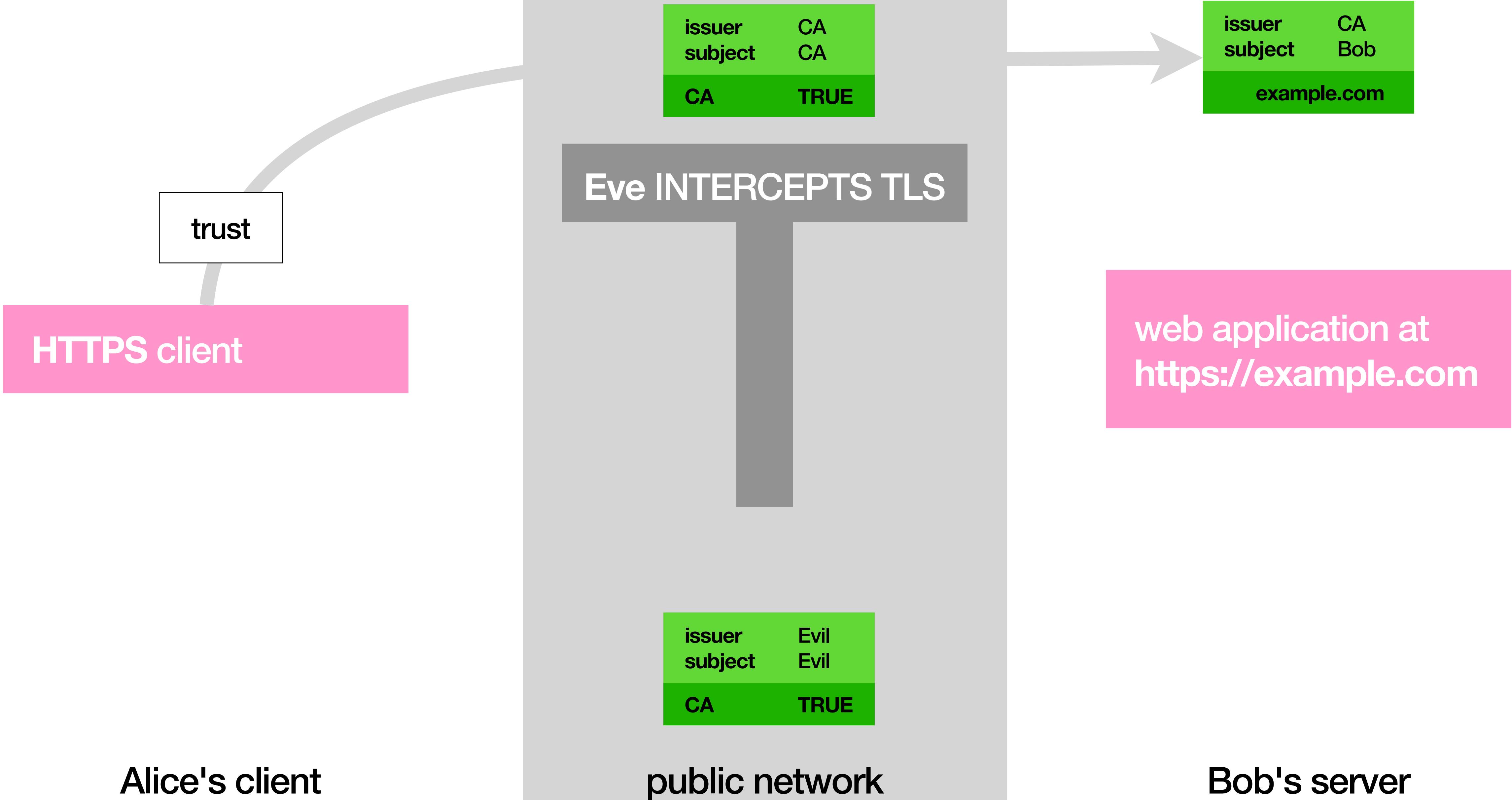
web application at  
<https://example.com>

Alice's client

public network

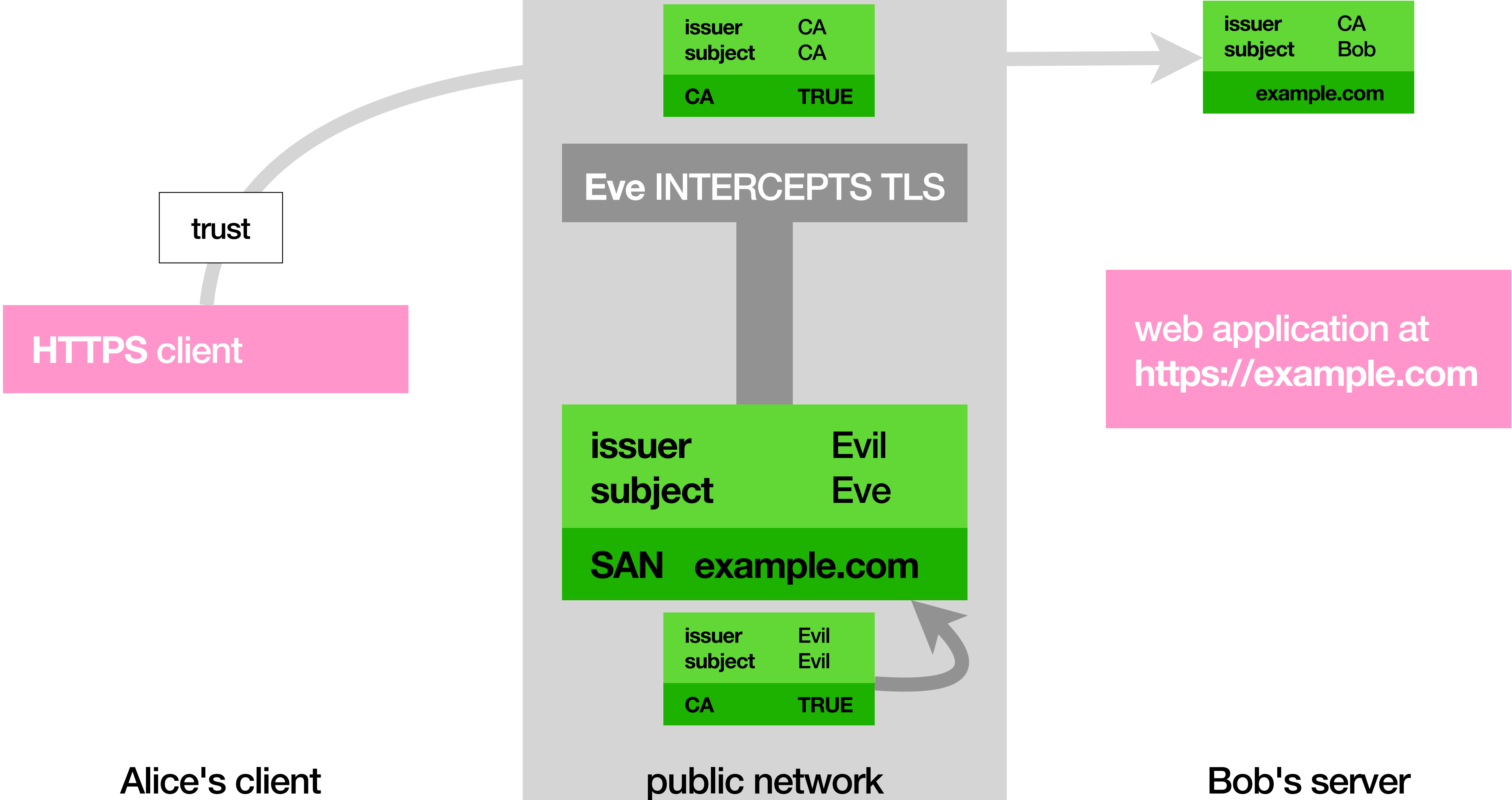
Bob's server

# MITM (aka Man-in-the-middle)

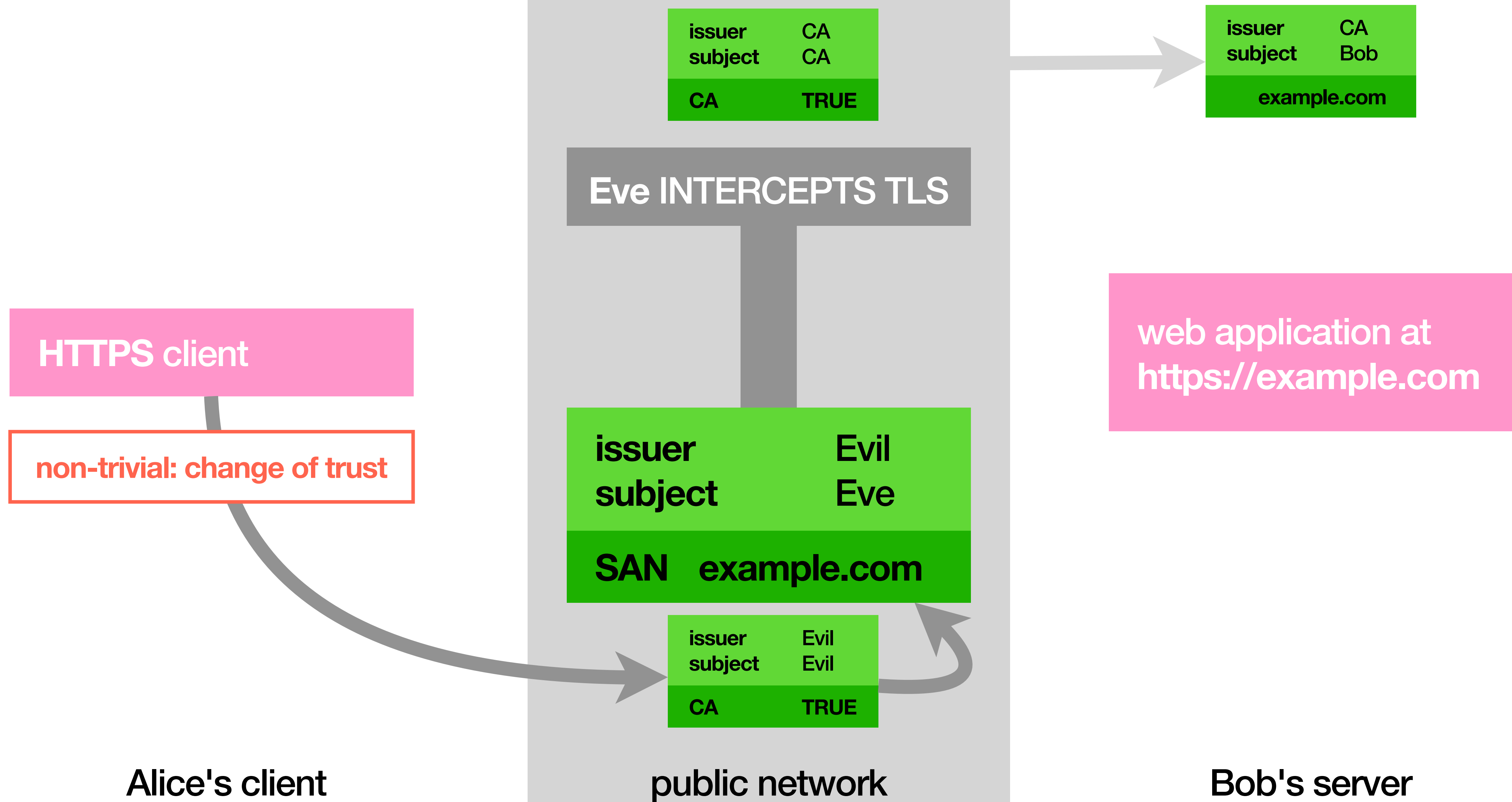




# MITM (aka Man-in-the-middle)

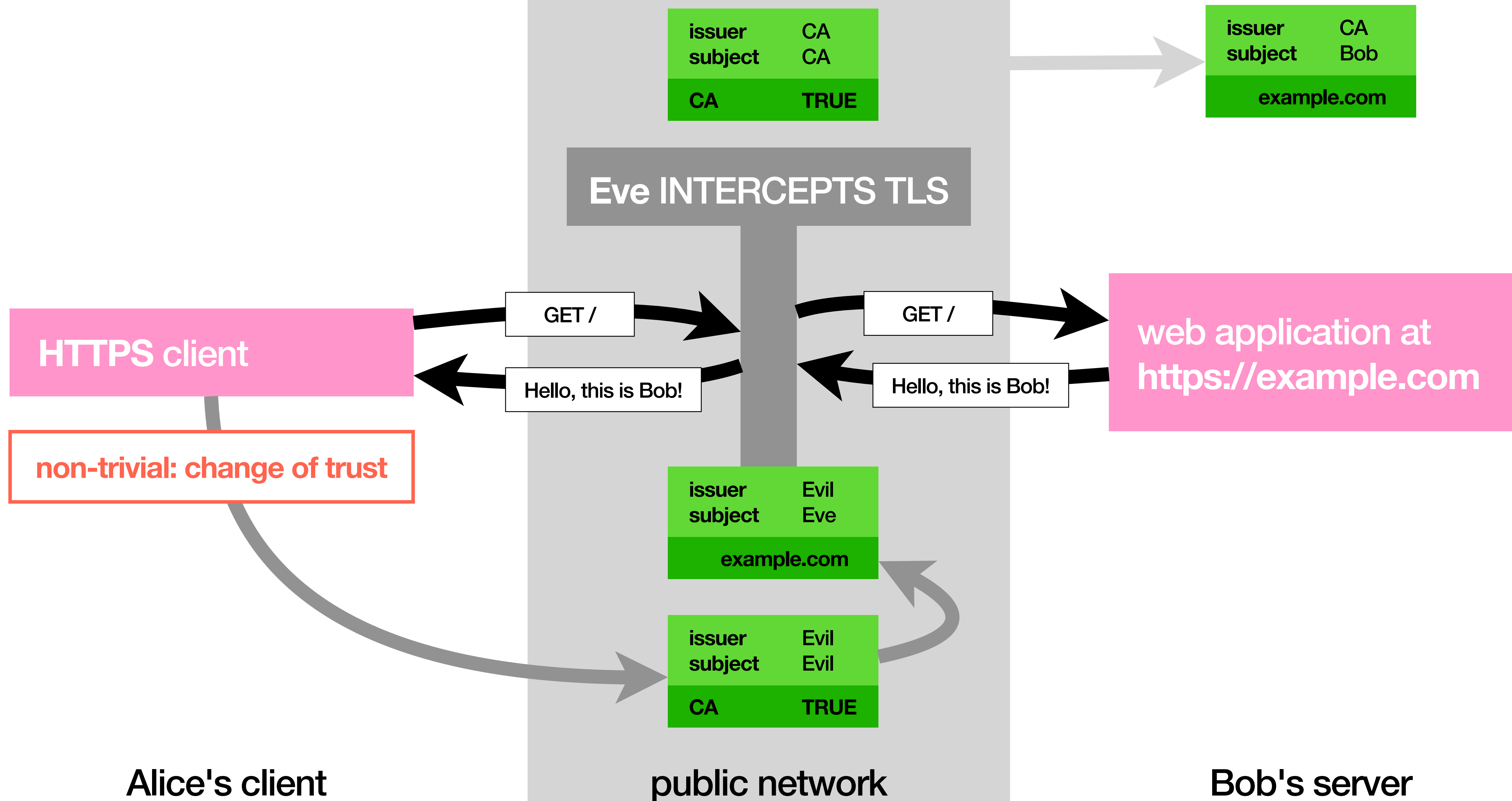


# MITM (aka Man-in-the-middle)

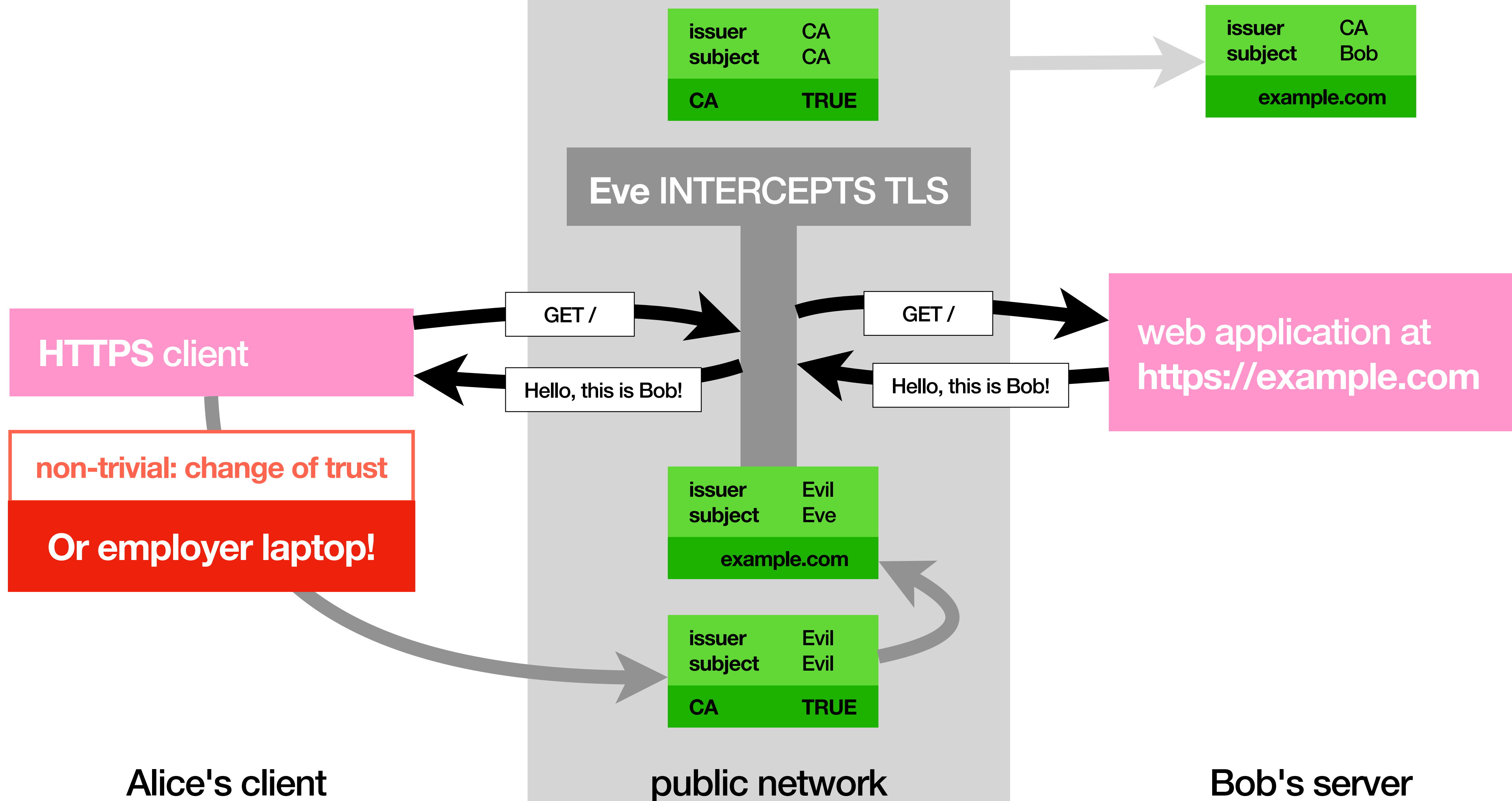




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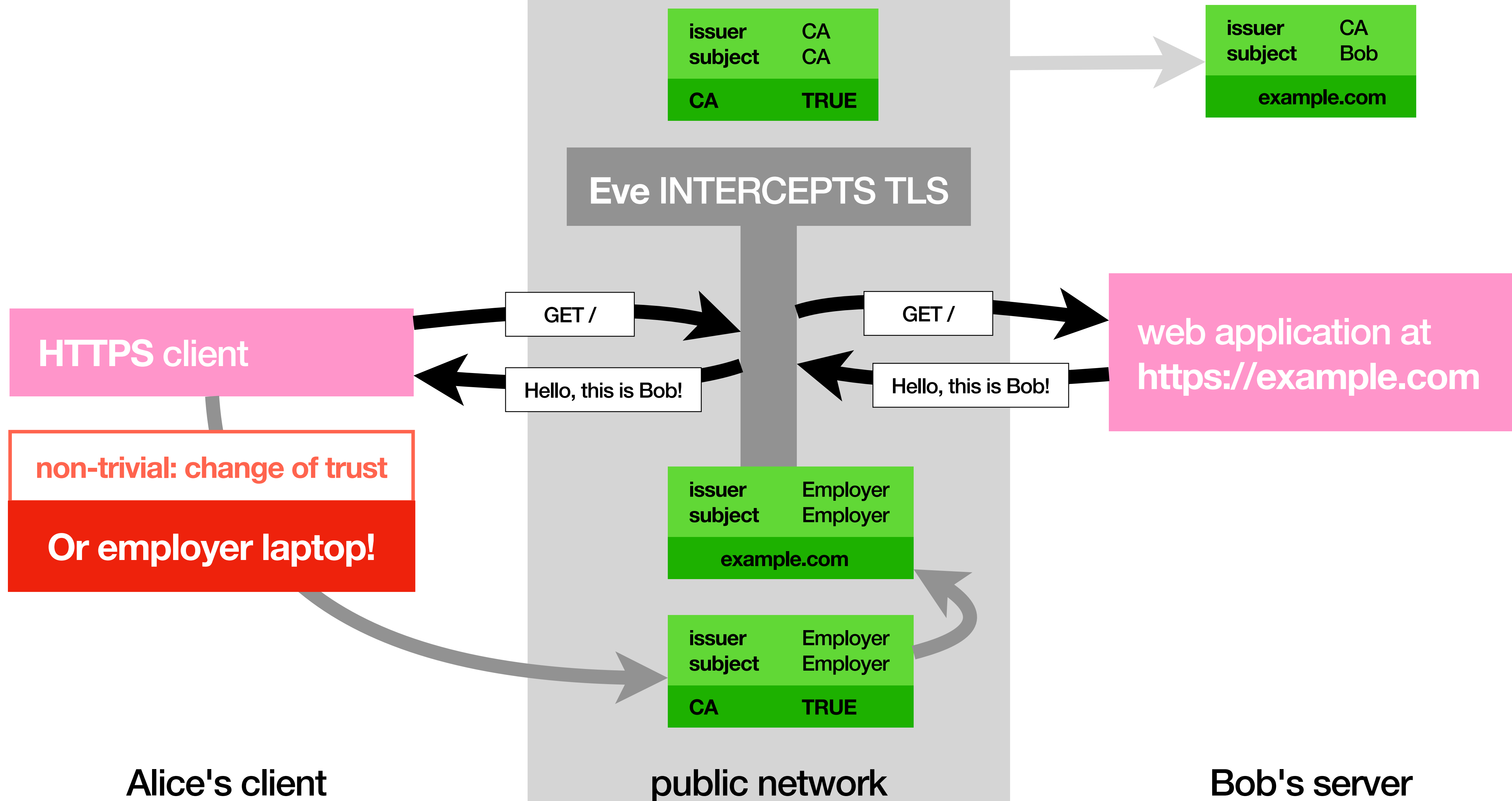


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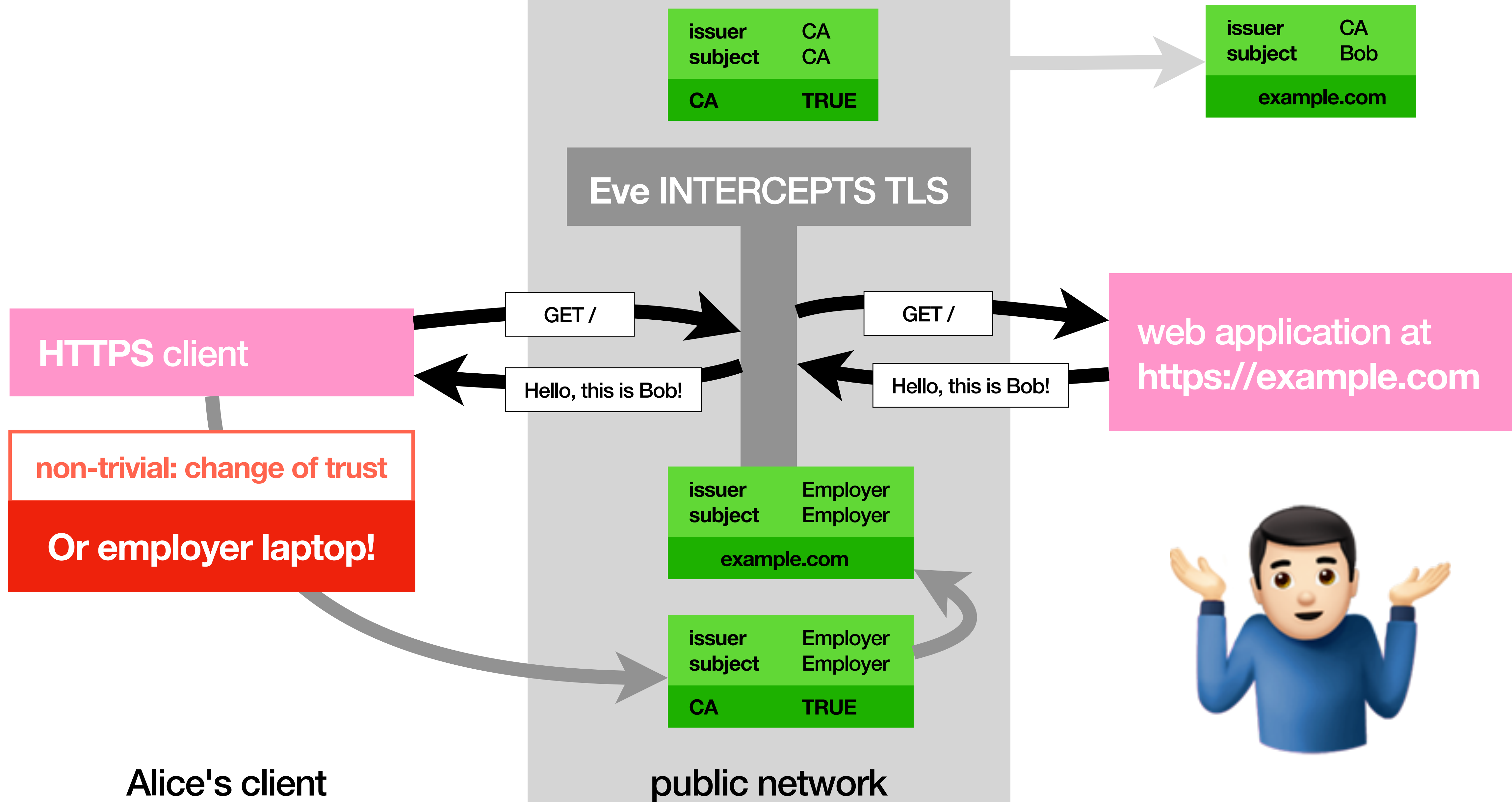




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# MITM (aka Man-in-the-middle)





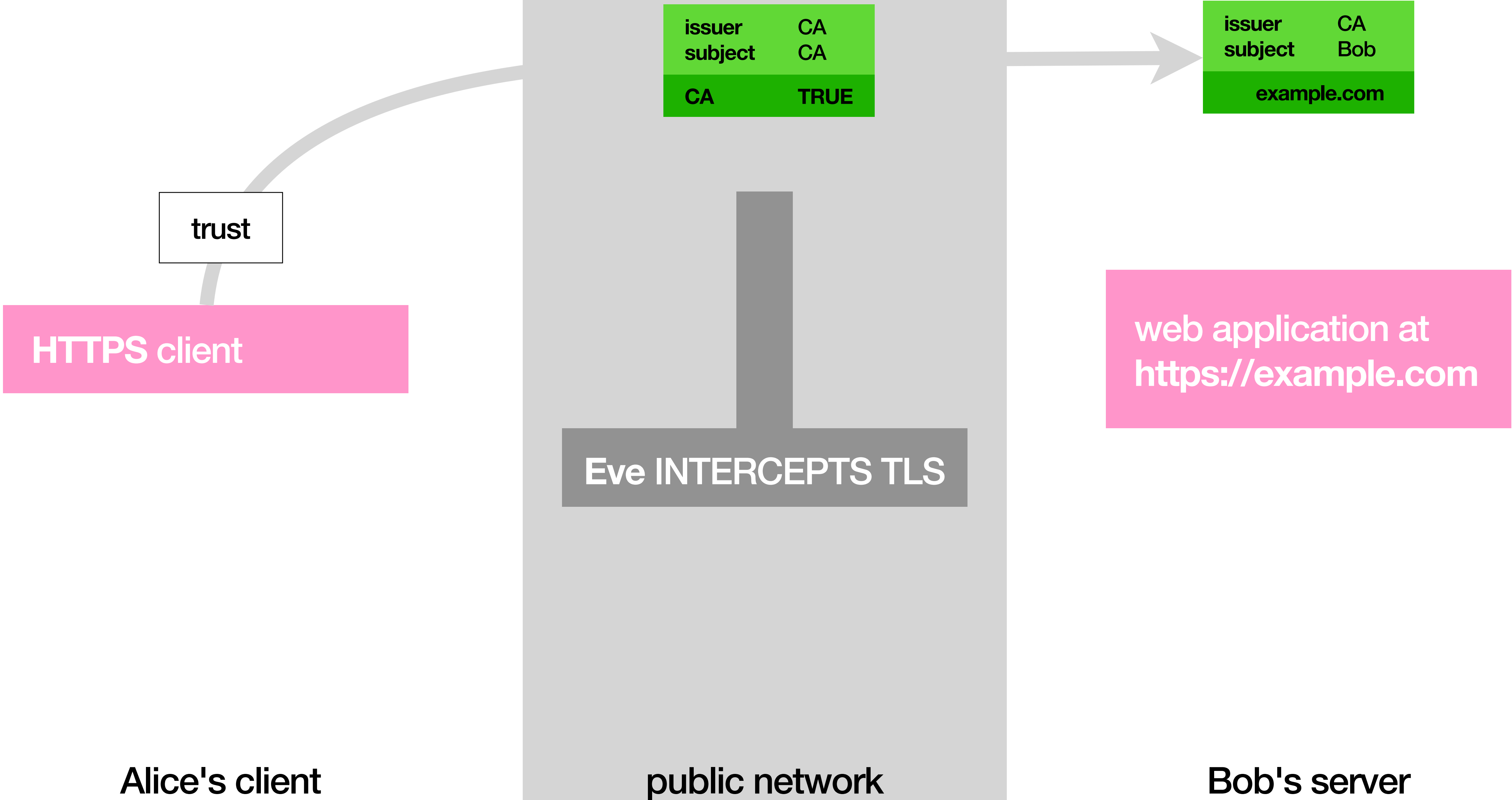
# MITM tk2 (aka the now-be-afraid version)

Alice's client

public network

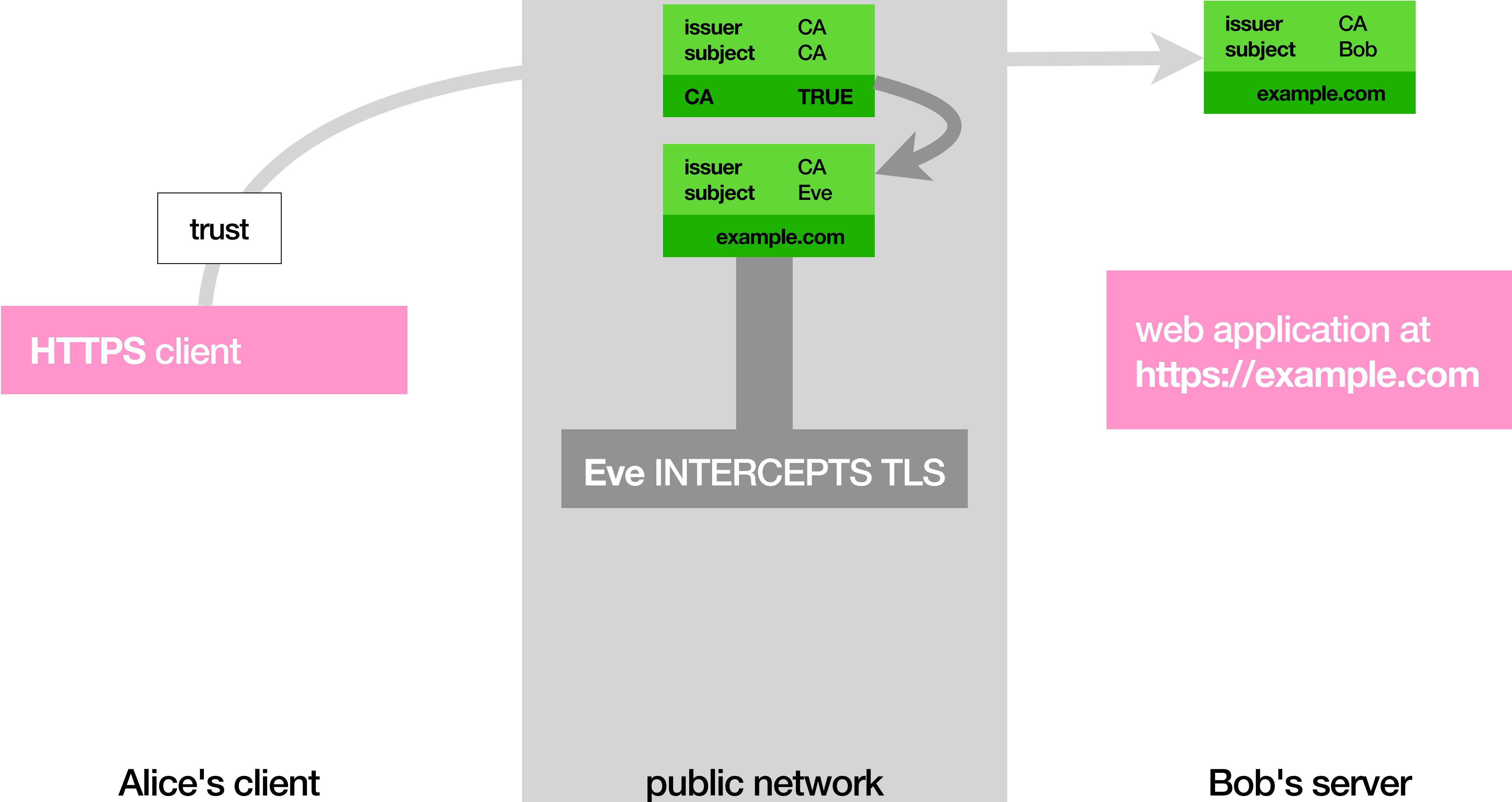
Bob's server

# MITM tk2 (aka the now-be-afraid version)

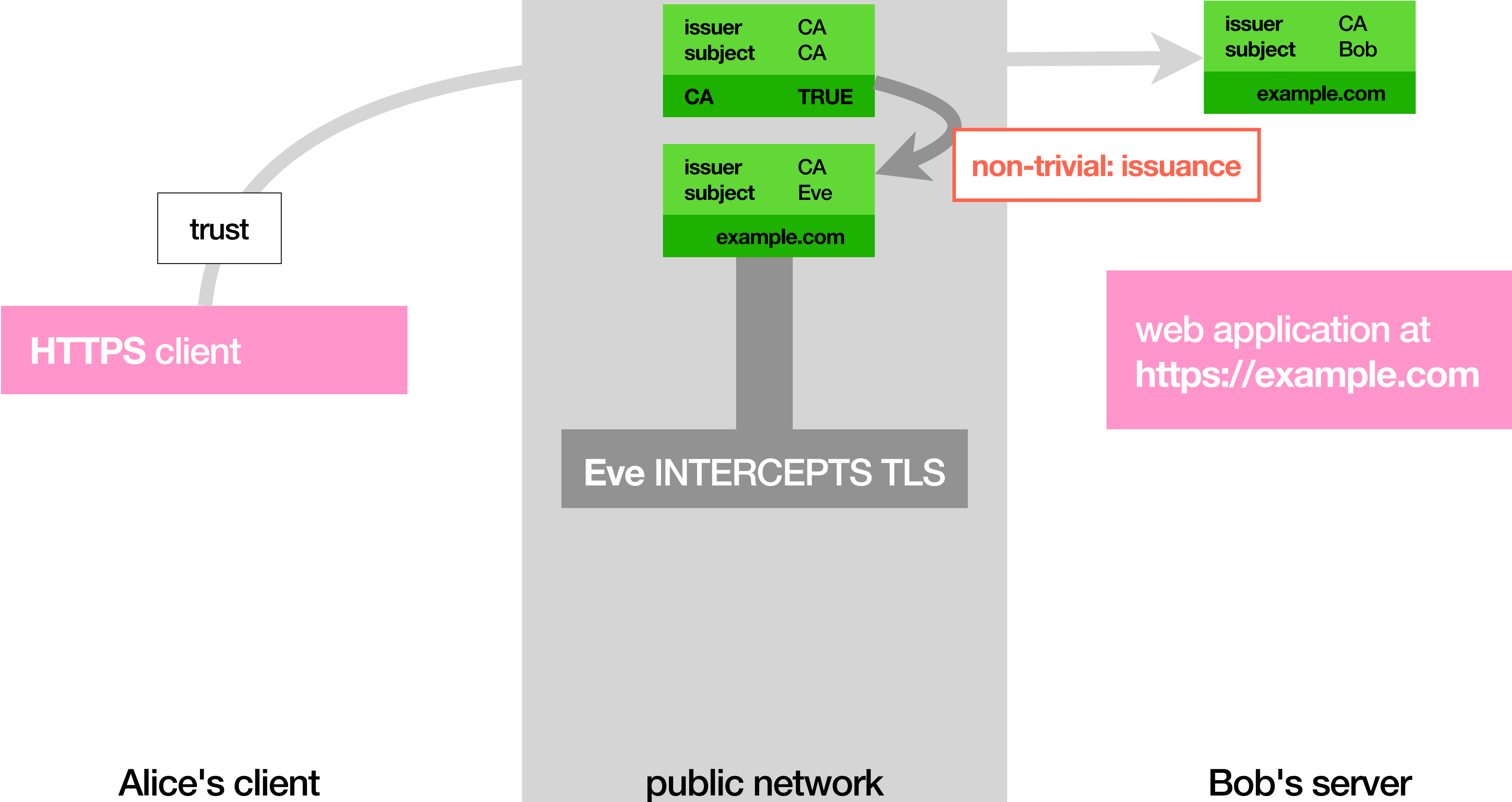




# MITM tk2 (aka the now-be-afraid version)

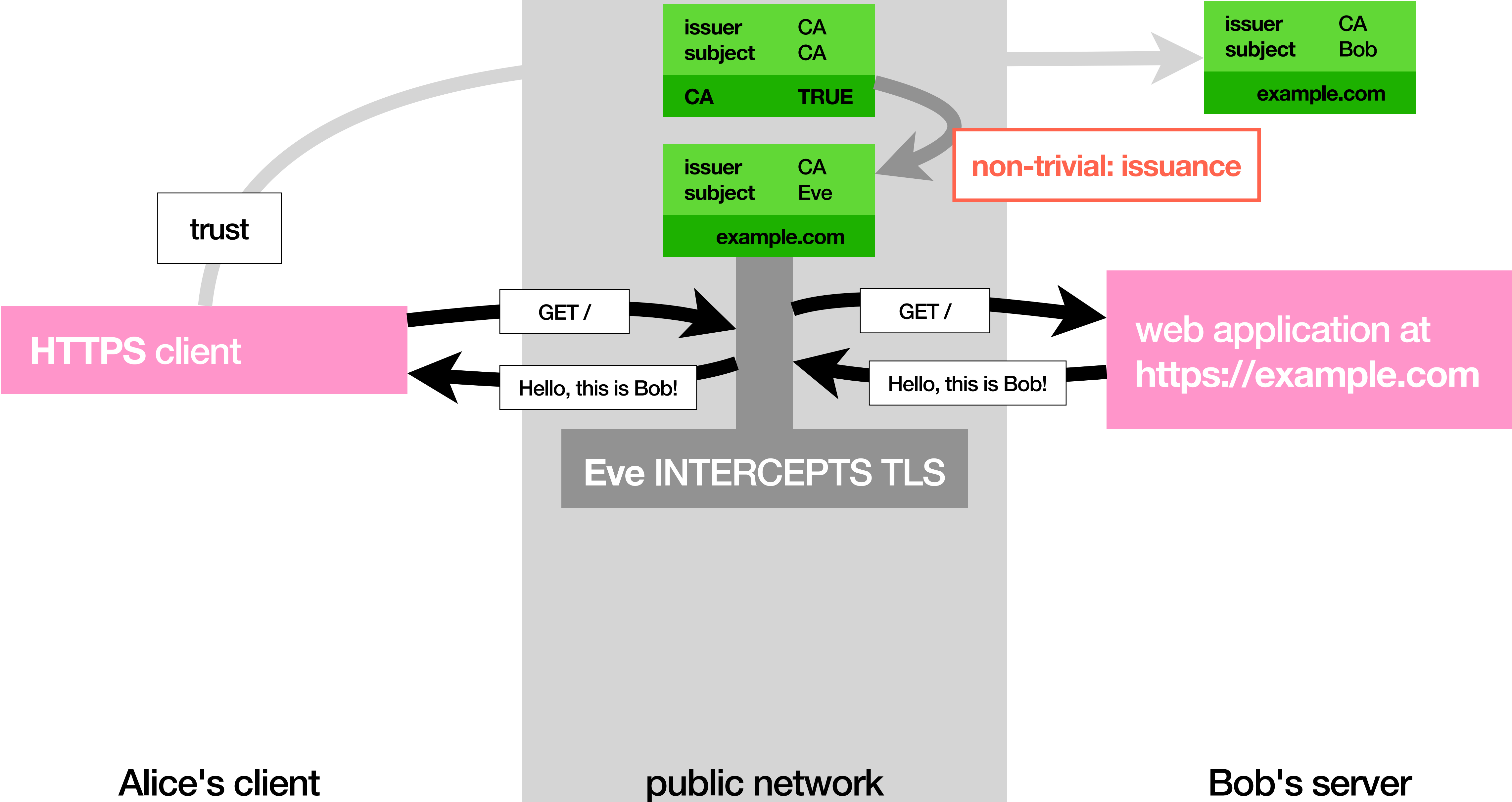


# MITM tk2 (aka the now-be-afraid version)

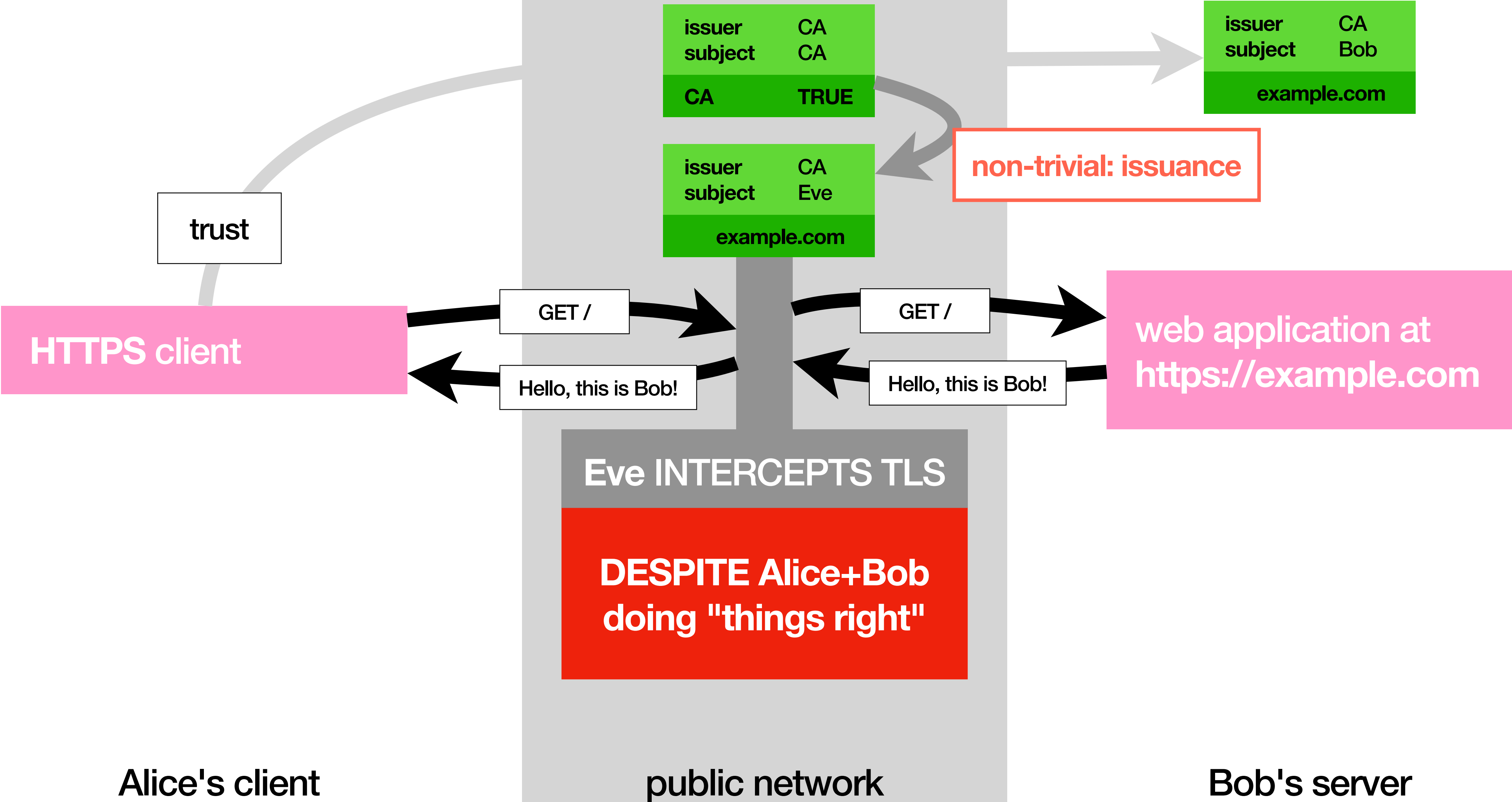




# MITM tk2 (aka the now-be-afraid version)

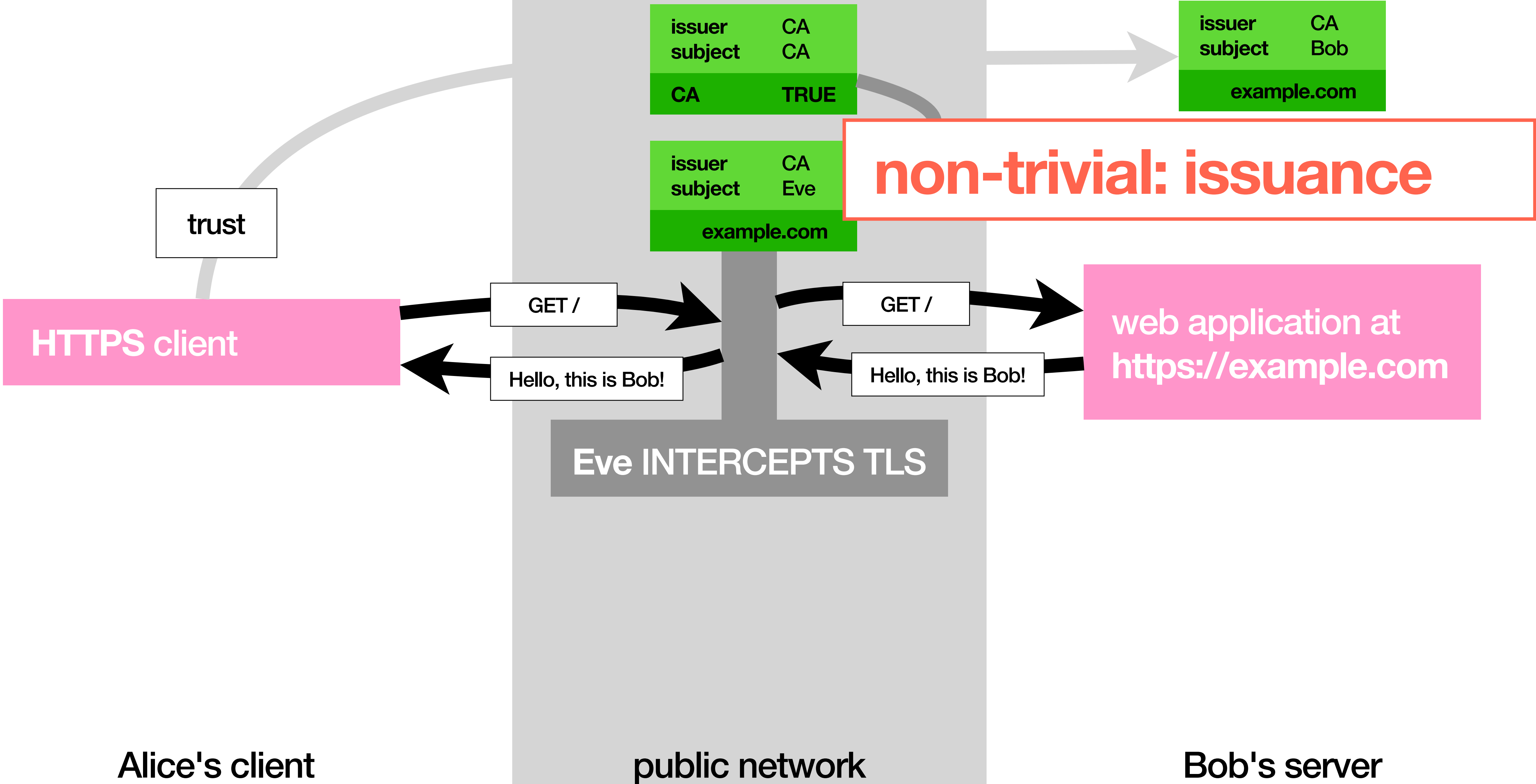


# MITM tk2 (aka the now-be-afraid version)

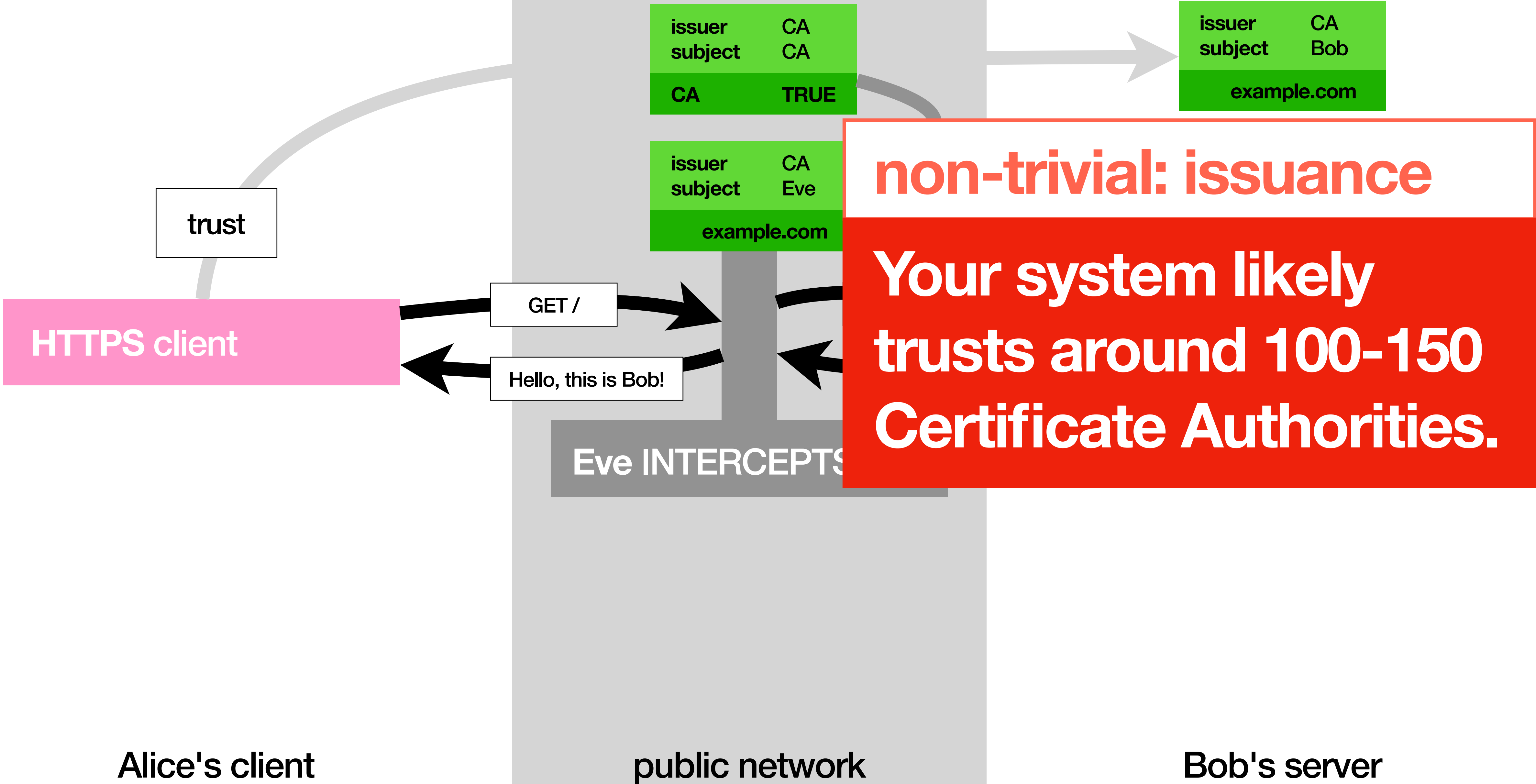




# MITM tk2 (aka the now-be-afraid version)

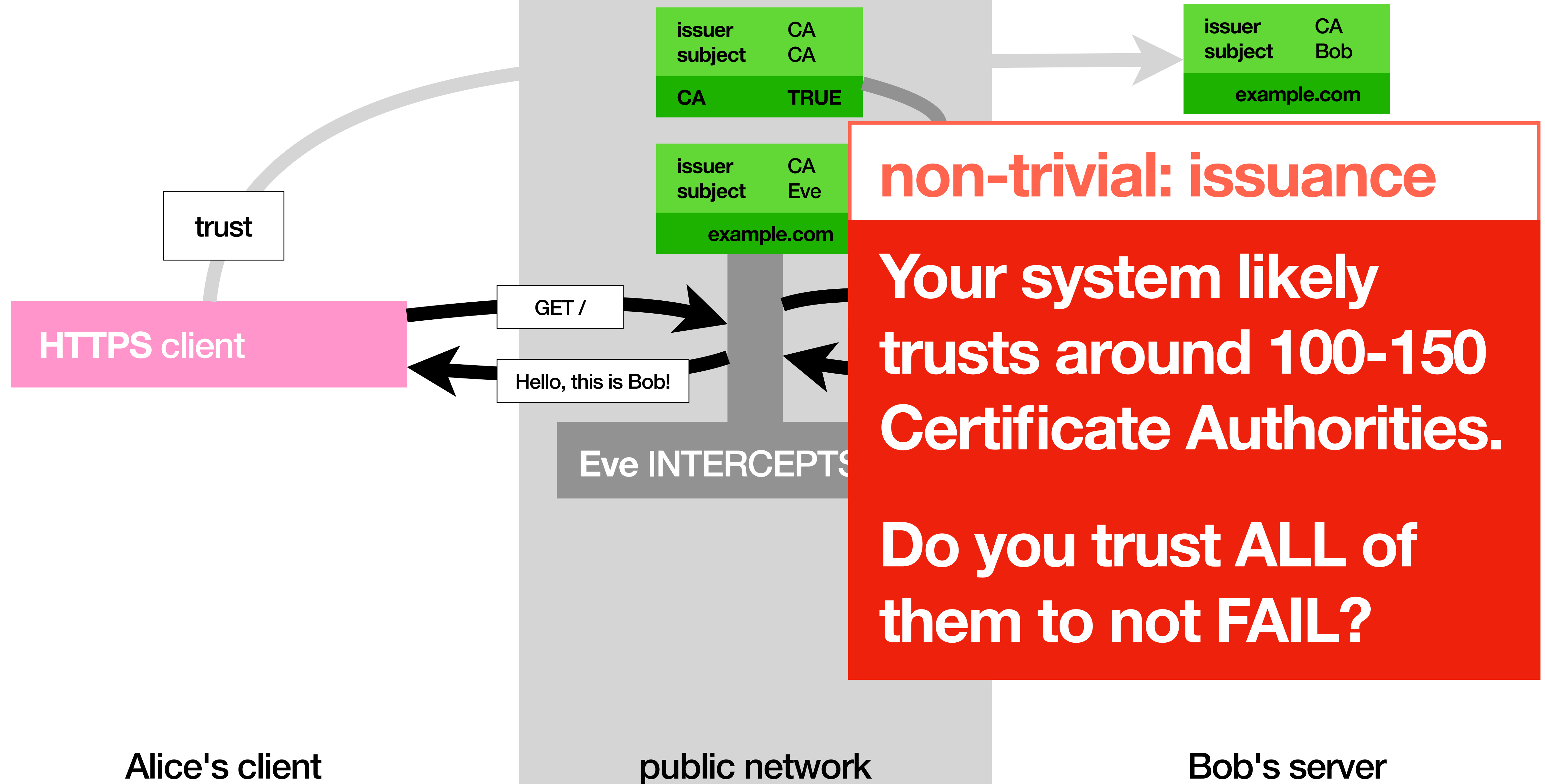


# MITM tk2 (aka the now-be-afraid version)

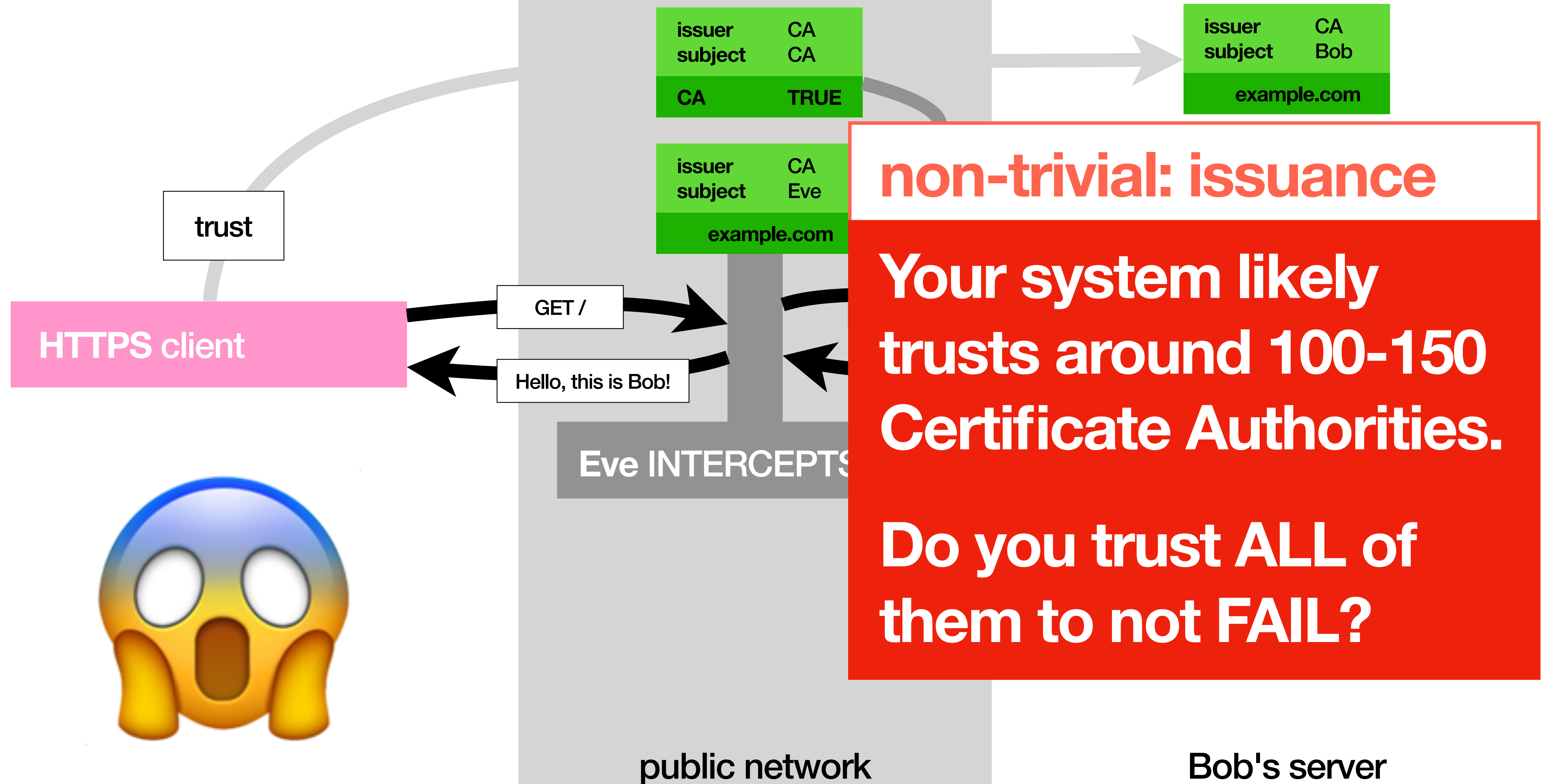




# MITM tk2 (aka the now-be-afraid version)



# MITM tk2 (aka the now-be-afraid version)





**MITM**

# MITM

**who do you trust?**





# Breaking It



# Breaking It

Exploring ways of breaking certificate validation.

**Break It**



# Break It

**who do you trust?**

**we're done**



**thanks**

# **Trust Me, I'm a Certificate Authority**

**PyCon PT 2026 Tutorial**

**Tiago Montes**



**DON'T**

# **Trust Me, I'm a Certificate Authority**

**PyCon PT 2026 Tutorial**

**Tiago Montes**